

**UNIVERSITY
OF TWENTE.**

**MODELLING, DYNAMICS AND
KINEMATICS**

LECTURE NOTES

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Preface

Sections marked with an asterisk (*) are considered background material.

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Part I

Mathematical Foundations

Chapter 1

Sets, Maps and Notation *

1.1 Logic and sets

The expression $P := Q$ reads as ‘ P is defined as Q ’. Note that the symbol ‘ $:=$ ’ is not the same as ‘ $=$ ’ (‘ P is defined as Q ’ is not the same as ‘ Q is defined as P ’).

Logical expressions are written using the following notation:

- $\Rightarrow$ Therefore / Implies
- $\Leftrightarrow$ If and only if / Equivalent
- $\forall$ Any / For all
- $\exists$ Exists
- $\nexists$ Does not exist
- $\exists!$ Exists unique

A *set* is informally a collection of *elements*. This is obviously not a proper definition since it immediately raises new questions: what is a ‘collection’ and what is an ‘element’? Instead of defining sets in terms of something else, they are defined axiomatically, i.e. by a list of properties that sets should satisfy. This, however, is beyond the scope of this course, and the curious reader is referred to Zermelo¹–Fraenkel² set theory.

Relations between sets are denoted by the following symbols:

- $\in$ Belongs to / Element of
- $\notin$ Does not belong to / Not an element of
- $\subseteq$ Subset
- $=$ Equal
- $\subset$ Proper subset

¹Ernst Zermelo (1871–1953), German mathematician

²Abraham Fraenkel (1891–1965), German-Israeli mathematician

The *empty set* is the unique set that has no elements and is denoted by $\emptyset$.

A possible way of defining a particular set is by listing its elements. Curly brackets (or braces) $\{\dots\}$ denote a set consisting of the elements listed inside the brackets, while parentheses $(\dots)$ denote an *ordered set* of the elements listed inside them.

Another way of defining a set is by specifying a property that its elements should satisfy. The set of elements in M satisfying property P is denoted as

$$\{x \in M \mid P(x)\}.$$

Example 1.1. Positive real numbers can be defined as a restriction of all real numbers:

$$\mathbb{R}^+ := \{x \in \mathbb{R} \mid x > 0\}.$$

Definition 1.2. The *Cartesian³ product* of two sets A and B is the set

$$A \times B := \{(a, b) \mid a \in A, b \in B\}$$

of all ordered pairs (a, b) with $a \in A$ and $b \in B$, denoted by $A \times B$.

Since we can identify $A \times (B \times C) \cong (A \times B) \times C$, it makes sense to consider the Cartesian product of more than two sets and write $A \times B \times C$.

Example 1.3.

$$\mathbb{R}^n := \underbrace{\mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R}}_{n \text{ times}}$$

$\mathbb{R}^n$ is thus a set of n -tuples of real numbers.

1.2 Maps

Definition 1.4. A *map* or a *function* f from a set A to a set B is a rule that assigns to each element $a \in A$ a unique element $f(a) \in B$, denoted by

$$\begin{aligned} f: A &\rightarrow B \\ a &\mapsto f(a). \end{aligned}$$

Set A is called the *domain* of f and set B is called the *target* of f .

Definition 1.5. A *composition* of two maps $f: A \rightarrow B$ and $g: B \rightarrow C$ is the map

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ & \searrow & & \nearrow & \\ & & g \circ f & & \end{array}$$

where

$$\begin{aligned} g \circ f: A &\rightarrow C \\ a &\mapsto g(f(a)). \end{aligned}$$

The expression $g \circ f$ can be read as ‘ g after f ’.

³René Descartes (1596–1650), French philosopher and mathematician. In Latin, his name is *Renatus Cartesius*, from which ‘Cartesian’ is derived.

The composition of maps is associative, i.e. for maps $f: A \rightarrow B$, $g: B \rightarrow C$ and $h: C \rightarrow D$ the following holds:

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

This property allows writing composition chains without parentheses, i.e. $h \circ g \circ f$.

Example 1.6.

$$\sqrt{(x+y)^2} := (\sqrt{\circ} \circ (\cdot)^2 \circ +)(x, y),$$

where

$$\begin{array}{ccccccc} \mathbb{R}^2 & \xrightarrow{+} & \mathbb{R} & \xrightarrow{(\cdot)^2} & \mathbb{R}_{\geq 0} & \xrightarrow{\sqrt{\cdot}} & \mathbb{R}_{\geq 0} \\ (x, y) & \mapsto & x + y & \mapsto & (x + y)^2 & \mapsto & \sqrt{(x + y)^2} \end{array}$$

Definition 1.7. The *identity map* on a set A is the map

$$\begin{aligned} \text{id}_A: A &\rightarrow A \\ a &\mapsto a. \end{aligned}$$

Definition 1.8. A map is called *injective* if it assigns distinct elements of the domain to distinct elements of the target, i.e. for a map $f: A \rightarrow B$:

$$\forall a_1, a_2 \in A : a_1 \neq a_2 \quad \Rightarrow \quad f(a_1) \neq f(a_2).$$

Definition 1.9. A map is called *surjective* if every element of the target is the image of some element of the domain, i.e. for a map $f: A \rightarrow B$:

$$\forall b \in B : \exists a \in A : f(a) = b.$$

A map can be injective but not surjective, surjective but not injective, both injective and surjective, or neither injective nor surjective. Come up with examples for each case.

Definition 1.10. A map is called *bijective* or *one-to-one* if it is both injective and surjective. Alternatively, a map $f: A \rightarrow B$ is bijective if

$$\forall b \in B : \exists! a \in A : f(a) = b.$$

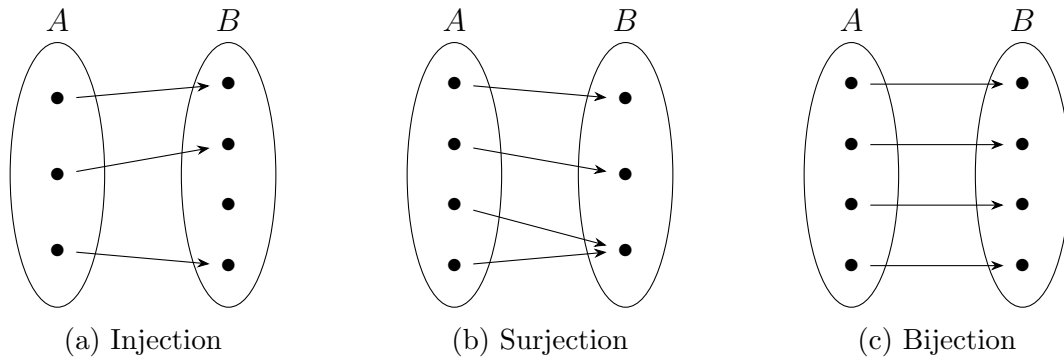


Figure 1.1: Map types

A map is bijective if and only if it is invertible, that is for a map $f: A \rightarrow B$ there exists a map $f^{-1}: B \rightarrow A$ such that $f^{-1} \circ f = \text{id}_A$ and $f \circ f^{-1} = \text{id}_B$. The map f^{-1} is called the *inverse* of f .

The only structure of a set has is its cardinality, i.e. the number of elements it contains. Therefore, the structure-preserving maps are the maps that preserve the cardinality, that is the bijective maps. Moreover, any bijection has the inverse which is also bijective. Thus, we call a bijective map an *isomorphism* of sets and the sets are called *isomorphic* if there exists a bijective map between them. An isomorphism of sets A and B is indicated by $A \simeq B$. In case there is a canonical choice of an isomorphism between two sets, we write $A \cong B$.

Chapter 2

Linear Algebra

2.1 Vector spaces *

2.1.1 Definition

Definition 2.1. A real *vector space* is a triple $(V, \oplus, \odot)$ consisting of a set V and two maps

$$\begin{aligned} \oplus: V \times V &\rightarrow V & \text{and} & & \odot: \mathbb{R} \times V &\rightarrow V \\ (v, w) &\mapsto v \oplus w & & & (\lambda, v) &\mapsto \lambda \odot v \end{aligned}$$

such that for all $u, v, w \in V$ and $\lambda, \mu \in \mathbb{R}$ the following eight axioms are satisfied:

$C^\oplus$	$u \oplus v = v \oplus u$	(Commutativity of $\oplus$)
$A^\oplus$	$u \oplus (v \oplus w) = (u \oplus v) \oplus w$	(Associativity of $\oplus$)
$N^\oplus$	$\exists O \in V : \forall v \in V : O \oplus v = v$	(Neutral element for $\oplus$)
$I^\oplus$	$\forall v \in V : \exists (-v) \in V : v \oplus (-v) = O$	(Inverse of element for $\oplus$)
$A^\odot$	$\lambda \odot (\mu \odot v) = (\lambda \cdot \mu) \odot v$	(Associativity of $\odot$)
$D^{\odot+}$	$(\lambda + \mu) \odot v = \lambda \odot v \oplus \mu \odot v$	(Distributivity of $\odot$ and $+$)
$D^{\odot\oplus}$	$\lambda \odot (u \oplus v) = \lambda \odot u \oplus \lambda \odot v$	(Distributivity of $\odot$ and $\oplus$)
$U^\odot$	$1 \odot v = v$	(Unit element for $\odot$)

Elements of a vector space are called *vectors*, whereas the elements of (the field) $\mathbb{R}$ are called *scalars*. The map $\oplus$ is called the *addition* and the map $\odot$ is called the *scalar multiplication*.

Note that in axioms $A^\odot$ and $D^{\odot+}$ we used addition $+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ and multiplication $\cdot: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ of real numbers. Vector space operations $\oplus$ and $\odot$ are different operations defined on different spaces, therefore, we use modified symbols to highlight this distinction. In practice, however, we usually drop the modified notation and use $+$ and $\cdot$ for the vector space operations as well.

Example 2.2. We can make the set $\mathbb{R}^n$ into a vector space by defining the addition

and scalar multiplication component-wise as:

$$\begin{aligned}(u^1, \dots, u^n) \oplus (v^1, \dots, v^n) &:= (u^1 + v^1, \dots, u^n + v^n) \\ \lambda \odot (v^1, \dots, v^n) &:= (\lambda \cdot v^1, \dots, \lambda \cdot v^n).\end{aligned}$$

Such a choice of the operations on $\mathbb{R}^n$ is called *standard* or *canonical* vector space structure on $\mathbb{R}^n$, and we usually denote $\oplus$ and $\odot$ simply by $+$ and $\cdot$, respectively.

Example 2.3. The set of all smooth functions on $\mathbb{R}^n$

$$\mathcal{C}^\infty(\mathbb{R}^n, \mathbb{R}) := \{f: \mathbb{R}^n \rightarrow \mathbb{R} \mid f \text{ is infinitely many times differentiable}\}$$

equipped with the pointwise addition and scalar multiplication

$$\begin{aligned}(f \oplus g)(x) &:= f(x) + g(x) \\ (\lambda \odot f)(x) &:= \lambda \cdot f(x)\end{aligned}$$

for all $x \in \mathbb{R}^n$ is a vector space. Again, thus defined operations are considered *standard* and the circles around $\oplus$ and $\odot$ are usually omitted.

Example 2.4. The set $\mathbb{R}^{m \times n}$ of all $m \times n$ matrices with real entries equipped with the entry-wise addition and scalar multiplication is a vector space.

In all three previous examples, we defined the operations using the addition and multiplication of real numbers. One could also choose the operations differently and obtain different vector space structures on the same sets.

In each case, one should check that the operations are well-defined and the axioms of a vector space are satisfied.

What are the structure-preserving maps between vector spaces?

Definition 2.5. Given two real vector spaces $(V, \oplus, \odot)$ and $(W, \boxplus, \boxodot)$ a map $\varphi: V \rightarrow W$ that satisfies

$$\begin{aligned}\varphi(u \oplus v) &= \varphi(u) \boxplus \varphi(v) \\ \varphi(\lambda \odot v) &= \lambda \boxodot \varphi(v)\end{aligned}$$

for all $u, v \in V$ and $\lambda \in \mathbb{R}$ is called a *linear map*.

If the map φ is also bijective, it is called a *linear isomorphism* and the vector spaces are called *isomorphic*.

Proposition 2.6. A composition $\varphi \circ \psi$ of two linear maps ψ and φ is a linear map.

Example 2.7. Consider the space of all continuously differentiable functions $\mathcal{C}^1(\mathbb{R}, \mathbb{R})$ and the space of all continuous functions $\mathcal{C}(\mathbb{R}, \mathbb{R})$ equipped with the pointwise addition and scalar multiplication. The derivative map

$$\begin{aligned}\mathbf{D}: \mathcal{C}^1(\mathbb{R}, \mathbb{R}) &\rightarrow \mathcal{C}(\mathbb{R}, \mathbb{R}) \\ f &\mapsto \mathbf{D} f := \frac{df}{dx}\end{aligned}$$

is a linear map.

2.1.2 Basis and components of vectors

Definition 2.8. A *basis*¹ of a vector space $(V, +, \cdot)$ is any subset $B \subseteq V$ such that

(i) any finite subset $\{e_1, \dots, e_n\} \subseteq B$ is linearly independent, i.e.

$$\lambda^1 \cdot e_1 + \dots + \lambda^n \cdot e_n = \sum_{i=1}^n \lambda^i \cdot e_i = 0 \quad \Rightarrow \quad \lambda^1 = \dots = \lambda^n = 0$$

(ii) for any $v \in V$, there is a finite subset $\{e_1, \dots, e_n\} \subseteq B$ and scalars $v^1, \dots, v^n \in \mathbb{R}$ such that v decomposes as

$$v = \sum_{i=1}^n v^i \cdot e_i.$$

One can prove that every vector space admits a basis.

Definition 2.9. The *dimension* of a vector space V is the cardinality (number of elements) in any of its bases, denoted by $\dim V := |B|$. If $\dim V < \infty$ the vector space is called *finite-dimensional*, otherwise it is called *infinite-dimensional*.

The definition above is well-defined since one can prove that for any two bases B and B' of a vector space, $|B| = |B'|$.

Example 2.10. On the space $\mathcal{C}([0, 2\pi], \mathbb{R})$ of all continuous functions on the interval $[0, 2\pi]$ equipped with the pointwise addition and scalar multiplication, the set of functions $\{1, \sin(nx), \cos(nx) \mid n \in \mathbb{N}\}$ forms a basis called the *Fourier*² *basis*. This is, therefore, an example of an infinite-dimensional vector space.

Definition 2.11. In an n -dimensional vector space V , the *components* (or *coordinates*) of a vector $v \in V$ with respect to an ordered basis $(e_1, \dots, e_n)$ are n scalars $(v^1, \dots, v^n)$ such that

$$v = \sum_{i=1}^n v^i \cdot e_i.$$

Having chosen a basis, components of a vector are uniquely determined due to the linear independence of the basis.

The i -th component of a vector v is commonly denoted by v^i with a superscript, whereas the i -th basis vector in an ordered basis is denoted by e_i with a subscript. This convention will become especially useful when dealing with components of tensors.

Later, we will give another, more sophisticated definition of the components of a vector using so-called *dual basis*.

¹This defines what is called a *Hamel basis* of a vector space. Note that only finite sums were considered here. Instead, one could allow infinite series and define a so-called *Schauder basis*, however, this would require a topology on the vector space to deal with series convergence.

²Joseph Fourier (1768–1830), French mathematician and physicist.

Example 2.12. Consider a vector space $(\mathbb{R}^2, +, \cdot)$ with the standard addition and scalar multiplication. What components does a vector $v = (3, 5) \in \mathbb{R}^2$ has? This question is meaningless unless a basis is chosen. We only speak of components with respect to a basis. Let us choose a basis $e_1 = (3, 1)$, $e_2 = (0, 2)$. Then the vector v decomposes as

$$v = (3, 5) = (3, 1) + 2 \cdot (0, 2) = 1 \cdot e_1 + 2 \cdot e_2,$$

thus the components of $(3, 5)$ in this basis are $(1, 2)$.

Definition 2.13. Let $(V, \oplus, \odot)$ be an n -dimensional vector space and $(e_1, \dots, e_n)$ be its basis. The *coordinatisation map* associated with the basis is the map

$$\begin{aligned} \psi_e: V &\rightarrow \mathbb{R}^n \\ v &\mapsto \psi_e(v) := (v^1, \dots, v^n), \end{aligned}$$

where $(v^1, \dots, v^n)$ are the components of v in that basis.

Proposition 2.14. *The coordinatisation map ψ_e is a linear isomorphism between $(V, \oplus, \odot)$ and $(\mathbb{R}^n, +, \cdot)$ with the standard addition and scalar multiplication.*

As a consequence, all n -dimensional vector spaces are isomorphic to $\mathbb{R}^n$ and each other.

This property also allows one to choose a basis and work in components using $\mathbb{R}^n$ instead of the abstract vector space V . However, this must be done with caution, since the choice of basis is not unique and the results one obtains in components depend on this choice. Some people say:

‘A gentleman only chooses a basis if he must.’

2.1.3 Components of linear maps

Let $\varphi: V^{(n)} \rightarrow W^{(m)}$ be a linear map from an n -dimensional vector space V to an m -dimensional vector space W . Let us choose bases $(e_1, \dots, e_n)$ and $(f_1, \dots, f_m)$ in V and W , respectively.

Suppose φ applied to a vector $v \in V$ is $w := \varphi(v) \in W$. Then decomposing v and using the linearity of φ we obtain

$$w = \varphi(v) = \varphi\left(\sum_{i=1}^n v^i \cdot e_i\right) = \sum_{i=1}^n \varphi(v^i \cdot e_i) = \sum_{i=1}^n v^i \cdot \varphi(e_i).$$

Note that for each i , $\varphi(e_i)$ is some vector in W , and therefore can be decomposed with respect to the basis $(f_1, \dots, f_m)$:

$$\forall i \in \{1, \dots, n\}: \quad \varphi(e_i) = \sum_{j=1}^m (\varphi(e_i))^j \cdot f_j =: \sum_{j=1}^m \varphi^j_i \cdot f_j,$$

where for all $i \in \{1, \dots, n\}$ and $j \in \{1, \dots, m\}$, $\varphi^j_i \in \mathbb{R}$ is scalar. Substituting this into the expression for w we obtain

$$w = \varphi(v) = \sum_{i=1}^n v^i \cdot \left(\sum_{j=1}^m \varphi^j_i \cdot f_j \right) = \sum_{j=1}^m \left(\sum_{i=1}^n \varphi^j_i v^i \right) \cdot f_j = \sum_{j=1}^m w^j \cdot f_j.$$

Thus, for all $j \in \{1, \dots, m\}$, the numbers

$$w^j = \sum_{i=1}^n \varphi^j_i v^i \in \mathbb{R} \quad (2.1)$$

are the components of $\varphi(v) = w$ in the basis of W . As a result, having chosen bases in V and in W , the action of φ on an arbitrary $v \in V$ is encoded in $n \cdot m$ numbers φ^j_i . These numbers are called the *components of the linear map φ* with respect to the chosen bases.

Being delighted by this result, people came up with some ‘witchcraft’ called matrix multiplication. We can write components of vectors by stacking them into columns and arranging components of linear maps into rectangular tables of numbers called *matrices*. Then, the action of a linear map on a vector

$$\varphi(v) = w$$

can be written as

$$\begin{pmatrix} \varphi^1_1 & \varphi^1_2 & \cdots & \varphi^1_n \\ \vdots & \vdots & \ddots & \vdots \\ \varphi^m_1 & \varphi^m_2 & \cdots & \varphi^m_n \end{pmatrix} \cdot \begin{pmatrix} v^1 \\ v^2 \\ \vdots \\ v^n \end{pmatrix} = \begin{pmatrix} w^1 \\ \vdots \\ w^m \end{pmatrix},$$

where we invent the matrix multiplication rule (‘row times column’) to replicate the component expression in (2.1). This means that whenever one represents a linear map by a matrix, one has chosen bases and is working in components.

The matrix $\Phi := (\varphi^j_i)_{j \in \{1, \dots, m\}, i \in \{1, \dots, n\}}$ composed of the components φ^j_i of the linear map φ is called the *coordinate representation* of the linear map φ with respect to the chosen bases.

We now introduce one more notation. Let us denote the columns with the components of vectors with the bold font, e.g.

$$\mathbf{v} := \begin{pmatrix} v^1 \\ \vdots \\ v^n \end{pmatrix} \in \mathbb{R}^n.$$

This notation helps to write the expressions in components more compactly, however, it hides a lot of information.

Using this notation, the action of a linear map φ on a vector v in components can be written as

$$\Phi \cdot v = w.$$

Another point of view on it is obtained by considering coordinatisation maps ψ_e and ψ_f . By Proposition 2.14 they are linear isomorphisms and hence preserve all the vector-space properties. It is therefore natural to require that the following diagram commutes (going along the arrows both ways gives the same result):

$$\begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ \psi_e \downarrow & & \downarrow \psi_f \\ \mathbb{R}^n & \xrightarrow{\Phi} & \mathbb{R}^m \end{array}$$

The diagram says that first going from V to components in $\mathbb{R}^n$ and then applying Φ gives the same result as first applying φ and then going from W to components in $\mathbb{R}^m$. Then $\Phi = \psi_f \circ \varphi \circ \psi_e^{-1}$ is a linear map between $\mathbb{R}^n$ and $\mathbb{R}^m$ which is represented by a matrix.

Exercise 2.15. Consider a composition of two linear maps $\theta: U \rightarrow V$ and $\varphi: V \rightarrow W$ with U , V and W having the dimensions k , n and m , respectively. Let $(g_1, \dots, g_k)$, $(e_1, \dots, e_n)$ and $(f_1, \dots, f_m)$ be bases in U , V and W . Show that the coordinate representation of the composition $\varphi \circ \theta$ is given by the matrix product $\Phi \cdot \Theta$ of the coordinate representations Φ of φ and Θ of θ . That is, first express the equation $w = (\varphi \circ \theta)(u)$ in components and then verify that the matrix multiplication rule ‘row times column’ replicates this expression.

2.1.4 Change of basis

Let $(V, +, \cdot)$ be an n -dimensional vector space with a chosen basis $(e_1, \dots, e_n)$. Let us now choose a new basis $(\tilde{e}_1, \dots, \tilde{e}_n)$. Each element $\tilde{e}_j$ of the new basis is a vector in V and hence can be expressed in components in the old basis:

$$\tilde{e}_j = \sum_{i=1}^n (\tilde{e}_j)^i \cdot e_i =: \sum_{i=1}^n A^i_j \cdot e_i.$$

The transformation between two bases can thus be encoded in the n^2 numbers $A^i_j \in \mathbb{R}$. We can compose these numbers into a square matrix $A \in \mathbb{R}^{n \times n}$ and write the basis elements in rows of length n :

$$(\tilde{e}_1 \quad \dots \quad \tilde{e}_n) = (e_1 \quad \dots \quad e_n) \cdot \begin{pmatrix} A^1_1 & \dots & A^1_n \\ \vdots & \ddots & \vdots \\ A^n_1 & \dots & A^n_n \end{pmatrix}$$

and use the witchcraft of matrix multiplication (‘row times column’) to the transformation of basis. Employing even more concealed notation, we can write this as

$$\tilde{e} = e \cdot A,$$

where $\mathbf{e} := (e_1 \ \cdots \ e_n)$ is the boldface notation for the basis vector stacked into a row.

Note that the old basis can also be expressed in terms of the new one, therefore, there must exist some matrix that we denote with A^{-1} such that

$$\mathbf{e} = \tilde{\mathbf{e}} \cdot A^{-1}.$$

Let us now see how a basis transformation affects the components of vectors. Consider an arbitrary vector $v \in V$ and decompose it with respect to the new basis:

$$v = \sum_{i=1}^n v^i \cdot e_i = \sum_{i=1}^n v^i \cdot \left(\sum_{j=1}^n (A^{-1})^j_i \cdot \tilde{e}_j \right) = \sum_{j=1}^n \left(\sum_{i=1}^n (A^{-1})^j_i v^i \right) \tilde{e}_j = \sum_{j=1}^n \tilde{v}^j \cdot \tilde{e}_j.$$

Its components in the old and the new bases are thus related by

$$\tilde{v}^j = \sum_{i=1}^n (A^{-1})^j_i v^i.$$

Writing the same equation using matrix multiplication we obtain

$$\begin{pmatrix} \tilde{v}^1 \\ \vdots \\ \tilde{v}^n \end{pmatrix} = \begin{pmatrix} (A^{-1})^1_1 & \cdots & (A^{-1})^1_n \\ \vdots & \ddots & \vdots \\ (A^{-1})^n_1 & \cdots & (A^{-1})^n_n \end{pmatrix} \cdot \begin{pmatrix} v^1 \\ \vdots \\ v^n \end{pmatrix},$$

or in the boldface notation,

$$\tilde{\mathbf{v}} = A^{-1} \cdot \mathbf{v}.$$

Analogously, it can be shown that

$$\mathbf{v} = A \cdot \tilde{\mathbf{v}}.$$

We observe that matrix A transforms the old basis into the new one, at the same time it acts in another direction with components: it transforms the components of vectors in the new basis into the components in the old basis. As a result, the transformation from the old basis to the new one is implemented as:

$$\tilde{\mathbf{e}} = \mathbf{e} \cdot A \quad \text{and} \quad \tilde{\mathbf{v}} = A^{-1} \cdot \mathbf{v}.$$

Remark 2.16. The basis transformation matrix A can be seen as the coordinate representation of the identity linear map $\text{id}_V: V \rightarrow V$ with respect to the new and the old bases. The identity id_V map is invertible (its inverse is also id_V), so the matrix A^{-1} is the coordinate representation of the identity map with respect to the old and the new bases:

$$\begin{array}{ccc} V & \xrightarrow{\text{id}_V} & V \\ \psi_{\tilde{\mathbf{e}}} \downarrow & & \downarrow \psi_{\mathbf{e}} \\ \mathbb{R}^n & \xrightarrow{A} & \mathbb{R}^n \end{array} \qquad \begin{array}{ccc} V & \xleftarrow{\text{id}_V} & V \\ \psi_{\tilde{\mathbf{e}}} \downarrow & & \downarrow \psi_{\mathbf{e}} \\ \mathbb{R}^n & \xleftarrow{A^{-1}} & \mathbb{R}^n \end{array}$$

As a result, the matrix A indeed represents the change of coordinates from the new basis to the old one, and A^{-1} represents the change of coordinates from the old basis to the new one:

$$\begin{aligned} A &= \psi_e \circ \text{id}_V \circ \psi_{\tilde{e}}^{-1} = \psi_e \circ \psi_{\tilde{e}}^{-1}: \mathbb{R}^n \rightarrow \mathbb{R}^n \\ A^{-1} &= \psi_{\tilde{e}} \circ \text{id}_V \circ \psi_e^{-1} = \psi_{\tilde{e}} \circ \psi_e^{-1}: \mathbb{R}^n \rightarrow \mathbb{R}^n. \end{aligned}$$

Finally, $A^{-1} \cdot A = A \cdot A^{-1} = \text{id}_{\mathbb{R}^n}$, therefore, A^{-1} is indeed the inverse matrix of A .

Transformation of components of linear maps

Let us now consider a linear map $\varphi: V \rightarrow W$ between two vector spaces V and W with bases $(e_1, \dots, e_n)$ and $(f_1, \dots, f_m)$, respectively. Let us also choose new bases $(\tilde{e}_1, \dots, \tilde{e}_n)$ and $(\tilde{f}_1, \dots, \tilde{f}_m)$ in V and W , respectively. Suppose the components of vectors in these bases are related by the transformation matrices B and C , respectively:

$$\tilde{\mathbf{v}} = B \cdot \mathbf{v} \quad \text{and} \quad \tilde{\mathbf{w}} = C \cdot \mathbf{w}.$$

Let Φ be the coordinate representation matrix of φ with respect to the old bases, and $\tilde{\Phi}$ its coordinate representation with respect to the new bases.

Then, for any vector $v \in V$ and its image $w = \varphi(v) \in W$,

$$\Phi \cdot \mathbf{v} = \mathbf{w} \quad \text{and} \quad \tilde{\Phi} \cdot \tilde{\mathbf{v}} = \tilde{\mathbf{w}}.$$

Plugging in the transformation of vectors' components we obtain

$$\tilde{\Phi} \cdot \tilde{\mathbf{v}} = \tilde{\mathbf{w}} \quad \Rightarrow \quad \tilde{\Phi} \cdot B \cdot \mathbf{v} = C \cdot \mathbf{w} \quad \Rightarrow \quad (C^{-1} \cdot \tilde{\Phi} \cdot B) \cdot \mathbf{v} = \mathbf{w}.$$

Therefore, the components of the linear map φ transform as

$$\tilde{\Phi} = C \cdot \Phi \cdot B^{-1}.$$

Keep in mind that under a change of basis, components of vectors and linear maps transform, not the vectors and linear maps themselves.

2.2 Tensors

2.2.1 Dual space

Given a vector space, one can always construct another very useful vector space dual to the given one.

Definition 2.17. The *dual space* of a vector space $(V, \oplus, \odot)$ a vector space $(V^*, \boxplus, \boxdot)$ consisting of all linear maps from V to $\mathbb{R}$

$$V^* := \{f: V \rightarrow \mathbb{R} \mid f \text{ is linear}\}$$

equipped with the following vector space maps:

$$\begin{aligned} (\alpha \boxplus \beta)(v) &:= \alpha(v) + \beta(v) \\ (\lambda \boxdot \alpha)(v) &:= \lambda \cdot \alpha(v). \end{aligned} \quad \forall \alpha, \beta \in V^*, \forall v \in V, \forall \lambda \in \mathbb{R}$$

One should verify that the dual space is indeed a vector space by checking the axioms.

Hereafter, we will omit the boxes around $\boxplus$ and $\boxminus$ and write $\alpha + \beta$ and $\lambda\alpha$ instead, however, one should remember that these are different operations on different spaces.

The elements of a dual space are called *covectors* (sometimes also *linear functionals*, *dual vectors* or *one-forms*).

Thus, covectors can act on vectors to produce scalars. Given a vector $v \in V$ and a covector $\alpha \in V^*$ we call such an action the *dual pairing* and denote it by

$$\langle \alpha \mid v \rangle := \alpha(v) \in \mathbb{R}.$$

One should not be confused by the terminology – a covector is also a vector since it is an element of a vector space V^* . Then what is the dual space of V^* itself? The following proposition claims that in a finite-dimensional setting, it is the original vector space V .

Proposition 2.18. *A finite-dimensional vector space V is canonically isomorphic to the dual of its dual, i.e.*

$$(V^*)^* = V.$$

We see that in finite dimensions covectors are vectors and vectors are covectors. Therefore, the distinction is solely conceptual and is usually determined by the context. For instance, velocities will be considered vectors, while forces will be considered covectors dual to velocities.

Proposition 2.19. *The dimension of the dual space is the same as the dimension of the original vector space, $\dim V^* = \dim V$.*

Bases in a vector space V and in its dual V^* can be chosen independently. However, given a basis in V , there is a canonical choice of a basis in V^* , called the dual basis.

Definition 2.20. The *dual basis* to a basis $(e_1, \dots, e_n)$ of an n -dimensional vector space V is the basis $(\varepsilon^1, \dots, \varepsilon^n)$ of the dual space V^* such that

$$\langle \varepsilon^i \mid e_j \rangle = \varepsilon^i(e_j) = \delta_j^i := \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad \forall i, j \in \{1, \dots, n\}$$

The symbol δ_j^i is called the *Kronecker³ delta*.

Proposition 2.21. *For any basis of V , there is a unique dual basis in V^* .*

The use of the dual basis in V^* proves to be extremely convenient when working with components.

³Leopold Kronecker (1823–1891), German mathematician.

Consider a covector $\alpha \in V^*$ in the dual of an n -dimensional vector space V . Let $(\varepsilon^1, \dots, \varepsilon^n)$ be the basis of V^* dual to the basis $(e_1, \dots, e_n)$ of V . By Definition 2.11, the components of α are the coefficients in the decomposition

$$\alpha = \sum_{i=1}^n \alpha_i \cdot \varepsilon^i$$

The duality of the bases provides us with a convenient method to compute components. Let us apply α to the j -th basis vector e_j of V :

$$\alpha(e_j) = \left(\sum_{i=1}^n \alpha_i \cdot \varepsilon^i \right) (e_j) = \sum_{i=1}^n \alpha_i \varepsilon^i(e_j) = \sum_{i=1}^n \alpha_i \delta_j^i = \alpha_j,$$

where we used linearity and the definition of the dual basis. Thus, the components of a covector in the dual basis can be simply determined by evaluating the covector on the basis vectors.

Analogously, the components of a vector can be computed by evaluating the dual basis covectors on the vector:

$$\varepsilon^i(v) = \varepsilon^i \left(\sum_{j=1}^n v^j \cdot e_j \right) = \sum_{j=1}^n v^j \varepsilon^i(e_j) = \sum_{j=1}^n v^j \delta_j^i = v^i. \quad (2.2)$$

The action of a covector $\alpha \in V^*$ on a vector $v \in V$ expressed in components in the dual bases is then

$$\langle \alpha \mid v \rangle = \alpha(v) = \sum_{i=1}^n \alpha_i \varepsilon^i(v) = \sum_{i=1}^n \alpha_i v^i.$$

Stacking components of a vector into a column and components of a covector into a row, we can write the dual pairing as a matrix multiplication:

$$\langle \alpha \mid v \rangle = \alpha(v) = (\alpha_1 \quad \cdots \quad \alpha_n) \cdot \begin{pmatrix} v^1 \\ \vdots \\ v^n \end{pmatrix}.$$

The boldface notation for the components of covectors is defined as shorthand for the row of components:

$$\boldsymbol{\alpha} := (\alpha_1 \quad \cdots \quad \alpha_n) \in \mathbb{R}^{1 \times n}$$

Then the dual pairing can be written as

$$\langle \alpha \mid v \rangle = \boldsymbol{\alpha} \cdot \mathbf{v}.$$

Example 2.22. In the scope of this course, velocities v are considered vectors, while forces f and momenta p – covectors. Then their dual pairings are the scalars representing power and kinetic energy, respectively:

$$P = \langle f \mid v \rangle \quad \text{and} \quad K = \langle p \mid v \rangle.$$

Transformation of components of covectors

Recall the setting of Section 2.1.4. Let $(e_1, \dots, e_n)$ and $(\tilde{e}_1, \dots, \tilde{e}_n)$ be two bases of the vector space V related by a transformation matrix A , and $(\varepsilon^1, \dots, \varepsilon^n)$ and $(\tilde{\varepsilon}^1, \dots, \tilde{\varepsilon}^n)$ the corresponding dual bases of the dual space V^* . Then, in the boldface notation the components of a vector transform as

$$\mathbf{v} = A \cdot \tilde{\mathbf{v}}.$$

The dual pairing of an arbitrary vector v with an arbitrary covector α expressed in the dual bases is

$$\langle \alpha \mid v \rangle = \tilde{\alpha} \cdot \tilde{\mathbf{v}} = \alpha \cdot \mathbf{v} = \alpha \cdot A \cdot \tilde{\mathbf{v}}.$$

Thus, the components of a covector transform as

$$\tilde{\alpha} = \alpha \cdot A \quad \text{or} \quad \alpha = \tilde{\alpha} \cdot A^{-1}$$

In conclusion, the components of covectors transform with the same matrix in the opposite directions to the components of vectors:

$$\tilde{\mathbf{v}} \xrightarrow{A} \mathbf{v} \quad \text{and} \quad \tilde{\alpha} \xleftarrow{A} \alpha.$$

Remark 2.23. Sometimes one wants to work with components of covectors using not rows but columns. In this case, the transformation rule is:

$$\tilde{\alpha}^\top = A^\top \cdot \alpha^\top \quad \text{or} \quad \alpha^\top = (A^{-1})^\top \cdot \tilde{\alpha}^\top.$$

2.2.2 Dual linear maps

Definition 2.24. Given a linear map $\varphi: V \rightarrow W$ between two vector spaces, one can construct a *dual map* φ^* between the dual spaces defined in the opposite direction as

$$\begin{aligned} \varphi^*: W^* &\rightarrow V^* \\ \alpha &\mapsto \varphi^*(\alpha) := \alpha \circ \varphi. \end{aligned}$$

As a result, for any vector $v \in V$ and covector $\alpha \in W^*$ we have

$$\langle \varphi^*(\alpha) \mid v \rangle = \langle \alpha \mid \varphi(v) \rangle. \quad (2.3)$$

Let $(e_1, \dots, e_n)$ and $(f_1, \dots, f_m)$ be bases in V and W , respectively, and $(\varepsilon^1, \dots, \varepsilon^n)$ and $(\xi^1, \dots, \xi^m)$ be the corresponding dual bases in V^* and W^* . The expression $\beta = \varphi^*(\alpha)$ for any $\alpha \in W^*$ and $\beta \in V^*$ written in components is

$$\beta_i = \sum_{j=1}^m (\varphi^*)^j_i \alpha_j \quad \text{or} \quad (\beta_1 \ \cdots \ \beta_n) = (\alpha_1 \ \cdots \ \alpha_m) \cdot \begin{pmatrix} (\phi^*)^1_1 & \cdots & (\phi^*)^1_n \\ \vdots & \ddots & \vdots \\ (\phi^*)^m_1 & \cdots & (\phi^*)^m_n \end{pmatrix}$$

Denoting by Φ^* the matrix of components of the dual map φ^* in the dual bases, we can write this as

$$\beta = \alpha \cdot \Phi^*.$$

Thus, the equation (2.3) in components reads

$$\alpha \cdot \Phi^* \cdot v = \alpha \cdot \Phi \cdot v,$$

and therefore the matrix of components of the dual map is the same as the matrix of components of the original map: $\Phi^* = \Phi$.

However, if one chooses to work with components of covectors as columns, one transposes the expression to get

$$\beta^\top = (\Phi^*)^\top \cdot \alpha^\top = \Phi^\top \cdot \alpha^\top.$$

2.2.3 Tensors

Definition 2.25. A *tensor* of rank $\binom{p}{q}$ over a vector space V is a multilinear map

$$T: \underbrace{V^* \times \cdots \times V^*}_{p \text{ times}} \times \underbrace{V \times \cdots \times V}_{q \text{ times}} \rightarrow \mathbb{R}$$

Example 2.26. A covector $\alpha \in V^*$ is a $\binom{0}{1}$ -tensor on V . Indeed, it is a linear map $\alpha: V \rightarrow \mathbb{R}$.

Example 2.27. A vector $v \in V$ is a $\binom{1}{0}$ -tensor on V . One can think of it as a linear map

$$\begin{aligned} v: V^* &\rightarrow \mathbb{R} \\ \alpha &\mapsto \alpha(v). \end{aligned}$$

Example 2.28. An inner product $g: V \times V \rightarrow \mathbb{R}$ is a $\binom{0}{2}$ -tensor on V .

Example 2.29. A linear map on the same vector space $\varphi: V \rightarrow V$ is a $\binom{1}{1}$ -tensor on V . This is not immediately seen since it maps vectors to vectors, however, it can be seen as a linear map

$$\begin{aligned} \varphi: V^* \times V &\rightarrow \mathbb{R} \\ (\alpha, v) &\mapsto \langle \alpha \mid \varphi(v) \rangle. \end{aligned}$$

Exercise 2.30. Consider an inner product space (V, g) and define a map

$$\begin{aligned} g^\flat: V &\rightarrow V^* \\ v &\mapsto g^\flat(v) := g(v, \cdot). \end{aligned}$$

Prove that $g^\flat$ is well-defined and is a linear isomorphism, i.e. an inner product g induces an identification between the vector space and its dual.

Definition 2.31. Let $(e_1, \dots, e_n)$ be a basis of a vector space V , and $(\varepsilon^1, \dots, \varepsilon^n)$ be its dual basis in V^* . The *components* of a $\binom{p}{q}$ -tensor T are $n^p \cdot n^q$ numbers given by the evaluation of the tensor on the basis vectors and covectors:

$$T^{i_1 \dots i_p}_{j_1 \dots j_q} := T(\varepsilon^{i_1}, \dots, \varepsilon^{i_p}, e_{j_1}, \dots, e_{j_q}) \in \mathbb{R} \quad \forall i_1, \dots, i_p, j_1, \dots, j_q \in \{1, \dots, n\}.$$

Indices for p covector slots are put up and are called *contravariant indices*, while indices for q vector slots are put down and are called *covariant indices*.

The name of the indices comes from the way components of tensors transform under a change of basis. Recall that if a basis is transformed with a matrix A , then components of vectors v^i (with an upper, contravariant index) transform with the inverse matrix A^{-1} , or contravariantly with the change of basis, while the components of covectors in the dual bases α_i (with a lower, covariant index) transform with the matrix A , or covariantly with the change of basis.

Example 2.32. Let T be a $\binom{1}{2}$ -tensor on V . Then, using the linearity of T in each slot, for any vectors $v, w \in V$ and covector $\alpha \in V^*$ we have

$$\begin{aligned} T(\alpha, v, w) &= T\left(\sum_{i=1}^n \alpha_i \cdot \varepsilon^i, \sum_{j=1}^n v^j \cdot e_j, \sum_{k=1}^n w^k \cdot e_k\right) \\ &= \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \alpha_i v^j w^k T(\varepsilon^i, e_j, e_k) \\ &= \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \alpha_i v^j w^k T^i_{jk}. \end{aligned}$$

Note that this definition of components of a tensor is consistent with the definitions of components of vectors, covectors, linear maps, and inner products given earlier.

As seen in Example 2.32, expressions in components involve summations over repeated upper and lower indices. To simplify the notation, we introduce the *Einstein⁴ summation convention*: whenever an index i appears once up a^i and once down b_i in a term, the summation over i is implied, and the summation sign $\sum_{i=1}^n$ is omitted. E.g.,

$$T(\alpha, v, w) = \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \alpha_i v^j w^k T^i_{jk} =: \alpha_i v^j w^k T^i_{jk}.$$

Example 2.33. Now, the following expressions in components can be written more compactly:

- $v = v^i \cdot e_i$,
- $\alpha = \alpha_j \cdot \varepsilon^j$,
- $\alpha(v) = \alpha_j \varepsilon^j(v^i \cdot e_i) = \alpha_j v^i \varepsilon^j(e_i) = \alpha_j v^j \delta_i^j = \alpha_i v^i$,

⁴Albert Einstein (1879–1955), German-born theoretical physicist.

- $w = \varphi(v) = \varphi^i_j v^j \cdot e_i$,
- $w^j = \varphi^j_i v^i$,
- $g(v, w) = g_{ij} v^i w^j$,
- etc.

Transformation of tensor components

Proposition 2.34. *Let $(e_1, \dots, e_n)$ be a basis of V that is transformed into a new basis $(\tilde{e}_1, \dots, \tilde{e}_n)$ with a matrix A , i.e.*

$$\tilde{e}_j = A^i_j \cdot e_i \quad \text{or} \quad e_i = (A^{-1})^j_i \cdot \tilde{e}_j.$$

Then the components of a $\binom{p}{q}$ -tensor T transform from the old basis to the new basis with the following rule:

$$\tilde{T}^{k_1 \dots k_p}_{l_1 \dots l_q} = (A^{-1})^{k_1}_{i_1} \dots (A^{-1})^{k_p}_{i_p} A^{j_1}_{l_1} \dots A^{j_q}_{l_q} T^{i_1 \dots i_p}_{j_1 \dots j_q} \quad (2.4)$$

Proof. Firstly, we wish to deduce how the corresponding dual bases $(\varepsilon^1, \dots, \varepsilon^n)$ and $(\tilde{\varepsilon}^1, \dots, \tilde{\varepsilon}^n)$ are related. Recall that the i -th component of a vector is obtained by acting on the vector with the i -th dual basis covector, so for any vector $v \in V$ we have

$$\tilde{\varepsilon}^j(v) = \tilde{v}^j = (A^{-1})^j_i v^i = (A^{-1})^j_i \varepsilon^i(v)$$

Since this holds for any vector v , we have

$$\tilde{\varepsilon}^j = (A^{-1})^j_i \cdot \varepsilon^i.$$

Therefore, the components of a tensor in the new basis are

$$\begin{aligned} \tilde{T}^{k_1 \dots k_p}_{l_1 \dots l_q} &= T(\tilde{\varepsilon}^{k_1}, \dots, \tilde{\varepsilon}^{k_p}, \tilde{e}_{l_1}, \dots, \tilde{e}_{l_q}) \\ &= T((A^{-1})^{k_1}_{i_1} \cdot \varepsilon^{i_1}, \dots, (A^{-1})^{k_p}_{i_p} \cdot \varepsilon^{i_p}, A^{j_1}_{l_1} \cdot e_{j_1}, \dots, A^{j_q}_{l_q} \cdot e_{j_q}) \\ &= (A^{-1})^{k_1}_{i_1} \dots (A^{-1})^{k_p}_{i_p} A^{j_1}_{l_1} \dots A^{j_q}_{l_q} T(\varepsilon^{i_1}, \dots, \varepsilon^{i_p}, e_{j_1}, \dots, e_{j_q}) \\ &= (A^{-1})^{k_1}_{i_1} \dots (A^{-1})^{k_p}_{i_p} A^{j_1}_{l_1} \dots A^{j_q}_{l_q} T^{i_1 \dots i_p}_{j_1 \dots j_q} \end{aligned}$$

□

2.2.4 Inner product

An important structure a vector space can additionally be equipped with is an inner product.

Definition 2.35. Let $(V, +, \cdot)$ be a vector space. An *inner product* on V is a map:

$$g: V \times V \rightarrow \mathbb{R}$$

satisfying the following properties for all $u, v, w \in V$ and $\lambda \in \mathbb{R}$:

- (i) Symmetry: $g(u, v) = g(v, u)$
- (ii) (Bi)linearity: $g(\lambda \cdot u + v, w) = \lambda g(u, w) + g(v, w)$
- (iii) Positive definiteness: $g(v, v) \geq 0$ and $g(v, v) = 0 \Rightarrow v = 0$

Other common notations for the inner product are:

$$g(u, v) \equiv \langle u, v \rangle \equiv u \cdot v,$$

however, the notation $g(\cdot, \cdot)$ is the one that highlights that a choice of an additional structure has been made.

Example 2.36. As usual, we use $\mathbb{R}^n$ as the prototypical example of an n -dimensional vector space. The *Euclidean*⁵, *standard* or *canonical* inner product on $\mathbb{R}^n$ is defined for all $u = (u^1, \dots, u^n)$, $(v^1, \dots, v^n) \in \mathbb{R}^n$ as

$$g_{\text{can}}(u, v) := \sum_{i=1}^n u^i v^i.$$

Example 2.37. The space of all continuous functions $\mathcal{C}([0, 1], \mathbb{R})$ on the interval $[0, 1]$ can be equipped with the inner product defined as

$$g(f, g) := \int_0^1 dx f(x) \cdot g(x).$$

A bare vector space not carrying an inner product does not have such notions as lengths or angles. An inner product allows one to define them.

Definition 2.38. A *norm* associated with an inner product g on a vector space V is the map $\|\cdot\|_g: V \rightarrow \mathbb{R}$ defined as

$$\|v\|_g := \sqrt{g(v, v)}.$$

A *metric* or *distance* associated with an inner product g is the map $d_g: V \times V \rightarrow \mathbb{R}$ defined as

$$d_g(u, v) := \|u - v\|_g = \sqrt{g(u - v, u - v)}.$$

Exercise 2.39. Prove that the metric d_g satisfies the following properties:

- (i) Positivity: $d_g(u, v) \geq 0$ and $d_g(u, v) = 0 \Leftrightarrow u = v$
- (ii) Symmetry: $d_g(u, v) = d_g(v, u)$
- (iii) Triangle inequality: $d_g(u, v) \leq d_g(u, w) + d_g(w, v)$

In fact, any function on any set that satisfies these properties is called a *metric* or a *distance* on the set, and the set equipped with such a function is called a *metric space*.

⁵Euclid (around 300 BCE), ancient Greek mathematician.

Definition 2.40. An *angle* θ between two non-zero vectors v and w in an inner product space (V, g) is defined with

$$\cos \theta := \frac{g(v, w)}{\|v\| \|w\|}.$$

Definition 2.41. Two vectors v and w in an inner product space (V, g) are called *orthogonal* if $g(v, w) = 0$.

Definition 2.42. A set of k vectors $\{v_1, \dots, v_k\}$ in (V, g) is called an *orthonormal system* if all vectors are of unit length and are pairwise orthogonal, i.e.

$$g(v_i, v_j) = \delta_{ij} := \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad \forall i, j \in \{1, \dots, k\}.$$

A basis of (V, g) is called *orthonormal* if its elements form an orthonormal system.

Proposition 2.43. Any inner product space admits an orthonormal basis.

Proof. Look up the [Gram⁶–Schmidt⁷ process](#), which is a method to construct an orthonormal basis from an arbitrary basis. $\square$

Having introduced a new mathematical structure – an inner product space – we study structure-preserving maps between such spaces.

Definition 2.44. Given two inner product spaces (V, g_V) and (W, g_W) a linear map $\varphi: V \rightarrow W$ that preserves the inner product, i.e.

$$g_W(\varphi(u), \varphi(v)) = g_V(u, v) \quad \forall u, v \in V,$$

is called a *linear isometry* of inner product spaces. If φ is also invertible, it is called an *isometric isomorphism*.

The name ‘isometry’ etymologically comes from the Greek words (isos, ‘equal’) and (metron, ‘measure’) highlighting that the lengths are preserved under such maps.

Exercise 2.45. Show that a linear map φ between inner product spaces (V, g_V) and (W, g_W) is a linear isometry if and only if it preserves the associated metric (distance), i.e.

$$d_{g_W}(\varphi(u), \varphi(v)) = d_{g_V}(u, v) \quad \forall u, v \in V.$$

Definition 2.46. Given a basis $(e_1, \dots, e_n)$ of an inner product space (V, g) , the *components of the inner product* g in this basis are $n \times n$ numbers given by the evaluation of the inner product on the basis vectors:

$$g_{ij} := g(e_i, e_j) \in \mathbb{R} \quad \forall i, j \in \{1, \dots, n\}.$$

⁶Jørgen Pedersen Gram (1850–1916), Danish actuary and mathematician.

⁷Erhard Schmidt (1876–1959), German mathematician.

In components, we therefore have that the inner product of two vectors u and v is

$$g(u, v) = \sum_{i=1}^n \sum_{j=1}^n g_{ij} u^i v^j.$$

As with components of linear maps, we can stack components of an inner product into a square matrix, and replicate the previous expression using matrix multiplication as

$$g(u, v) = \begin{pmatrix} u^1 & \cdots & u^n \end{pmatrix} \cdot \begin{pmatrix} g_{11} & \cdots & g_{1n} \\ \vdots & \ddots & \vdots \\ g_{n1} & \cdots & g_{nn} \end{pmatrix} \cdot \begin{pmatrix} v^1 \\ \vdots \\ v^n \end{pmatrix}.$$

Note that here we had to stack components of a vector u into a row and components of a vector v into a column. Using the bold font notation, we can write this more compactly as

$$g(u, v) = \mathbf{u}^\top \cdot G \cdot \mathbf{v},$$

where the matrix $G = (g_{ij})_{i,j \in \{1, \dots, n\}}$ is the matrix of components of the inner product g in the chosen basis.

⚠ A matrix is merely a table of numbers! It therefore hides the nature of the object it represents. Both a linear map $\varphi: V \rightarrow V$ (see Section 2.1.3) and an inner product $g: V \times V \rightarrow \mathbb{R}$ can be represented by a square matrix. However, those matrices behave differently under a change of basis. Let the components of vectors in V transform under a change of basis with a matrix A :

$$\mathbf{v} = A \cdot \tilde{\mathbf{v}}$$

Then, denoting the matrix of components of the inner product in the new basis by $\tilde{G}$, for any $u, v \in V$ we have

$$\tilde{\mathbf{u}}^\top \cdot \tilde{G} \cdot \tilde{\mathbf{v}} = g(u, v) = \mathbf{u}^\top \cdot G \cdot \mathbf{v} = \tilde{\mathbf{u}}^\top \cdot A^\top \cdot G \cdot A \cdot \tilde{\mathbf{v}}.$$

Therefore, the components of the inner product transform as

$$\tilde{G} = A^\top \cdot G \cdot A.$$

On the other hand, the components of the linear map φ transform as

$$\tilde{\Phi} = A^{-1} \cdot \Phi \cdot A.$$

Remark 2.47. One is used to speak about eigenvalues and eigenvectors of a matrix, whereas in reality these notions are only defined for linear maps $\varphi: V \rightarrow V$ on a vector space V ($\varphi(v) = \lambda \cdot v$). When the linear map is represented by a matrix Φ in a chosen basis, the eigenvalues and eigenvectors can be computed from the matrix, and they are invariant under a change of basis, i.e. the matrix $A^{-1} \cdot \Phi \cdot A$ has the same eigenvalues and eigenvectors as the matrix Φ . However, it does not make any sense to speak about eigenvalues and eigenvectors of an inner product. For a matrix G , representing an inner product in one basis, they will not be the same for the matrix $A^\top \cdot G \cdot A$ representing the same inner product in another basis.

A direct consequence of the definition of an orthonormal basis is the following proposition.

Proposition 2.48. *Components of an inner product g in any orthonormal basis $(e_1, \dots, e_n)$ are given by the identity matrix.*

$$g_{ij} = g(e_i, e_j) = \delta_{ij} \quad \forall i, j \in \{1, \dots, n\}.$$

Therefore, the following holds.

Proposition 2.49. *Given an orthonormal basis $(e_1, \dots, e_n)$ of an inner product space (V, g) , the coordinatisation map $\psi_e: V \rightarrow \mathbb{R}^n$ is an isometric isomorphism between (V, g) and $(\mathbb{R}^n, g_{\text{can}})$ with the standard inner product.*

Proposition 2.50. *Coordinate representation Φ of a linear isometry $\varphi: V \rightarrow W$ from an inner product space (V, g_V) to an inner product space (W, g_W) with respect to orthonormal bases $(e_1, \dots, e_n)$ of V and $(f_1, \dots, f_m)$ of W is a linear isometry on $(\mathbb{R}^n, g_{\text{can}})$.*

Proof. The coordinate representation is defined as the composition of linear isometries:

$$\Phi = \psi_f \circ \varphi \circ \psi_e^{-1},$$

such that the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ \psi_e \downarrow & & \downarrow \psi_f \\ \mathbb{R}^n & \xrightarrow{\Phi} & \mathbb{R}^n \end{array}$$

□

Proposition 2.51. *The coordinatisation map ψ_e associated with an orthonormal basis $(e_1, \dots, e_n)$ of an inner product space (V, g) can be expressed as*

$$\begin{aligned} \psi_e: V &\rightarrow \mathbb{R}^n \\ v &\mapsto \psi_e(v) = (g(v, e_1), \dots, g(v, e_n)). \end{aligned}$$

Vector product

When calculating the cross product of two vectors in a three-dimensional space, the right-hand rule is typically applied. To properly define a vector product, a preferred orientation of space must be chosen to define the notion of right-handedness. Let V be a vector space. Recall that any two ordered bases of V are related by a unique invertible transformation matrix A (see Section 2.1.4). If the transformation matrix A has a positive determinant ($\det A > 0$), the bases are said to share the *same orientation*. Conversely, if $\det A < 0$, the bases have *opposite orientations*. This means that

the set of all ordered bases in V is divided into two distinct classes, each containing bases with the same orientation. Choosing a preferred *orientation* of V amounts to selecting one of these two classes and calling it *positively oriented*. In three-dimensional space, the bases in the selected class are typically called *right-handed*, while those in the opposite class are called *left-handed*. In $\mathbb{R}^3$, with the standard vector space structure, the standard basis $((1, 0, 0), (0, 1, 0), (0, 0, 1))$ is designated as right-handed, establishing the standard orientation of $\mathbb{R}^3$. An orientation is thus an additional structure on a vector space V .

Definition 2.52. Let (V, g) be an oriented inner product space (see Section 2.2.4). The *vector product* (or *cross product*) is a bilinear map

$$\times: V \times V \rightarrow V,$$

$$(u, v) \mapsto u \times v := \begin{cases} 0, & \text{if } u \text{ and } v \text{ are linearly dependent,} \\ \|u\| \|v\| \sin(\angle(u, v)) \cdot n, & \text{otherwise,} \end{cases}$$

where

- $\sin(\angle(u, v)) = \sqrt{1 - \cos^2(\angle(u, v))}$ is the sine of the angle between u and v ,
- $n \in V$ a unit vector: $\|n\| = 1$,
- n is orthogonal to both u and v : $g(u, n) = g(v, n) = 0$,
- n is directed such that the ordered basis (u, v, n) is right-handed.

The vector product is sometimes denoted by the wedge symbol: $u \wedge v \equiv u \times v$.

It can be shown that in dimension three, the vector product of two vectors always exists and is unique.

Proposition 2.53. In $\mathbb{R}^3$ with the standard vector space structure and the standard orientation, the vector product of two vectors $u = (u^1, u^2, u^3)$ and $v = (v^1, v^2, v^3)$ is given by

$$u \times v = \begin{pmatrix} u^2 v^3 - u^3 v^2 \\ u^3 v^1 - u^1 v^3 \\ u^1 v^2 - u^2 v^1 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & -u^3 & u^2 \\ u^3 & 0 & -u^1 \\ -u^2 & u^1 & 0 \end{pmatrix}}_{=: \tilde{u}} \cdot \begin{pmatrix} v^1 \\ v^2 \\ v^3 \end{pmatrix} = \tilde{u}v. \quad (2.5)$$

Note that the vector product is not associative. Instead, the following identities hold.

Proposition 2.54 (Jacobi identity). For any three vectors $u, v, w \in V$, the following identity holds:

$$u \times (v \times w) + v \times (w \times u) + w \times (u \times v) = 0. \quad (2.6)$$

Exercise 2.55. Using the Jacobi identity for the standard vector product of $\mathbb{R}^3$ show that for any $u, v \in \mathbb{R}^3$,

$$(\widetilde{uv}) = \tilde{u}\tilde{v} - \tilde{v}\tilde{u},$$

where $\tilde{u}, \tilde{v}$ are anti-symmetric matrices as in (2.5).

Proposition 2.56 (Triple product identity). *For any three vectors $u, v, w \in V$ in an oriented inner product space (V, g) , the following identity holds:*

$$u \times (v \times w) = g(u, w) \cdot v - g(u, v) \cdot w. \quad (2.7)$$

Proof. Idea: Choose a positively-oriented orthonormal basis (e_1, e_2, e_3) of V , then work in $\mathbb{R}^3$ with the standard orientation and inner product. The coordinatisation map is a linear orientation-preserving isometry between V and $\mathbb{R}^3$, hence it preserves the vector product, and the result from $\mathbb{R}^3$ can be transferred back to V . $\square$

2.3 The ‘witchcraft’ of matrix multiplication

Definition 2.57. A *matrix* of dimension $m \times n$ with entries in a set $\mathcal{A}$ is a rectangular array (a table) of elements from $\mathcal{A}$ arranged in m rows and n columns. The entry in the i -th row and j -th column of a matrix A is denoted by A_{ij} with $i \in \{1, \dots, m\}$ and $j \in \{1, \dots, n\}$.

$$A = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \cdots & A_{mn} \end{pmatrix}$$

The set of all $m \times n$ matrices with entries in $\mathcal{A}$ is denoted by $\mathcal{A}^{m \times n}$.

The matrix composed of the entries A_{ij} is denoted by $A = (A_{ij})_{i \in \{1, \dots, m\}, j \in \{1, \dots, n\}}$.

Definition 2.58. Let

$$\cdot: \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$$

be an operation called *multiplication* and

$$+: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$$

be an associative operation called *addition*. The *matrix product* is the map

$$\begin{aligned} \cdot: \mathcal{A}^{l \times m} \times \mathcal{B}^{m \times n} &\rightarrow \mathcal{C}^{l \times n} \\ (A, B) &\mapsto A \cdot B, \end{aligned}$$

where

$$(A \cdot B)_{ij} = \sum_{k=1}^m A_{ik} \cdot B_{kj}$$

for all $i = \{1, \dots, l\}$ and $j = \{1, \dots, n\}$.

Note that in each summand $A_{ik} \cdot B_{kj}$ the adjacent index must be the same, and is summed over.

Definition 2.59. *Transposition* is the map

$$\begin{aligned} \top: \mathcal{A}^{m \times n} &\rightarrow \mathcal{A}^{n \times m} \\ A &\mapsto A^\top, \end{aligned}$$

where

$$(A^\top)_{ij} = A_{ji}$$

Tensorial expressions can sometimes be conveniently written using matrices. For objects with 1 or 2 indices, we postulate the following conventions:

- Objects with a single upper index $i \in \{1, \dots, m\}$ are arranged into an $n \times 1$ (column) matrix:

$$v^i \rightsquigarrow \mathbf{v}_{i1} := v^i \quad \text{or} \quad \mathbf{v} = \begin{pmatrix} v^1 \\ \vdots \\ v^m \end{pmatrix}$$

- Objects with a single lower index $i \in \{1, \dots, n\}$ are arranged into a $1 \times n$ (row) matrix:

$$\alpha_i \rightsquigarrow \boldsymbol{\alpha}_{1i} := \alpha_i \quad \text{or} \quad \boldsymbol{\alpha} = (\alpha_1 \quad \cdots \quad \alpha_n)$$

- Two-indexed objects with the first index $i \in \{1, \dots, m\}$ and the second index $j \in \{1, \dots, n\}$ are arranged into an $m \times n$ matrix:

$$\begin{aligned} \varphi^i_j &\rightsquigarrow \Phi_{ij} := \varphi^i_j \quad \text{or} \quad \Phi = \begin{pmatrix} \varphi^1_1 & \cdots & \varphi^1_n \\ \vdots & \ddots & \vdots \\ \varphi^m_1 & \cdots & \varphi^m_n \end{pmatrix} \\ g_{ij} &\rightsquigarrow G_{ij} := g_{ij} \quad \text{or} \quad G = \begin{pmatrix} g_{11} & \cdots & g_{1n} \\ \vdots & \ddots & \vdots \\ g_{m1} & \cdots & g_{mn} \end{pmatrix} \\ q^{ij} &\rightsquigarrow Q_{ij} := q^{ij} \quad \text{or} \quad Q = \begin{pmatrix} q^{11} & \cdots & q^{1n} \\ \vdots & \ddots & \vdots \\ q^{m1} & \cdots & q^{mn} \end{pmatrix} \end{aligned}$$

As a result, in an n -dimensional vector space V , basis elements (labelled by lower indices) are arranged into $1 \times n$ (row) matrices, while dual basis elements (labelled by upper indices) are arranged into $n \times 1$ (column) matrices:

$$\begin{aligned} e_i &\rightsquigarrow \mathbf{e}_{1i} := e_i \quad \text{or} \quad \mathbf{e} = (e_1 \quad \cdots \quad e_n) \\ \varepsilon^i &\rightsquigarrow \boldsymbol{\varepsilon}_{i1} := \varepsilon^i \quad \text{or} \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon^1 \\ \vdots \\ \varepsilon^n \end{pmatrix}. \end{aligned}$$

Example 2.60. Let $\varphi: V \rightarrow W$ be a linear map between two vector spaces V and W of dimensions n and m , respectively. Suppose some bases in V and W are chosen, and let φ^i_j be the components of φ , v^i be the components of $v \in V$, and w^j be the components of $w = \varphi(v) \in W$ in these bases, which we arrange into matrices:

$$\mathbf{v} \in \mathbb{R}^{n \times 1} \quad \mathbf{w} \in \mathbb{R}^{m \times 1} \quad \Phi \in \mathbb{R}^{m \times n}.$$

Then the expression for the action of φ on v in components can be written using matrix multiplication (with the operations induced from $\mathbb{R}$):

$$w_{i1} = w^i = \varphi^i_j v^j = \sum_{j=1}^n \Phi_{ij} \cdot v_{j1} = (\Phi \cdot \mathbf{v})_{i1} \quad \forall i \in \{1, \dots, m\}$$

or

$$\mathbf{w} = \begin{pmatrix} w^1 \\ \vdots \\ w^m \end{pmatrix} = \begin{pmatrix} \varphi^1_1 & \varphi^1_2 & \cdots & \varphi^1_n \\ \vdots & \vdots & \ddots & \vdots \\ \varphi^m_1 & \varphi^m_2 & \cdots & \varphi^m_n \end{pmatrix} \cdot \begin{pmatrix} v^1 \\ v^2 \\ \vdots \\ v^n \end{pmatrix} = \Phi \cdot \mathbf{v}.$$

Example 2.61. Let $(V, +, \cdot)$ be a real vector space of dimension n . Put $\mathcal{A} = V$, $\mathcal{B} = \mathbb{R}$ and $\mathcal{C} = V$, let the multiplication be the scalar multiplication of V and addition be the addition of V . Let $(e_1, \dots, e_n)$ and $(\tilde{e}_1, \dots, \tilde{e}_n)$ be two bases of V related by $\tilde{e}_j = A^i_j \cdot e_i$. We compose the following matrices:

$$\mathbf{e} \in \mathbb{R}^{1 \times n} \quad \tilde{\mathbf{e}} \in \mathbb{R}^{1 \times n} \quad A \in \mathbb{R}^{n \times n}.$$

Then the change of basis can be written using matrix multiplication:

$$\tilde{e}_{1j} = \tilde{e}_j = A^i_j \cdot e_i = \sum_{i=1}^n \mathbf{e}_{1i} \cdot A_{ij} = (\mathbf{e} \cdot A)_{1j} \quad \forall j \in \{1, \dots, n\}$$

or

$$\tilde{\mathbf{e}} = (\tilde{e}_1 \quad \cdots \quad \tilde{e}_n) = (e_1 \quad \cdots \quad e_n) \cdot \begin{pmatrix} A^1_1 & \cdots & A^1_n \\ \vdots & \ddots & \vdots \\ A^n_1 & \cdots & A^n_n \end{pmatrix} = \mathbf{e} \cdot A.$$

Example 2.62. Let V be a real vector space of dimension n . Put $\mathcal{A} = V^*$, $\mathcal{B} = V$, $\mathcal{C} = \mathbb{R}$, and let the operations be the dual pairing and the addition of real numbers, respectively. Then, the duality condition reads

$$I_{ij} = \delta^i_j = \varepsilon^i(e_j) = \sum_1^1 \boldsymbol{\varepsilon}_{i1} \cdot \mathbf{e}_{1j} = (\boldsymbol{\varepsilon} \cdot \mathbf{e})_{ij} \quad \forall i, j \in \{1, \dots, n\}$$

for a basis $(\varepsilon^1, \dots, \varepsilon^n)$ of V^* and a basis $(e_1, \dots, e_n)$ of V can be written using matrix multiplication as

$$\boldsymbol{\varepsilon} \cdot \mathbf{e} = \begin{pmatrix} \varepsilon^1 \\ \vdots \\ \varepsilon^n \end{pmatrix} \cdot (e_1 \quad \cdots \quad e_n) = \begin{pmatrix} \varepsilon^1(e_1) & \cdots & \varepsilon^1(e_n) \\ \vdots & \ddots & \vdots \\ \varepsilon^n(e_1) & \cdots & \varepsilon^n(e_n) \end{pmatrix} = \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{pmatrix}.$$

Example 2.63. Let g be an inner product on a vector space V of dimension n and let $v, w \in V$ be two vectors. Upon choosing a basis in V , the components of the inner product g_{ij} and the components of the vectors v^i and w^i in this basis can be arranged into matrices:

$$G \in \mathbb{R}^{n \times n} \quad \mathbf{v} \in \mathbb{R}^{n \times 1} \quad \mathbf{w} \in \mathbb{R}^{n \times 1}.$$

Then the expression $g(v, w)$ is written in components as

$$g(v, w) = g_{ij} v^i w^j = \sum_{i=1}^n \mathbf{v}_{i1} \cdot \sum_{j=1}^n G_{ij} \cdot \mathbf{w}_{j1} = \sum_{i=1}^n \mathbf{v}_{i1} \cdot (G \cdot \mathbf{w})_{i1}.$$

Note that the adjacent indices in the summands $\mathbf{v}_{i1} \cdot (G \cdot \mathbf{w})_{i1}$ do not match, therefore, we employ transposition to be able to write the expression as a matrix multiplication:

$$g(v, w) = \sum_{i=1}^n \mathbf{v}_{i1} \cdot (G \cdot \mathbf{w})_{i1} = \sum_{i=1}^n (\mathbf{v}^\top)_{1i} \cdot (G \cdot \mathbf{w})_{i1} = \mathbf{v}^\top \cdot G \cdot \mathbf{w}.$$

Example 2.64. Under a change of basis in V , $\tilde{e}_j = A^j_i \cdot e_i$, the components of the inner product transform as

$$\begin{aligned} \tilde{G}_{kl} &= \tilde{g}_{kl} \\ &= A^i_k A^j_l g_{ij} \\ &= \sum_{i=1}^n A_{ik} \cdot \sum_{j=1}^n G_{ij} \cdot A_{jl} \\ &= \sum_{i=1}^n A_{ik} \cdot (G \cdot A)_{il} \\ &= \sum_{i=1}^n (A^\top)_{ki} \cdot (G \cdot A)_{il} \\ &= (A^\top \cdot G \cdot A)_{kl} \quad \forall k, l \in \{1, \dots, n\} \end{aligned}$$

or

$$\tilde{G} = A^\top \cdot G \cdot A.$$

Chapter 3

Groups and Manifolds

3.1 Groups

3.1.1 Definition

Definition 3.1. A *group* is a pair $(G, \bullet)$ consisting of a set G and a map (binary operation)

$$\begin{aligned}\bullet: G \times G &\rightarrow G \\ (g, h) &\mapsto g \bullet h\end{aligned}$$

such that the following three axioms are satisfied for all $f, g, h \in G$:

$$\begin{array}{ll}\text{A} & f \bullet (g \bullet h) = (f \bullet g) \bullet h \quad (\text{Associativity}) \\ \text{U} & \exists e \in G : \forall g \in G : e \bullet g = g \bullet e = g \quad (\text{Unit element}) \\ \text{I} & \forall g \in G : \exists g^{-1} \in G : g \bullet g^{-1} = g^{-1} \bullet g = e \quad (\text{Inverse of element})\end{array}$$

If the binary operation is commutative, i.e. $g \bullet h = h \bullet g$ for all $g, h \in G$, the group is called *Abelian*¹ or *commutative*.

One can prove that a unit element e is unique and is thus called *the unit element* or *the identity element* of the group $(G, \bullet)$.

Example 3.2. Any vector space $(V, +)$ together with the addition as the group operation is an Abelian group. The unit element is the zero vector $0 \in V$ and the inverse of a vector $v \in V$ is its additive inverse $-v$.

Example 3.3. One of the most significant applications of group theory is in the study of symmetries, particularly in the context of physical systems. In mathematics and physics, a symmetry refers to a transformation that leaves an object or system invariant, meaning that the system appears the same before and after the transformation. Groups provide a rigorous framework for understanding these symmetries, as they encapsulate the set of all possible transformations that a system can undergo without altering its fundamental properties. In this context, the group operation is the composition of transformations.

¹Niels Henrik Abel (1802–1829), Norwegian mathematician.

Example 3.4. Given a bare set M with no additional structure, the set of all bijective maps on M

$$\text{Aut}(M) := \{f: M \rightarrow M \mid f \text{ is bijective}\}$$

together with the composition $\circ$ of such maps is a group $(\text{Aut}(M), \circ)$ called the *automorphism group* of set M . Its identity element is the identity map id_M on M .

A notion of the automorphism group generalises to any mathematical structure and is defined as a set of all structure-preserving maps together with the composition. For instance, the automorphism group of a vector space V , i.e. all invertible linear maps on V , is called the *general linear group* of V and is denoted by $\text{GL}(V) \equiv \text{Aut}(V)$. The automorphism group of an inner product space (V, g) , consisting of all linear isomorphisms that preserve the inner product, is called the *isometry group* or the *orthogonal group* of V and is denoted by $\text{O}(V, g)$.

Definition 3.5. A map $\varphi: G \rightarrow H$ between two groups $(G, \bullet)$ and $(H, \star)$ is called a *group homomorphism* if it preserves the binary operation, i.e. for all $f, g \in G$:

$$\varphi(f \bullet g) = \varphi(f) \star \varphi(g).$$

If, in addition, φ is bijective, it is called a *group isomorphism*.

It is not difficult to verify that a group homomorphism preserves the identity and inverses, i.e.

$$\begin{aligned}\varphi(e_G) &= e_H \\ \varphi(g^{-1}) &= (\varphi(g))^{-1}\end{aligned}$$

Example 3.6. Consider the group $(\mathbb{R}, +)$ of real numbers with the addition and the group $(\mathbb{R}_{>0}, \cdot)$ of positive real numbers with the multiplication. The *exponential map*

$$\begin{aligned}\exp: \mathbb{R} &\rightarrow \mathbb{R}_{>0} \\ x &\mapsto \exp(x) \equiv e^x\end{aligned}$$

is a group isomorphism between these groups, It indeed preserves the group operation since for all $x, y \in \mathbb{R}$:

$$e^{x+y} = e^x \cdot e^y.$$

3.1.2 Group action

Definition 3.7. A *left group action* of a group $(G, \bullet)$ on a set M is a map

$$\begin{aligned}\triangleright: G \times M &\rightarrow M \\ (g, m) &\mapsto g \triangleright m\end{aligned}$$

satisfying

$$(i) \quad e \triangleright m = m,$$

$$(ii) \quad (g \bullet h) \triangleright m = g \triangleright (h \triangleright m)$$

for all $g, h \in G$ and $m \in M$.

Alternatively, one can define a right group action.

Definition 3.8. A *right group action* of a group $(G, \bullet)$ on a set M is a map

$$\begin{aligned} \triangleleft: M \times G &\rightarrow M \\ (m, g) &\mapsto m \triangleleft g \end{aligned}$$

satisfying

$$(i) \quad m \triangleleft e = m,$$

$$(ii) \quad m \triangleleft (g \bullet h) = (m \triangleleft g) \triangleleft h$$

for all $g, h \in G$ and $m \in M$.

Equivalently, one can define a left action of G on M as a group homomorphism from $(G, \bullet)$ to $(\text{Aut}(M), \circ)$. Given

$$\begin{aligned} \varphi: G &\rightarrow \text{Aut}(M) \\ g &\mapsto \varphi(g), \end{aligned}$$

we define $g \triangleright m := (\varphi(g))(m)$.

3.2 Manifolds

In the scope of this course, we will give only an intuitive definition of a manifold.

Definition 3.9. A *smooth manifold* of dimension n is a ‘smooth space’ that everywhere locally ‘looks like’ the set $\mathbb{R}^n$.

The dimension of a manifold is therefore the number of independent coordinates needed to coordinatise a local patch of the manifold.

In mechanics, manifolds are most often used to describe the configuration space of a mechanical system. For instance, the configuration space of a planar pendulum is a circle, which is a one-dimensional manifold. The configuration space of a double planar pendulum is a torus (the surface of a doughnut), which is a two-dimensional manifold. The configuration space of a single pendulum in 3D space is a two-dimensional sphere. Finally, the configuration space of a rigid body is a six-dimensional manifold $\text{SE}(3)$ called the special Euclidean group, which is one of the main objects of study in this course.

If the configuration of a system evolves in time, it is described by a trajectory in the configuration space, which is a curve on the manifold.

Definition 3.10. Given a smooth manifold M , a *smooth curve* is a ‘smooth’ map $\gamma: \mathbb{R} \rightarrow M$.

At each point of the curve, it is possible to define the velocity $v = \dot{\gamma}$ of the curve as a *tangent vector* to the manifold at that point. The set of tangent vectors of all possible curves passing through a point $p \in M$ forms a vector space of dimension n called the *tangent space* to the manifold at p and is denoted by $T_p M$. (One can meaningfully define the operation of addition and scalar multiplication of tangent vectors.) It is important to note that tangent spaces at different points of a manifold

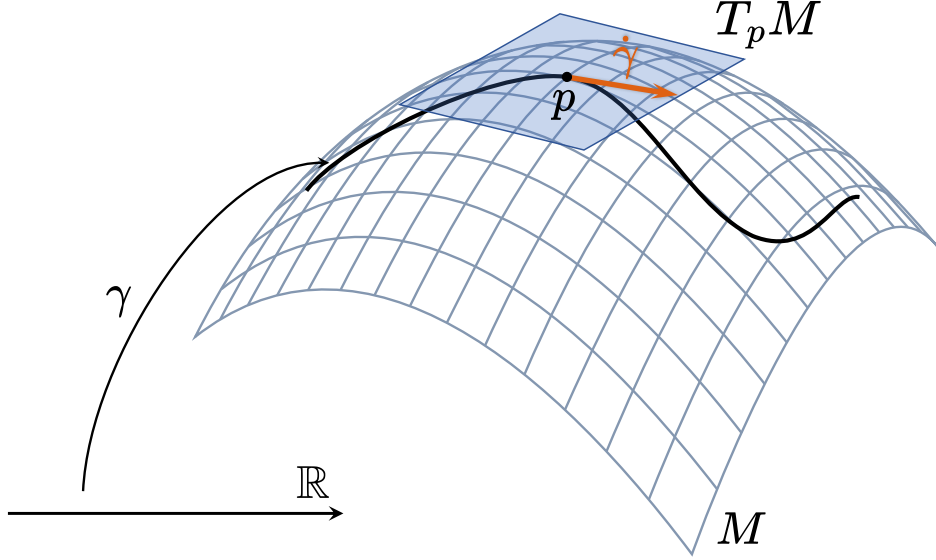


Figure 3.1: A vector $\dot{\gamma}$ tangent to a curve γ is an element of the tangent space $T_p M$ to a manifold M at the point p .

are distinct vector spaces; one cannot add a vector $v \in T_p M$ to a vector $w \in T_q M$ for $p \neq q$.

Any vector space has a dual space. The dual space to the tangent space $T_p M$ is called the *cotangent space* to the manifold at p and is denoted by $T_p^* M$. For mechanical systems, the elements of a tangent spaces are velocities v , while the elements of cotangent spaces are forces f and momenta p . The dual pairing of a cotangent vector with a tangent vector gives a scalar – power $P = \langle f | v \rangle$ for a force paired with velocity and twice the kinetic energy $2T = \langle p | v \rangle$ for a momentum paired with velocity, respectively.

The manifold setting allows working with physical quantities without using coordinates (which exist solely in our imagination, not in the physical world), obtaining results that hold independently of the choice of coordinates and then explicitly translating them into coordinates only when necessary.

A structure-preserving map between smooth manifolds is called a *smooth map*. In case of a bijective smooth map, whose inverse is also smooth, the map is called a *diffeomorphism*.

Definition 3.11. Given a map $\varphi: M \rightarrow N$ between two manifolds, one can define a map between the tangent spaces, called the *push-forward* or the *differential* of φ .

At each point $p \in M$, it is a linear map

$$\begin{aligned}\varphi_{*,p}: T_p M &\rightarrow T_{\varphi(p)} N \\ \dot{\gamma} &\mapsto \varphi_{*,p} \dot{\gamma}.\end{aligned}$$

If a vector $\dot{\gamma} \in T_p M$ is tangent to a curve γ on M at p , then $\varphi_{*,p} \dot{\gamma}$ is tangent to the curve $\varphi \circ \gamma$ on N at $\varphi(p)$.

Other common notations for the push-forward are $\varphi_{*,p} \equiv d_p \varphi \equiv T_p \varphi$.

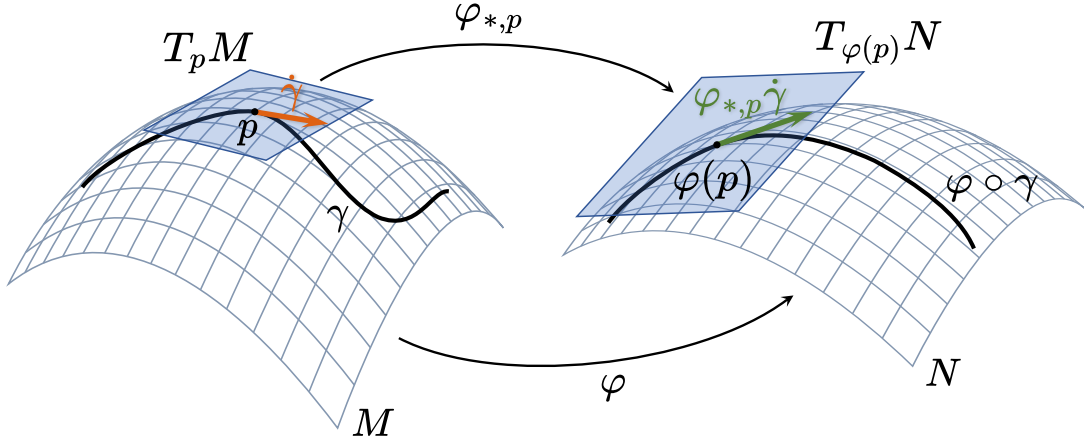


Figure 3.2: The push-forward $\varphi_{*,p}$ of a vector $\dot{\gamma}$ tangent to a curve γ at p is a vector tangent to the curve $\varphi \circ \gamma$ at $\varphi(p)$.

If φ is a diffeomorphism, the push-forward is a linear isomorphism between the tangent spaces at corresponding points.

3.3 Lie groups and their Lie algebras

Possibly the most important mathematical object in this course, on par with a vector space, is a Lie² group. Described informally, a Lie group is a group that is also a smooth manifold.

Definition 3.12. A *Lie group* is a smooth manifold G equipped with a group operation $\bullet$ such that the maps

$$\begin{aligned}\bullet: G \times G &\rightarrow G & \text{and} & & \iota: G &\rightarrow G \\ (g, h) &\mapsto g \bullet h & & & g &\mapsto g^{-1}\end{aligned}$$

are smooth.

²Sophus Lie (/li:/ LEE; 1842–1899), Norwegian mathematician.

As discussed in Section 3.1.1, groups are used to represent symmetries of physical systems. Lie groups are used to represent smooth symmetries. For instance, if a system is invariant under translations in space, rotations, or time translations, the corresponding symmetry groups are Lie groups.

Example 3.13. Consider a square and a circle on a plane. Rotational symmetries of the square form a discrete group of four elements, which is not a Lie group (it fails to be a smooth manifold), while rotational symmetries of the circle forms a Lie group, which as a manifold looks like a circle.

Example 3.14. The symmetry group of a bare vector space V is the general linear group $\mathrm{GL}(V)$, which is a Lie group consisting of all invertible linear maps on V . Taking as the vector space $(\mathbb{R}^n, +, \cdot)$ with the standard addition and scalar multiplication, the general linear group $\mathrm{GL}(n) := \mathrm{GL}(\mathbb{R}^n)$ is a Lie group of invertible $n \times n$ matrices. If a vector space is equipped with additional structure, such as an inner product, its symmetry group will be a Lie subgroup of the general linear group that preserves the structure.

Definition 3.15. The *Lie algebra* $\mathfrak{g}$ of a Lie group G is the tangent space to G at the identity element $e \in G$:

$$\mathfrak{g} := T_e G.$$

Definition 3.16. *Left* and *right translations* by an element $h \in G$ on a Lie group G are the diffeomorphisms:

$$\begin{aligned} \mathrm{l}_h: G &\rightarrow G & \text{and} & & \mathrm{r}_h: G &\rightarrow G \\ g &\mapsto h \bullet g & & & g &\mapsto g \bullet h. \end{aligned}$$

Finally, any tangent vector to a Lie group G can be left- or right-translated to the tangent space at the identity, i.e. to the Lie algebra $\mathfrak{g}$. Given a tangent vector $\dot{h} \in T_h G$ at a point $h \in G$, its left, respectively right, representation in the Lie algebra is obtained using the push-forward (Definition 3.11) of the left, respectively right, translation by h^{-1} :

$$\begin{aligned} (\mathrm{l}_{h^{-1}})_{*,h}: T_h G &\rightarrow T_{h^{-1}h} G = \mathfrak{g} & \text{and} & & (\mathrm{r}_{h^{-1}})_{*,h}: T_h G &\rightarrow T_{hh^{-1}} G = \mathfrak{g} \\ \dot{h} &\mapsto (\mathrm{l}_{h^{-1}})_{*,h} \dot{h} & & & \dot{h} &\mapsto (\mathrm{r}_{h^{-1}})_{*,h} \dot{h}. \end{aligned}$$

Definition 3.17. The *conjugation* by an element $g \in G$ on a Lie group G is the diffeomorphism:

$$\begin{aligned} C_g = \mathrm{l}_g \circ \mathrm{r}_{g^{-1}}: G &\rightarrow G \\ h &\mapsto g \bullet h \bullet g^{-1}. \end{aligned}$$

Note the conjugation of the identity element by any element g : $C_g(e) = g \bullet e \bullet g^{-1} = g \bullet g^{-1} = e$ is the identity element e itself. Therefore, the push-forward of the conjugation by g at the identity element is a linear map: $(C_g)_{*,e}: T_e G \rightarrow T_e G$, which is a linear isomorphism of the Lie algebra $\mathfrak{g} = T_e G$.

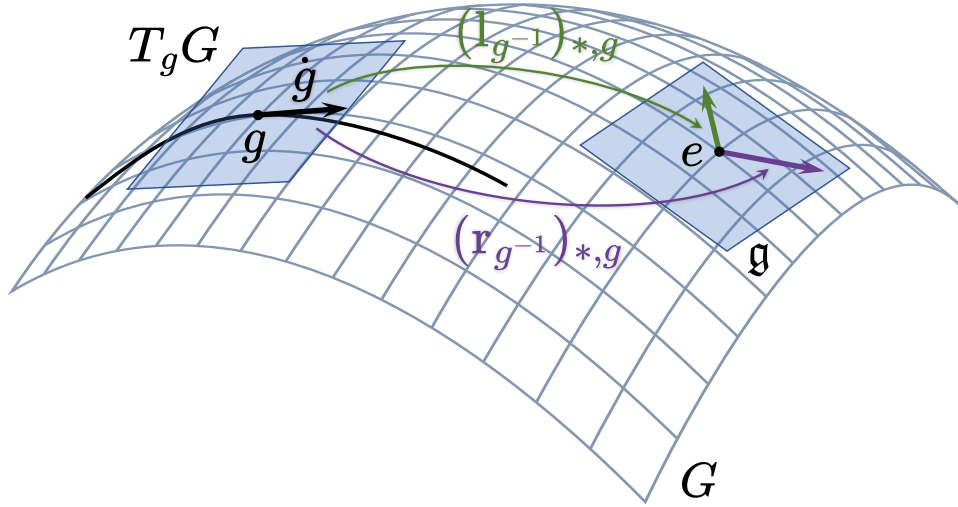


Figure 3.3: Left and right translation of a tangent vector $\dot{g}$ at a point g to the Lie algebra $\mathfrak{g}$ of a Lie group G .

Definition 3.18. The *adjoint action* of a Lie group G on its Lie algebra $\mathfrak{g}$ is the map:

$$\begin{aligned} \text{Ad}: G \times \mathfrak{g} &\rightarrow \mathfrak{g} \\ (g, X) &\mapsto \text{Ad}_g(X) := (C_g)_{*,e} X. \end{aligned}$$

Chapter 4

Linear Differential Equations *

One of the simplest examples of a differential equation is a so-called 1st-order linear homogeneous ordinary differential equation (ODE) with constant coefficients. It has a form

$$\dot{x} = ax(t),$$

where x is a differentiable function $x \in \mathcal{C}^1(\mathbb{R} \rightarrow \mathbb{R})$, $\dot{x} := \frac{dx}{dt}$ and $a \in \mathbb{R}$.

Given an initial condition (x_0, t_0) such that $x(t_0) = x_0$, the unique solution to this equation is

$$x(t) = e^{a(t-t_0)}x_0.$$

Extending to systems, a homogeneous system of first-order linear ODEs with constant coefficients is given by:

$$\dot{x} = Ax, \tag{4.1}$$

where $x \in \mathcal{C}^1(\mathbb{R} \rightarrow \mathbb{R}^n)$ and $A \in \mathbb{R}^{n \times n}$ is an $n \times n$ matrix of constant coefficients.

Given an initial condition (x_0, t_0) such that $x(t_0) = x_0$, the unique solution to this equation is

$$x(t) = e^{A(t-t_0)}x_0.$$

Here we used the matrix exponential, which is defined as:

$$e^A := I + A + \frac{A^2}{2} + \frac{A^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{A^k}{k!}. \tag{4.2}$$

The matrix exponential has the familiar properties of the usual exponential, although with some restrictions.

- $\frac{d}{dt}e^{At} = Ae^{At}$
- $(e^A)^{-1} = e^{-A}$

- if $A = TJT^{-1}$, then $e^A = Te^JT^{-1}$
- if $AB = BA$, then $e^{A+B} = e^Ae^B$
- if $A = \text{diag}(\lambda_1, \dots, \lambda_n)$, then $e^A = \text{diag}(e^{\lambda_1}, \dots, e^{\lambda_n})$

Part II

Kinematics

Chapter 5

Euclidean Spaces and Rigid Motion

5.1 Affine and Euclidean spaces

Before studying kinematics and dynamics of mechanical systems, we need to properly understand the physical space in which these systems are embedded. And the understanding comes once we establish the minimal mathematical structure that describes it. Note that in the 3-dimensional space of physical points, there is no special point that we might call the origin. It also does not make sense to talk about the sum of two points. So, a vector space already assumes more structure than we need. As we will see in this chapter, the appropriate structure to describe the physical space is the Euclidean space¹.

The Euclidean space is a special case of a more general structure called an affine space when we equip it with a notion of distance. Informally, an affine space is a collection of points, connected by vectors that represent translations between the points. Formally, an affine space is defined using a group action (see Section 3.1.2) of a vector space (see Section 2.1) on a set of points.

Definition 5.1. An *affine space* is triple $(\mathcal{A}, \vec{\mathcal{A}}, \oplus)$ consisting of a set $\mathcal{A}$ together with a vector space $(\vec{\mathcal{A}}, +_{\vec{\mathcal{A}}}, \cdot_{\vec{\mathcal{A}}})$ and a right free and transitive action $\oplus$ of $(\vec{\mathcal{A}}, +_{\vec{\mathcal{A}}})$ as an additive group on $\mathcal{A}$. Elements of $\mathcal{A}$ are called *points*, and elements of $\vec{\mathcal{A}}$ are called *vectors* or *translations*.

Explicitly the axioms of the action $\oplus$ are:

$$\begin{aligned}\oplus: \mathcal{A} \times \vec{\mathcal{A}} &\rightarrow \mathcal{A} \\ (p, v) &\mapsto \oplus(p, v) := p \oplus v,\end{aligned}$$

such that

$$(i) \text{ Right identity } p \oplus 0_{\vec{\mathcal{A}}} = p, \quad \forall p \in \mathcal{A},$$

¹We are working in the realm of Newtonian mechanics assuming existence of absolute time and absolute space.

- (ii) Right associativity $(p \oplus v) \oplus w = p \oplus (v +_{\vec{\mathcal{A}}} w)$, $\forall p \in \mathcal{A}, \forall v, w \in \vec{\mathcal{A}}$
- (iii) Free and transitive action $\forall p, q \in \mathcal{A} \exists! v \in \vec{\mathcal{A}}: p \oplus v = q$.

The first two axioms are the axioms of a right group action. Axiom (i) states that a translation by the zero vector $0_{\vec{\mathcal{A}}} \in \vec{\mathcal{A}}$ does nothing. Axiom (ii) states that a composition of translations is equivalent to a single translation by the sum of the vectors as pictured in Figure 5.1. Axiom (iii) ensures that for any two points p and q , there exists a unique vector that translates p to q .

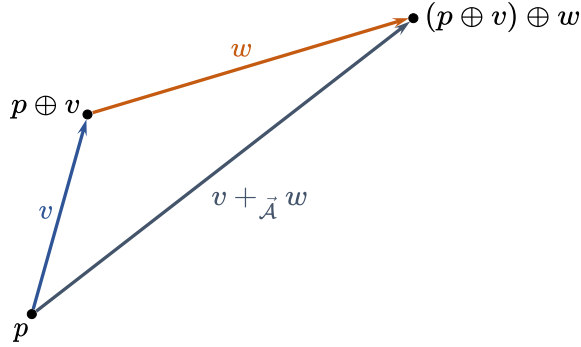


Figure 5.1: Associativity of the action $\oplus$ in an affine space.

As a consequence of axiom (iii), the following map (returning this unique vector) is well-defined:

$$\begin{aligned} - : \mathcal{A} \times \mathcal{A} &\rightarrow \vec{\mathcal{A}} \\ (p, q) &\mapsto -(p, q) \equiv p - q := v, \quad \text{s.t. } q \oplus v = p. \end{aligned}$$

The *dimension* of an affine space is defined as the dimension of the vector space $\vec{\mathcal{A}}$.

An affine space is sometimes referred to as a vector space without an origin. Indeed, it contains no special point. We can, however, manually choose an ‘origin’ which will allow us to identify the affine space with its associated vector space as follows.

Proposition 5.2. *Fix a point $o \in \mathcal{A}$, then the map*

$$\begin{aligned} (\cdot - o) : \mathcal{A} &\rightarrow \vec{\mathcal{A}} \\ p &\mapsto p - o \end{aligned}$$

with the inverse map

$$\begin{aligned} (o \oplus \cdot) : \vec{\mathcal{A}} &\rightarrow \mathcal{A} \\ v &\mapsto o \oplus v \end{aligned}$$

is a bijection between $\mathcal{A}$ and $\vec{\mathcal{A}}$.

Indeed, when one speaks of points in space as vectors, one always implicitly chooses an origin.

Look at the definition of an affine space and verify that it is indeed the mathematical structure that underlies the physical space. It is important to keep in mind that physical space does not have a special point that we might call the origin and that points in space are not vectors – one cannot add them or multiply them by a scalar. Differences between points, however, are vectors that can be added and multiplied by scalars in a meaningful way.

The physical space we wish to describe has one more bit of structure – a notion of distance. Equipping an affine space with such a notion turns it into a Euclidean space.

Definition 5.3. A *Euclidean vector space* is a (real) inner product space of finite dimension $(\vec{\mathcal{E}}, g)$. (See Section 2.2.4)

Definition 5.4. A *Euclidean space* is an affine space $(\mathcal{E}, (\vec{\mathcal{E}}, g), \oplus)$, where $(\vec{\mathcal{E}}, g)$ is a Euclidean vector space.

Similarly to how a vector space $(V, \oplus, \odot)$ is denoted by the underlying set V , we typically denote a Euclidean space by $\mathcal{E}$.

Using the inner product of the associated vector space $\vec{\mathcal{E}}$ and its associated norm we can define the *distance* or *metric* on a Euclidean space $\mathcal{E}$ as

$$d: \mathcal{E} \times \mathcal{E} \rightarrow \mathbb{R}$$

$$(p, q) \mapsto d(p, q) := \|p - q\|_g = \sqrt{g((p - q), (p - q))}$$

making $(\mathcal{E}, d)$ a metric space.

Example 5.5. A prototypical example of an n -dimensional Euclidean space is built using $\mathbb{R}^n$. We take the set $\mathbb{R}^n$ as the set of points, the vector space $(\mathbb{R}^n, g_{\text{can}})$ with the canonical inner product as the associated vector space, and the addition $+$ of $\mathbb{R}^n$ as the action. Thus, we obtain the Euclidean space $(\mathbb{R}^n, (\mathbb{R}^n, g_{\text{can}}), +)$. This is precisely the structure that is often implicitly used when one uses n coordinates to describe points in space.

As usual, having introduced a new mathematical structure, we study the maps that preserve it. In the case of a Euclidean space, such maps will play a crucial role since they will be used to describe rigid body motions.

Definition 5.6. An *isometry* is a map $h: (\mathcal{E}_1, d_1) \rightarrow (\mathcal{E}_2, d_2)$ between two Euclidean spaces that preserves the distance, i.e.

$$d_2(h(p), h(q)) = d_1(p, q) \quad \forall p, q \in \mathcal{E}_1.$$

If h is bijective, and its inverse h^{-1} is an isometry as well, then h is called an *isometric isomorphism* or an *isomorphism of Euclidean spaces*.

An isometry h induces a map $\vec{h}$ between the associated vector spaces

$$\begin{aligned}\vec{h}: \vec{\mathcal{E}}_1 &\rightarrow \vec{\mathcal{E}}_2 \\ v &\mapsto \vec{h}(v) := h(o \oplus v) - h(o)\end{aligned}\tag{5.1}$$

for some $o \in \mathcal{E}_1$.

It can be shown that $\vec{h}$ is uniquely defined by h and that it is a linear map preserving the inner product (*linear isometry*) [1]. Moreover, if h is bijective, then $\vec{h}$ is a linear isomorphism between the vector spaces and the inverse h^{-1} is an isometry as well. Therefore, any bijective isometry is an isometric isomorphism.

Definition 5.7. An isometry $h: \mathcal{E} \rightarrow \mathcal{E}$ on the same Euclidean space is called *proper* or *positive* if $\vec{h}$ has a positive determinant².

Thus, an isometry preserves distances between points and angles between vectors. A positive isometry additionally preserves the orientation of the space, i.e. it does not allow for reflections.

Exercise 5.8. Show that an isometry h together with the associated map $\vec{h}$ preserves the action $\oplus$ of the affine space, i.e. for two Euclidean spaces $(\mathcal{E}_1, \vec{\mathcal{E}}_1, \oplus_1)$ and $(\mathcal{E}_2, \vec{\mathcal{E}}_2, \oplus_2)$ and an isometry $h: \mathcal{E}_1 \rightarrow \mathcal{E}_2$ the following holds:

$$h(p \oplus_1 v) = h(p) \oplus_2 \vec{h}(v) \quad \forall p \in \mathcal{E}_1, \forall v \in \vec{\mathcal{E}}_1.$$

In other words, conclude that h is indeed a structure-preserving map.

As with vector spaces, it can be more convenient in practice to work with numbers instead of abstract elements of the space. In Euclidean spaces, it is achieved by choosing a coordinate frame.

Definition 5.9. A *coordinate frame* in an n -dimensional Euclidean space $\mathcal{E}$ is a tuple $\Psi = (o, e_1, \dots, e_n)$, where $o \in \mathcal{E}$ is a point in space, and $e_i \in \vec{\mathcal{E}}$ constitute an ordered basis of $\vec{\mathcal{E}}$. The point o is called the *origin* of the frame. The frame is called *orthonormal* if $g(e_i, e_j) = \delta_{ij}$, where δ_{ij} is the Kronecker delta.

Definition 5.10. The *coordinatisation map* associated to a frame $\Psi = (o, e_1, \dots, e_n)$ is defined as

$$\begin{aligned}\psi: \mathcal{E} &\rightarrow \mathbb{R}^n \\ p &\mapsto \psi(p) := \psi_e(p - o),\end{aligned}$$

where $\psi_e: \vec{\mathcal{E}} \rightarrow \mathbb{R}^n$ is the coordinatisation map of the vector space $\vec{\mathcal{E}}$ associated with the basis $(e_1, \dots, e_n)$.

One can see the coordinatisation map as a composition of two maps

$$\psi = \psi_e \circ (\cdot - o), \quad \mathcal{E} \xrightarrow{(\cdot - o)} \vec{\mathcal{E}} \xrightarrow{\psi_e} \mathbb{R}^n,$$

²Determinant is defined for any linear map $\varphi: V \rightarrow V$ and is a notion independent of the choice of a basis. Naively, it is the usual matrix determinant of the coordinate representation of φ in some basis.

where we first identify $\mathcal{E}$ with $\vec{\mathcal{E}}$ using the point o as the origin and then take the components of the vector in the basis $(e_1, \dots, e_n)$ of $\vec{\mathcal{E}}$.

Remark 5.11. Technically, the map $\psi_e: \vec{\mathcal{E}} \rightarrow \mathbb{R}^n$ brings us to the vector space $\mathbb{R}^n$, but not the Euclidean space $\mathbb{R}^n$. So, we need to compose it with the map $(0 + \cdot): \mathbb{R}^n \rightarrow \mathbb{R}^n$, which is the identity map on $\mathbb{R}^n$ as a set, to obtain $\psi = (0 + \cdot) \circ \psi_e \circ (\cdot - o)$ as a map from $\mathcal{E}$ to $\mathbb{R}^n$ as Euclidean spaces.

Proposition 5.12. *The coordinatisation map ψ associated with an orthonormal frame Ψ in an n -dimensional Euclidean space $\mathcal{E}$ is an isometric isomorphism between $\mathcal{E}$ and $\mathbb{R}^n$ as the canonical Euclidean n -space.*

Proof. Let us represent the coordinatisation map as the composition $\psi = \psi_e \circ (\cdot - o)$. The vector space coordinatisation map $\psi_e: \vec{\mathcal{E}} \rightarrow \mathbb{R}^n$ is an isometric isomorphism between the inner product space $(\vec{\mathcal{E}}, g)$ and $(\mathbb{R}^n, g_{\text{can}})$ by Proposition 2.49. Therefore, by Exercise 2.45, ψ_e preserves the distance. Thus, it is left to show that the bijective map $(\cdot - o)$ preserves distances. Let $d_{\mathcal{E}}$ and $d_{\vec{\mathcal{E}}}$ be the metrics on $\mathcal{E}$ and $\vec{\mathcal{E}}$, respectively. Then, for any $p, q \in \mathcal{E}$ we have

$$d_{\vec{\mathcal{E}}}(p - o, q - o) = \|(p - o) - (q - o)\|_{\vec{\mathcal{E}}} = \|p - q\|_{\vec{\mathcal{E}}} = d_{\mathcal{E}}(p, q).$$

Thus, the coordinatisation map ψ is a bijective isometry between $\mathcal{E}$ and $\mathbb{R}^n$ as a composition of two bijections preserving distances. $\square$

We stress that even though the isomorphism between $\mathcal{E}$ and $\mathbb{R}^n$ always exists, there is no canonical choice of such an isomorphism since it depends on the choice of a frame. This is completely analogous to the isomorphism between an n -dimensional vector space and $\mathbb{R}^n$, which is not canonical either and is established by choosing a basis. The reason to work with an abstract Euclidean space $\mathcal{E}$ instead of $\mathbb{R}^n$ despite the isomorphism is to keep conceptual clarity: in $\mathbb{R}^n$ one may be tempted to impose more structure than is actually present in the physical space, such as the origin or the preferred directions of the axes.

Remark 5.13 (Smooth structure of Euclidean spaces). A Euclidean n -space can always be equipped with a structure of an n -dimensional smooth manifold. The coordinatisation maps ψ provide homeomorphisms between $\mathcal{E}$ and $\mathbb{R}^n$ forming a smooth atlas. Inner product on $\vec{\mathcal{E}}$ gives rise to so-called flat Riemannian³ metric on $\mathcal{E}$ and allows to canonically identify all tangent spaces $T_p\mathcal{E}$ with $\vec{\mathcal{E}}$.

5.2 Rigid motion

Now, as we have studied the structure of Euclidean spaces and isometries, we are fully equipped to describe the motion of rigid bodies in space. Suppose that we have a rigid body in a physical space. Then we expect that after a rigid motion, the body is not deformed, i.e. all the distances between points will be preserved, as well as that the body is not mirrored. This is precisely the properties of a positive

³Bernhard Riemann (1826–1866), German mathematician.

isometry. (By motion we mean a single change between two states of the body, not a continuous process.)

Definition 5.14. A *rigid motion* or *rigid transformation* is a positive isometry $h: \mathcal{E} \rightarrow \mathcal{E}$ of a Euclidean space $\mathcal{E}$.

Thus, a rigid motion is a transformation of the entire Euclidean space. The isometry property ensures that the distances between points are preserved and, therefore, shapes and sizes, hence the name *rigid*. The positive property excludes reflections, such as transformations that map a right hand to a left hand. If a rigid body is embedded in a Euclidean space, then a rigid transformation would model the motion of the body in space. Note that in this context, the term ‘motion’ is used to describe a single change between two states of the body, not a continuous process.

Proposition 5.15. An isometry $h: \mathcal{E} \rightarrow \mathcal{E}$ on the same Euclidean space is a positive isometric isomorphism, i.e. it is bijective and the inverse h^{-1} of a positive isometry is a positive isometry as well.

Exercise 5.16. Prove Proposition 5.15.

Remark 5.17. Together with the fact that the composition of positive isometries $h_2 \circ h_1$ is a positive isometry, this proposition implies that the set of all rigid transformations of a Euclidean space forms a group. The group is called the *special Euclidean group* of $\mathcal{E}$ and is denoted by $\text{SE}(\mathcal{E})$. Furthermore, it can be shown that the group $\text{SE}(\mathcal{E})$ is a Lie group and that the application of a transformation h to the Euclidean space $\mathcal{E}$ is a Lie group action on the space.

5.2.1 Rotation

Before studying the general motion of a rigid body in space, we specialise to the case of rigid body rotation about a fixed point $o \in \mathcal{E}$.

Definition 5.18. A *rotation* of a Euclidean space $\mathcal{E}$ is a rigid motion $r: \mathcal{E} \rightarrow \mathcal{E}$ for which there exists a fixed point $o \in \mathcal{E}$, i.e. $r(o) = o$.

Remark 5.19. The set of all rotations r of a Euclidean space $\mathcal{E}$ about a fixed point $o \in \mathcal{E}$ together with the composition forms a Lie group, called the *special orthogonal group* of $\mathcal{E}$ at o and denoted by $\text{SO}(\mathcal{E}, o)$.

We shall now describe rotations of a Euclidean space in coordinates. Consider an orthonormal frame $\Psi = (o, e_1, \dots, e_n)$ centred at o . Then the *coordinate representation of rotation* r in that frame is given by a map $R: \mathbb{R}^n \rightarrow \mathbb{R}^n$, which replicates the action of r on the coordinates of points in $\mathcal{E}$. Precisely, the replication is defined as first mapping coordinates in $\mathbb{R}^n$ to the actual points in $\mathcal{E}$, then applying r , and finally going back to coordinates:

$$R := \psi \circ r \circ \psi^{-1}. \quad (5.2)$$

Equivalently, we require that the following diagram commutes⁴:

⁴A diagram is said to commute if all paths along the arrows between two vertices give the same result. They are therefore just a graphical representation of equalities between map compositions.

$$\begin{array}{ccc}
\mathcal{E} & \xrightarrow{r} & \mathcal{E} \\
\psi^{-1} \uparrow & & \downarrow \psi \\
\mathbb{R}^n & \xrightarrow{R} & \mathbb{R}^n
\end{array}$$

We wish to deduce the properties of the coordinate representation R of a rigid body rotation r .

Recall that the coordinatisation map is a composition of two maps $\psi = \psi_e \circ (\cdot - o)$. Therefore, the diagram above can be rewritten as follows.

$$\begin{array}{ccc}
\mathcal{E} & \xrightarrow{r} & \mathcal{E} \\
(\cdot - o) \downarrow & & \uparrow (o \oplus \cdot) \\
\vec{\mathcal{E}} & \xrightarrow{\vec{r}} & \vec{\mathcal{E}} \\
\psi_e \downarrow & & \uparrow \psi_e^{-1} \\
\mathbb{R}^n & \xrightarrow{R} & \mathbb{R}^n
\end{array}$$

By Definition 5.6, the linear map $\vec{r}$ acts on vectors $v = p - o \in \vec{\mathcal{E}}$ as

$$\vec{r}(p - o) = r(p) - r(o) \quad \forall p \in \mathcal{E}.$$

Since the rotation r preserves the centre of rotation o , we get that

$$\vec{r}(\cdot - o) = r(\cdot) - o,$$

or equivalently, that

$$r = (o \oplus \cdot) \circ \vec{r} \circ (\cdot - o),$$

meaning that the upper square of the diagram commutes. Note that this does not hold for a general rigid motion h and its associated linear map $\vec{h}$. Since the upper and the outer squares commute, the lower square also commutes, hence, R is the coordinate representation of the linear map $\vec{r}$ in the orthonormal basis $(e_1, \dots, e_n)$. R is therefore an $n \times n$ matrix. By Proposition 2.50, R is then a linear isometry on $(\mathbb{R}^n, g_{\text{can}})$ with a positive determinant. As a result, the positive isometry condition of r descends to orthonormal coordinates and imposes the following conditions on the matrix R :

$$(i) \quad g_{\text{can}}(v, w) = g_{\text{can}}(Rv, Rw) \quad \forall v, w \in \mathbb{R}^n;$$

$$(ii) \quad \det R > 0.$$

Note that the fact that the origin of the frame is the centre of rotation is crucial for this construction to work. If the frame is not centred at o , the rotation cannot be represented by a matrix R in these coordinates.

The first condition can be written using matrix multiplication as

$$v^\top R^\top I R w = v^\top I w \quad \forall v, w \in \mathbb{R}^n,$$

where we used the fact that the canonical inner product g_{can} on $\mathbb{R}^n$ is represented by the identity matrix I in an orthonormal basis. Since this equation holds for all $v, w \in \mathbb{R}^n$, we obtain the condition

$$R^\top R = I. \quad (5.3)$$

Matrices R of dimension $n \times n$ satisfying (5.3) are called *orthogonal matrices*⁵. The name ‘orthogonal’ is not accidental – the condition (5.3) is equivalent to the condition that the columns as well as the rows of an orthogonal matrix are pairwise orthonormal vectors in $(\mathbb{R}^n, g_{\text{can}})$.

Taking the determinant of (5.3), we notice that the determinant of an orthogonal matrix is either 1 or -1 :

$$\det(R^\top R) = \det R^\top \det R = (\det R)^2 = \det I = 1 \quad \Rightarrow \quad \det R = \pm 1.$$

Therefore, the second condition becomes

$$\det R = 1.$$

As a result, coordinate representations of rotations in an orthonormal frame with the origin at the centre of rotation are given by orthogonal matrices with determinant 1, called *rotation matrices*. The set of all rotation matrices is denoted by

$$\text{SO}(n) := \{R \in \mathbb{R}^{n \times n} \mid R^\top R = I, \det R = 1\}.$$

Together with the matrix multiplication, $\text{SO}(n)$ forms a Lie group called the *special orthogonal group* of $\mathbb{R}^n$.

Exercise 5.20. Verify that the axioms of a group (Section 3.1.1) are satisfied for $\text{SO}(n)$ with the matrix multiplication.

Remark 5.21. Using the notation of Remark 5.19, $\text{SO}(n)$ is the group $\text{SO}(\mathbb{R}^n, 0)$ of rotations of $\mathbb{R}^n$ about the origin, which is the coordinate representation of the abstract special orthogonal group $\text{SO}(\mathcal{E}, o)$ of the Euclidean space $\mathcal{E}$ at $o \in \mathcal{E}$.

Remark 5.22. The name ‘special’ comes from the restriction of the determinant to 1. The space of all orthogonal matrices with determinant ± 1 , that is, all linear isometries of $\mathbb{R}^n$ which include both rotations and reflections, also forms a Lie group called *orthogonal group* of $\mathbb{R}^n$ and is denoted by $\text{O}(n)$. $\text{SO}(n)$ is a subgroup of $\text{O}(n)$ and is a connected component of the identity, while $\text{O}(n)$ has two connected components. That means that starting from the identity matrix, one can continuously reach any other rotation in $\text{SO}(n)$, while a reflection cannot be reached by a continuous path from the identity since it belongs to the other connected component of $\text{O}(n)$.

We conclude that the action of a rotation r on an arbitrary point $q \in \mathcal{E}$ in coordinates of a frame Ψ is given by the matrix multiplication of the rotation matrix R on the

⁵In fact, orthogonal matrices with the matrix multiplication form a Lie group $\text{O}(n)$ called the *orthogonal group* of dimension n .

coordinates of q .

$$\begin{bmatrix} | \\ \psi(r(q)) \\ | \end{bmatrix} = R \cdot \begin{bmatrix} | \\ \psi(q) \\ | \end{bmatrix}.$$

A practically useful consequence of the orthogonality condition (5.3) is that the inverse of a rotation matrix (coordinate representation of the inverse rotation) is its transpose:

$$R^{-1} = R^{\top},$$

which is computationally a much lighter operation than the general matrix inversion.

What is the dimension of $\text{SO}(n)$ itself? That is, how many independent parameters are needed to describe a rotation in an n -dimensional space? In dimension 2, a plane rotation is described by a single angle, hence $\text{SO}(2)$ is one-dimensional. In dimension 3, a rotation is described by three angles, and hence $\text{SO}(3)$ is three-dimensional. In general, a rotation in n -dimensional space is described by $\frac{n(n-1)}{2}$ independent parameters, which can be seen by counting the number of independent entries in an orthogonal matrix R .

Example 5.23. Consider a rotation in 2D space by an angle θ . The rotation matrix is given by

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

The inverse rotation, that is the rotation by the angle $-\theta$, is given by the transpose of R .

Example 5.24. Consider a rotation in 3D space about the z -axis by an angle θ . The rotation matrix is given by

$$R = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Finally, consider a sequence of rotations $r_1, r_2, \dots, r_k$ about the same fixed point $o \in \mathcal{E}$. The composition r of these rotations is a rotation as well.

$$r = r_k \circ \dots \circ r_2 \circ r_1$$

The coordinate representation R of the total rotation in some fixed orthonormal frame Ψ centred at o is then given by the matrix product of the coordinate representations R_i of the individual rotations:

$$R = R_k \cdot \dots \cdot R_2 \cdot R_1. \quad (5.4)$$

This constitutes what is called the *chain rule* for rotations.

We therefore showed that rigid body rotations about a fixed point can be described by orthogonal matrices with determinant 1 once an orthonormal coordinate frame is chosen. Moreover, the set of all such matrices forms a Lie group $\text{SO}(n)$, which is a smooth manifold with a group structure.

5.2.1.1 Other representations of rotations

At this point, one may ask: why do we need to use Lie groups and matrices to describe rotations if we can use angles that parametrise it? Turns out that the use of Lie groups is crucial. Already in dimension 2, if one uses an angle θ to describe a rotation, one runs into a problem that θ is not a real number in $\mathbb{R}$, but a point on a circle. The topology of the space is different! So, for example, we have to identify the angles θ and $\theta + 2\pi$ as the same rotation. The Lie group $\text{SO}(2)$ as a manifold is precisely the circle.

Things get even worse in dimension 3. One may be familiar with other descriptions of rotations in 3D space using the Euler⁶ angles or aircraft angles (roll, pitch, yaw). They attempt to describe a rotation with three numbers, however, they suffer from the problem of singularities and do not cover the entire space of rotations globally.

Remark 5.25. In fact, such representations form only a local chart on the manifold $\text{SO}(3)$, and since $\text{SO}(3)$ is not homeomorphic to $\mathbb{R}^3$, it is impossible to cover it with a single global chart, that is, describe it with three numbers. 3×3 rotation matrices describe the entire space of rotations globally. They have 9 entries, which are, however, constrained by the orthogonality condition to only 3 independent parameters.

Another common representation of rotations is the use of quaternions⁷. This representation is geometric, and it does describe the entire space of rotations globally. Unit quaternions form a Lie group

$$\text{SU}(2) := \left\{ \begin{pmatrix} a + bi & c + di \\ -c + di & a - bi \end{pmatrix} \in \mathbb{C}^{2 \times 2} \mid a^2 + b^2 + c^2 + d^2 = 1 \right\},$$

called the *special unitary group* in dimension 2.

Remark 5.26. $\text{SU}(2)$ is a double cover of $\text{SO}(3)$, which means that each rotation in the 3D space is represented by two unit quaternions. As manifolds, the group $\text{SU}(2)$ is simply connected, while $\text{SO}(3)$ is not. This fact is visually illustrated by the Dirac's belt trick.

5.2.2 General rigid motion

We now consider a general rigid motion given by a positive isometry $h: \mathcal{E} \rightarrow \mathcal{E}$ and study its coordinate representation in an arbitrary orthonormal frame.

Exercise 5.27. Recall that a translation of an affine space is defined as the addition of a vector from the associated vector space $\vec{\mathcal{E}}$. Show that the translation $t = \cdot \oplus v$ by any vector $v \in \vec{\mathcal{E}}$ is a positive isometry of the Euclidean space.

Remark 5.28. The composition of two translations $t_1 = \cdot \oplus v_1$ and $t_2 = \cdot \oplus v_2$ is a translation by the sum of the vectors: $t_2 \circ t_1 = (\cdot \oplus v_1) \oplus v_2 = \cdot \oplus (v_1 + v_2)$. Therefore, the set of all translations forms a group isomorphic to the additive group $\vec{\mathcal{E}}$.

⁶Leonhard Euler (1707–1783), Swiss mathematician.

⁷A great [video by 3Blue1Brown](#) explaining quaternions and 3d rotations.

Theorem 5.29. *Any rigid motion $h: \mathcal{E} \rightarrow \mathcal{E}$ can be expressed as a composition of a rotation and a translation of $\mathcal{E}$.*

Proof. Consider an arbitrary point $p \in \mathcal{E}$ and its image $h(p)$ after the transformation. Now consider the difference vector $v = h(p) - p \in \vec{\mathcal{E}}$. Then the translation

$$t = (\cdot \oplus v): \mathcal{E} \rightarrow \mathcal{E}$$

by v is a positive isometry that, in particular, maps p to $h(p)$.

The inverse of t (translation by $-v$) is also a positive isometry, which maps $h(p)$ back to p . Therefore, the composition $h \circ t^{-1}$ is a positive isometry that fixes the point p , which by definition is a rotation about p . The total action of h can thus be decomposed as the composition $h = t \circ r$ of a rotation r and a translation t . $\square$

Note that the decomposition is not unique as it depends on the choice of the point p – the centre of rotation.

Remark 5.30. In dimensions 2 and 3 the result can be further refined. In 2D, any rigid transformation is either a rotation about a point or a pure translation. In 3D, the following theorems hold.

Theorem 5.31 (Euler’s rotation theorem). *In a 3-dimensional Euclidean space, any rotation $r: \mathcal{E} \rightarrow \mathcal{E}$ about a fixed point o is a rotation about an axis passing through o , that is, there always exists not just a point but an entire line that is fixed by the rotation. Furthermore, if the rotation is not the identity transformation, the axis is unique.*

Theorem 5.32 (Mozzi⁸–Chasles⁹ theorem). *In a 3-dimensional Euclidean space, any general rigid motion $h: \mathcal{E} \rightarrow \mathcal{E}$ can be represented by a composition of a rotation about an axis and a translation in the direction of the axis. Furthermore, if the motion is not an identity transformation this direction is unique, and if the motion is not a pure translation, the axis is unique.*

Such motion is called a *screw motion*, and the axis is called the *screw axis*.

We shall now describe general rigid motions in coordinates. Consider some orthonormal frame $\Psi = (o, e_1, \dots, e_n)$ centred at an arbitrary point $o \in \mathcal{E}$ and construct a coordinate representation of the motion h in that frame given by a map $H: \mathbb{R}^n \rightarrow \mathbb{R}^n$ replicating the action of h on the coordinates of points in $\mathcal{E}$. Replication means that

$$h = \psi^{-1} \circ H \circ \psi,$$

or equivalently, that the following diagram commutes:

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{h} & \mathcal{E} \\ \psi \downarrow & & \uparrow \psi^{-1} \\ \mathbb{R}^n & \xrightarrow{H} & \mathbb{R}^n \end{array}$$

⁸Giulio Mozzi (1730–1813), Italian mathematician.

⁹Michel Chasles (1793–1880), French mathematician.

Given an orthonormal frame Ψ in $\mathcal{E}$, the coordinatisation of rigid motions is given by the map

$$\begin{aligned} (\psi \circ \cdot \circ \psi^{-1}): \text{SE}(\mathcal{E}) &\rightarrow \text{SE}(n) \\ h &\mapsto \psi \circ h \circ \psi^{-1} =: H. \end{aligned} \quad (5.5)$$

It can be shown that the map (5.5) is a group isomorphism.

We wish to deduce the properties of the coordinate representation H of a general rigid motion h . Decomposing $h = t \circ r$ into a translation t and a rotation r about o , we expand the diagram as follows:

$$\begin{array}{ccccc} \mathcal{E} & \xrightarrow{r} & \mathcal{E} & \xrightarrow{t} & \mathcal{E} \\ \psi \downarrow & & \uparrow \psi^{-1} & & \uparrow \psi^{-1} \\ \mathbb{R}^n & \xrightarrow{R} & \mathbb{R}^n & \xrightarrow{T} & \mathbb{R}^n \end{array}$$

where T is the coordinate representation of the translation t and R is the coordinate representation of the rotation r . As we have shown for rotations, in a frame centred at o , $R \in \text{SO}(n)$ is an orthogonal matrix with determinant 1 acting on the coordinates of points. The coordinate representation T of the translation $t = (\cdot \oplus v)$ is given by the addition of the coordinate of v in the basis $(e_1, \dots, e_n)$ since the coordinatisation ψ is an isomorphism of Euclidean spaces and therefore preserves the action of translations. Denoting the components of v by $T \in \mathbb{R}^n$, we have

$$\begin{bmatrix} \psi(h(q)) \\ | \\ | \end{bmatrix} = R \cdot \begin{bmatrix} \psi(q) \\ | \\ | \end{bmatrix} + \begin{bmatrix} T \\ | \\ | \end{bmatrix} \quad (5.6)$$

for any point $q \in \mathcal{E}$. Since T represents the components of the vector $v = h(o) - o$, it is equal to the coordinates of a point $h(o)$ in the frame Ψ , which is centred at o (see Definition 5.10).

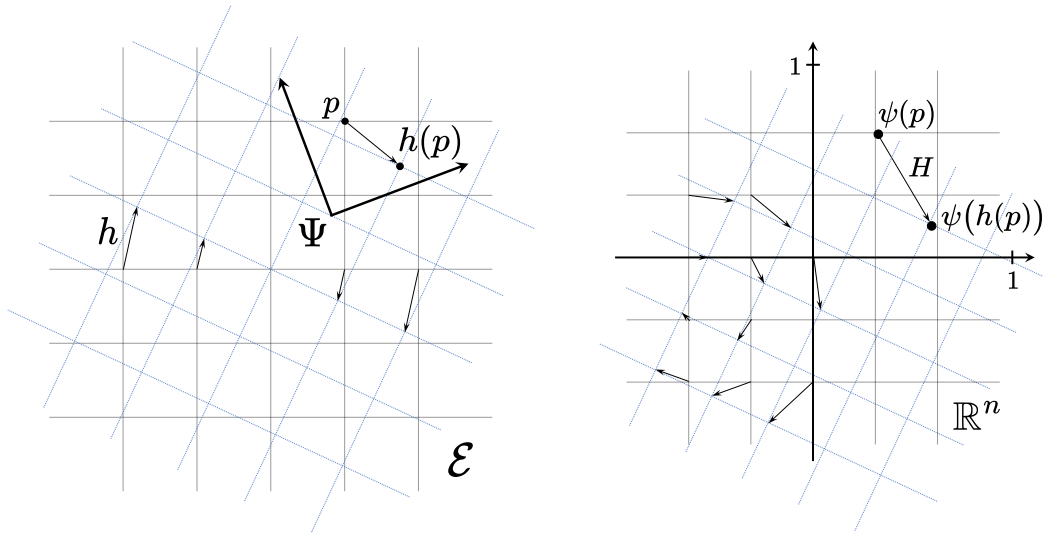


Figure 5.2: Coordinate representation of a general rigid motion h in an orthonormal frame Ψ .

5.2.2.1 Homogeneous matrices and the Lie group $\text{SE}(n)$

Note that in general the action of H of the coordinates of points is not given by a single matrix multiplication, but by a sum of a matrix multiplication and a vector addition. The coordinate representation of a composition of general rigid motions cannot, therefore, be directly written as a product of matrices. One can, however, employ a little trick to obtain the chain rule for general rigid motions.

Let us append all coordinates of with a 1 to obtain a column of length $n + 1$:

$$\psi(q) \longrightarrow \begin{pmatrix} \psi(q) \\ 1 \end{pmatrix}$$

Note that this is only a notational change, they still denote coordinates in an n -dimensional space, not $n + 1$ -dimensional. We can then construct the matrix H as:

$$H := \begin{pmatrix} R & T \\ \mathbf{0}_{1 \times n} & 1 \end{pmatrix} \in \mathbb{R}^{(n+1) \times (n+1)},$$

where $\mathbf{0}_{1 \times n}$ denotes a row of n zeros. Now, the matrix multiplication of H on the augmented coordinates of a point q precisely replicates the coordinate representation of the rigid motion h :

$$H \cdot \begin{pmatrix} \psi(q) \\ 1 \end{pmatrix} = \begin{pmatrix} R & T \\ \mathbf{0}_{1 \times n} & 1 \end{pmatrix} \cdot \begin{pmatrix} \psi(q) \\ 1 \end{pmatrix} = \begin{pmatrix} R \cdot \psi(q) + T \\ 1 \end{pmatrix} = \begin{pmatrix} \psi(h(q)) \\ 1 \end{pmatrix}.$$

Such matrices are called *homogeneous matrices*, and they play a pivotal role in this course as they will represent configurations of rigid bodies in space.

Consider a sequence of rigid motions $h_1, h_2, \dots, h_k$ and their coordinate representations with homogeneous matrices $H_1, H_2, \dots, H_k$ in the same orthonormal frame Ψ . The composition $h = h_k \circ \dots \circ h_1$ of these motions is a rigid motion as well, and its coordinate representation H is given by the matrix product of the homogeneous matrices:

$$H = H_k \cdot \dots \cdot H_2 \cdot H_1. \quad (5.7)$$

This constitutes the *chain rule* for general rigid motions.

The inverse h^{-1} of a rigid motion h is also a rigid motion, and its coordinate representation is given by the inverse of the homogeneous matrix H . Moreover, since H is a homogeneous matrix, and R is an orthogonal matrix (and therefore $R^\top = R^{-1}$), the inverse of H is given by

$$H^{-1} = \begin{pmatrix} R & T \\ \mathbf{0}_{1 \times n} & 1 \end{pmatrix}^{-1} = \begin{pmatrix} R^\top & -R^\top T \\ \mathbf{0}_{1 \times n} & 1 \end{pmatrix}. \quad (5.8)$$

Note that this operation is computationally much lighter than the general matrix inversion.

As a result, the set of all homogeneous matrices together with the matrix multiplication forms a group

$$\text{SE}(n) := \left\{ \begin{pmatrix} R & T \\ \mathbf{0}_{1 \times n} & 1 \end{pmatrix} \in \mathbb{R}^{(n+1) \times (n+1)} \mid R \in \text{SO}(n), T \in \mathbb{R}^n \right\},$$

called the *special Euclidean group* in dimension n . The name ‘special’, again, denotes the restriction to the orthogonal matrices R of determinant $+1$. In fact, $\text{SE}(n)$ is a Lie group of dimension $n(n+1)/2$. The group $\text{SE}(2)$ of rigid transformations of a plane is three-dimensional, while $\text{SE}(3)$ of rigid transformations of 3D space is six-dimensional. In this course, we will be mostly interested in the group $\text{SE}(3)$ as it describes the general rigid motions of 3D space.

Remark 5.33. Using the notation of Remark 5.17, $\text{SE}(n)$ is the group $\text{SE}(\mathbb{R}^n)$ of positive isometries of $\mathbb{R}^n$, which is the coordinate representation of the abstract special Euclidean group $\text{SE}(\mathcal{E})$ of the Euclidean space $\mathcal{E}$.

5.3 Dual viewpoint: coordinate transformations

So far, we used homogeneous matrices $H \in \text{SE}(n)$ to represent rigid motions in one coordinate frame Ψ . In this section, we show that homogeneous matrices also represent coordinate transformations between two orthonormal frames in Euclidean spaces thus establishing the duality between rigid motions and coordinate transformations.

Consider a rigid motion $h_2^1: \mathcal{E} \rightarrow \mathcal{E}$ and an arbitrary orthonormal frame $\Psi_1 = (o_1, e_1, \dots, e_n)$ in $\mathcal{E}$. After motion h_2^1 , the frame Ψ_1 is mapped to a new frame $\Psi_2 = (h_2^1(o_1), \vec{h}_2^1(e_1), \dots, \vec{h}_2^1(e_n))$, which is also orthonormal, since $\vec{h}_2^1$ (5.1) is a linear isometry and therefore maps orthonormal bases to orthonormal bases.

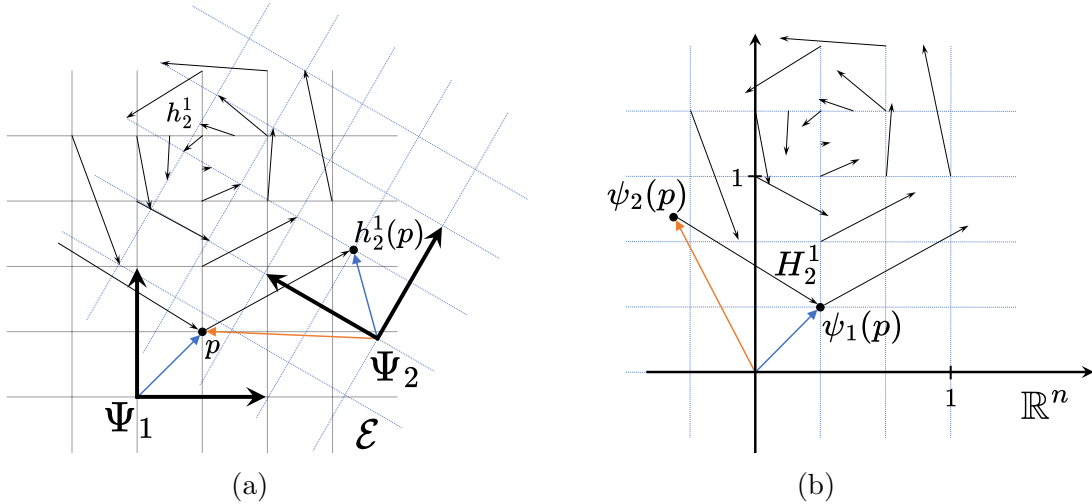


Figure 5.3: Motion h_2^1 from Ψ_1 to Ψ_2 and change of coordinates from Ψ_2 to Ψ_1 .

The following proposition states an expected property of rigid motions.

Proposition 5.34. *Given a rigid motion $h_2^1: \mathcal{E} \rightarrow \mathcal{E}$ that maps an orthonormal frame Ψ_1 to an orthonormal frame Ψ_2 , the coordinates of the moved points in the*

moved frame Ψ_2 coincide with the coordinates of the original points in the original frame Ψ_1 . Precisely,

$$\psi_1 = \psi_2 \circ h_2^1. \quad (5.9)$$

Exercise 5.35. Prove Proposition 5.34 using the definition of the coordinatisation map (Definition 5.10) and the properties of isometries.

Consider now the coordinate representation H_2^1 of the motion h_2^1 first in the frame Ψ_1 and then in the frame Ψ_2 . Using the definition of the coordinate representation and rewriting the identity (5.9) as $h_2^1 = \psi_2^{-1} \circ \psi_1$, we obtain

$$\begin{aligned} \text{In coordinates } \Psi_1: \quad H_2^1 &= \psi_1 \circ h_2^1 \circ \psi_1^{-1} = \psi_1 \circ \psi_2^{-1}; \\ \text{In coordinates } \Psi_2: \quad H_2^1 &= \psi_2 \circ h_2^1 \circ \psi_2^{-1} = \psi_1 \circ \psi_2^{-1}. \end{aligned}$$

Quite remarkably, the coordinate representation of a rigid motion in a frame is the same in the image of that frame under the motion. That is the motion looks the same from both the original and the moved frame which can be seen in Figure 5.3a. The expression on the right-hand side is the map $\psi_1 \circ \psi_2^{-1}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ that transforms coordinates from the frame Ψ_2 to the frame Ψ_1 . Equivalently, it is the inverse $(\psi_2 \circ \psi_1^{-1})^{-1}$ of the coordinate transformation from the frame Ψ_1 to the frame Ψ_2 . Therefore, we conclude with the following theorem underscoring the duality of rigid motions and coordinate transformations.

Theorem 5.36. *Given a rigid motion $h_2^1: \mathcal{E} \rightarrow \mathcal{E}$ that maps an orthonormal frame Ψ_1 to an orthonormal frame Ψ_2 , the coordinate representation H_2^1 of the motion in the frame Ψ_1 or the frame Ψ_2 is the inverse of the coordinate transformation from the frame Ψ_1 to the frame Ψ_2 .*

Hereafter, by a homogeneous matrix $H_b^a \in \text{SE}(n)$ we will denote the coordinate representation of a rigid motion from the frame Ψ_a to the frame Ψ_b and the coordinate transformation from the frame Ψ_b to the frame Ψ_a . It has the form

$$H_b^a = \begin{pmatrix} R_b^a & o_b^a \\ \mathbf{0}_{1 \times n} & 1 \end{pmatrix} \in \text{SE}(n), \quad (5.10)$$

where $R_b^a \in \text{SO}(n)$ is the coordinate representation of the rotation r_b^a , which co-orientates the frame Ψ_a with the frame Ψ_b , expressed in either of the two frames, and $o_b^a \in \mathbb{R}^n$ is the coordinate representation of the translation vector $o_b - o_a$ from the origin of the frame Ψ_a to the origin of the frame Ψ_b , or equivalently, $o_b^a \in \mathbb{R}^n$ is the coordinates of the origin o_b in the frame Ψ_a (see (5.6)).

Let $p \in \mathcal{E}$ be a point in space, and let p^a and p^b be its coordinates in the frames Ψ_a and Ψ_b , respectively. Moreover, let us denote with capital letters the coordinates augmented with a 1:

$$P^a = \begin{pmatrix} p^a \\ 1 \end{pmatrix}, \quad P^b = \begin{pmatrix} p^b \\ 1 \end{pmatrix}.$$

Then the relation between the coordinates of p is given by

$$P^a = H_b^a \cdot P^b. \quad (5.11)$$

As a rule of thumb, the upper index of a homogeneous matrix denotes the resulting coordinate frame of the expression.

Explicitly, the relation (5.11) is read as

$$\begin{pmatrix} p^a \\ 1 \end{pmatrix} = \begin{pmatrix} R_b^a & o_b^a \\ \mathbf{0}_{1 \times n} & 1 \end{pmatrix} \cdot \begin{pmatrix} p^b \\ 1 \end{pmatrix} = \begin{pmatrix} R_b^a \cdot p^b + o_b^a \\ 1 \end{pmatrix}.$$

Remark 5.37. This results also apply to the case of rotation about a fixed point. The rotation matrix can be thought of as the change of coordinates: $R_b^a = \psi_a \circ \psi_b^{-1}$. Then for any point $p \in \mathcal{E}$

$$p^a = R_b^a \cdot p^b.$$

Remark 5.38 (Frame orientation). Note that the coordinate transformation between two orthonormal frames is described by a homogeneous matrix H_b^a in the special Euclidean group $\text{SE}(n)$ only if there is a positive isometry h_b^a mapping the frame Ψ_b to the frame Ψ_a , i.e. the determinant of the linear isometry $\vec{h}_b^a$ is $+1$. In general, an isometric transformation between two orthonormal bases can have a determinant ± 1 . This means that all orthonormal frames in $\mathcal{E}$ can be divided into two classes, within each of which the frames are related by a positive isometry, while the frames from different classes are related by a negative isometry with determinant -1 , i.e. a reflection. One can always make a choice and say one class to have a *positive orientation* and the other to have a *negative orientation* and then restrict to one of the classes so that all coordinate transformations are described by homogeneous matrices in $\text{SE}(n)$. In our 3-dimensional physical space, an orthonormal frame $\Psi = (o, x, y, z)$ is called *right-handed* if the vectors x, y, z are oriented as the thumb, index, and middle fingers of the right hand, respectively, and *left-handed* otherwise. In a two-dimensional plane, an orthonormal frame is right-handed if x is rotated counterclockwise to y , where the counterclockwise direction is defined by how we view the plane. By convention, we assign right-handed frames to the positive orientation class and restrict to them in this course.

The chain rule (5.7) for rigid motions is equivalent to the chain rule for coordinate transformations. Given k orthonormal frames $\Psi_1, \dots, \Psi_k$ of the same orientation, the coordinate transformation from the frame Ψ_1 to the frame Ψ_k is given by the chain

$$H_1^k = H_{k-1}^k \cdot \dots \cdot H_2^3 \cdot H_1^2. \quad (5.12)$$

Note that in two consecutive coordinate transformations, the lower index of the first matrix and the upper index of the second matrix must always coincide.

$$H_c^a = H_b^a \cdot H_c^b.$$

Finally, swapping the indices of a homogeneous matrix gives the inverse coordinate transformation:

$$H_a^b = (H_b^a)^{-1}.$$

5.3.1 Rigid body configuration

We shall now introduce another role of a homogeneous matrix, possibly the most important within this course. We show that homogeneous matrices $H \in \text{SE}(n)$ also represent a relative position and orientation (pose) of the frames in $\mathcal{E}$ and therefore can be used to describe the configuration of a rigid body in space.

Proposition 5.39. *Given two orthonormal frames Ψ_1 and Ψ_2 in $\mathcal{E}$ with the same orientation, the motion $h_2^1 = \psi_2^{-1} \circ \psi_1$ is the unique positive isometry mapping the frame Ψ_1 to the frame Ψ_2 .*

The coordinate representation of the motion h_2^1 relating the two frames expressed in either of the two frames is then given by the unique homogeneous matrix

$$H_2^1 = \psi_1 \circ \psi_2^{-1}.$$

Consider a rigid body in the 3-dimensional Euclidean space $\mathcal{E}$ and a right-handed orthonormal frame Ψ_b fixed to the body. Consider also some reference right-handed orthonormal frame Ψ_r . Then the relative position and orientation of the body-fixed frame with respect to the reference frame are uniquely described by the homogeneous matrix $H_b^r \in \text{SE}(3)$ relating the two frames. Therefore, the configuration of the rigid body relative to a reference frame is given by the homogeneous matrix H_b^r , and the Lie group $\text{SE}(3)$ is the (relative) configuration space of a rigid body in space.

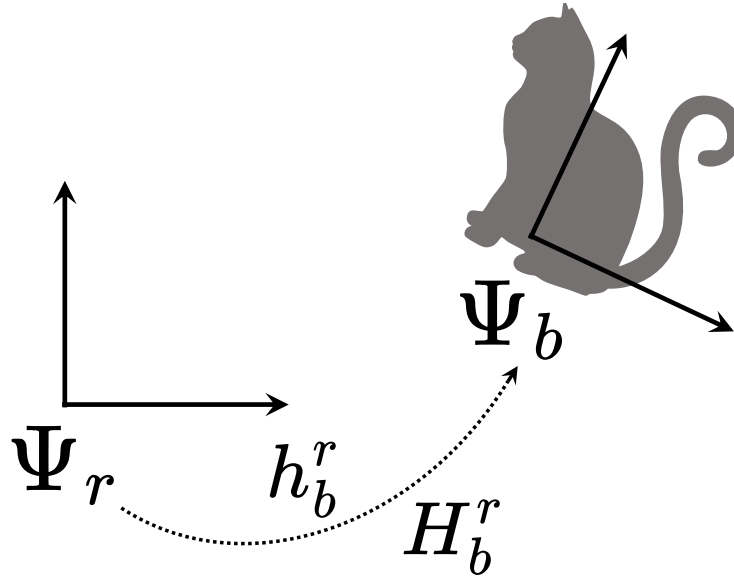


Figure 5.4: Configuration of a rigid cat (Ψ_b) relative to Ψ_r described by the homogeneous matrix H_b^r .

Analogously, in the case of a rigid body rotation about a fixed point $o \in \mathcal{E}$, the relative configuration space of the body is the special orthogonal group $\text{SO}(3)$.

The role of a homogeneous matrix $H_b^r \in \text{SE}(3)$ is thus threefold:

1. rigid motion from Ψ_r to Ψ_b expressed in either Ψ_r or Ψ_b ;
2. configuration of Ψ_b relative to Ψ_r ;
3. change of coordinates from Ψ_b to Ψ_r .

Chapter 6

Rigid Body Velocity

In this chapter we progress from a rigid body configuration at a single instant in time to a time evolution of rigid body configurations, which would enable us to introduce a notion of velocity. Therefore, we need a proper mathematical object to describe time. Formally, it is a one-dimensional Euclidean space $\mathcal{E}(1)$ with a preferred orientation (direction of time). The metric on $\mathcal{E}(1)$ allows measuring time intervals and to define the unit of time (say, seconds). Therefore, if we additionally choose a reference point in time (a zero), we can identify $\mathcal{E}(1)$ with the real numbers $\mathbb{R}$ and use the real number line as the time axis.

In the previous chapter we have shown that the configuration space of a rigid body with a body-fixed frame Ψ_b relative to a reference frame Ψ_r is the Lie group $\text{SE}(3)$ consisting of homogeneous matrices H_b^r . Likewise, the relative configuration space of a rigid body rotating about a fixed point is the Lie group $\text{SO}(3)$ consisting of rotation matrices R_b^r .

6.1 Angular velocity

6.1.1 Definition

For a moment, let us consider the case of a rotating rigid body. We now employ the smooth manifold structure (see Section 3.2) of the Lie group $\text{SO}(3)$. Let the configuration of the body smoothly evolve in time. This evolution is described by a smooth curve on the Lie group parametrised by time $t \in \mathbb{R}$:

$$\begin{aligned} R_b^r : \mathbb{R} &\rightarrow \text{SO}(3) \\ t &\mapsto R_b^r(t). \end{aligned}$$

At each configuration $R_b^r(t)$ through which the curve passes, the velocity of the curve is a tangent vector $\dot{R}_b^r \in T_{R_b^r(t)}\text{SO}(3)$ in the tangent vector space at the point $R_b^r(t) \in \text{SO}(3)$. Numerically, this tangent vector can indeed be represented by the time derivative of the rotation matrix:

$$\dot{R}_b^r(t) = \frac{d}{dt} R_b^r(t) \in T_{R_b^r(t)}\text{SO}(3).$$

This object does represent the angular velocity of the body, however, such a representation has a disadvantage: one cannot add or compare angular velocities of the body at different configurations (or of two different bodies at different configurations). This is because the tangent space $T_{R_b^r} \text{SO}(3)$ at each different configuration R_b^r is a different vector space. The Lie group structure, however, provides a way to fix this issue. The tangent space at the identity element $I \in \text{SO}(3)$ of the group – the Lie algebra $\mathfrak{so}(3)$ of $\text{SO}(3)$ – can be used as a common tangent space for all angular velocities.

There are two ways to translate a tangent vector $\dot{R}_b^r \in T_{R_b^r} \text{SO}(3)$ to the Lie algebra $\mathfrak{so}(3)$: the left translation and the right translation (see Section 3.3 for details). Note that the multiplication by $(R_b^r)^{-1}$ either from the left or the right maps R_b^r to the identity matrix:

$$l_{(R_b^r)^{-1}}(R_b^r) = (R_b^r)^{-1} R_b^r = I, \quad r_{(R_b^r)^{-1}}(R_b^r) = R_b^r (R_b^r)^{-1} = I.$$

As a result, the push-forward (see Definition 3.11) of the left- and right translation at R_b^r maps vectors from the tangent space $T_{R_b^r} \text{SO}(3)$ to the tangent space at the identity, i.e. to the Lie algebra $\mathfrak{so}(3) = T_I \text{SO}(3)$:

$$\begin{aligned} (l_{(R_b^r)^{-1}})_{*, R_b^r} : T_{R_b^r} \text{SO}(3) &\rightarrow \mathfrak{so}(3), & (r_{(R_b^r)^{-1}})_{*, R_b^r} : T_{R_b^r} \text{SO}(3) &\rightarrow \mathfrak{so}(3), \\ \dot{R}_b^r &\mapsto (R_b^r)^{-1} \dot{R}_b^r, & \dot{R}_b^r &\mapsto \dot{R}_b^r (R_b^r)^{-1}. \end{aligned}$$

A convenient property of matrix Lie groups is that such operations of left- and right translations can be written simply as a matrix multiplication from the left and right, respectively.

Definition 6.1. The *angular velocity* of a rotating rigid body with a body-fixed frame Ψ_b relative to a reference frame Ψ_r expressed in the body-fixed frame Ψ_b is the Lie algebra element

$$\tilde{\omega}_b^{b,r} := (R_b^r)^{-1} \dot{R}_b^r \in \mathfrak{so}(3). \quad (6.1)$$

The *angular velocity* of a rotating rigid body with a body-fixed frame Ψ_b relative to a reference frame Ψ_r expressed in the reference frame Ψ_r is the Lie algebra element

$$\tilde{\omega}_b^{r,r} := \dot{R}_b^r (R_b^r)^{-1} \in \mathfrak{so}(3). \quad (6.2)$$

Recall that the inverse $(R_b^r)^{-1}$ of a rotation matrix R_b^r is the rotation matrix R_r^b , i.e. the configuration of the reference frame Ψ_r relative to the body-fixed frame Ψ_b . The angular velocity is thus

$$\tilde{\omega}_b^{b,r} = R_r^b \dot{R}_b^r \quad \text{and} \quad \tilde{\omega}_b^{r,r} = \dot{R}_b^r R_r^b.$$

The index notation we use for angular velocity reads as follows:

$\tilde{\omega}_a^{c,b}$ – the angular velocity of Ψ_a relative to Ψ_b expressed in Ψ_c .

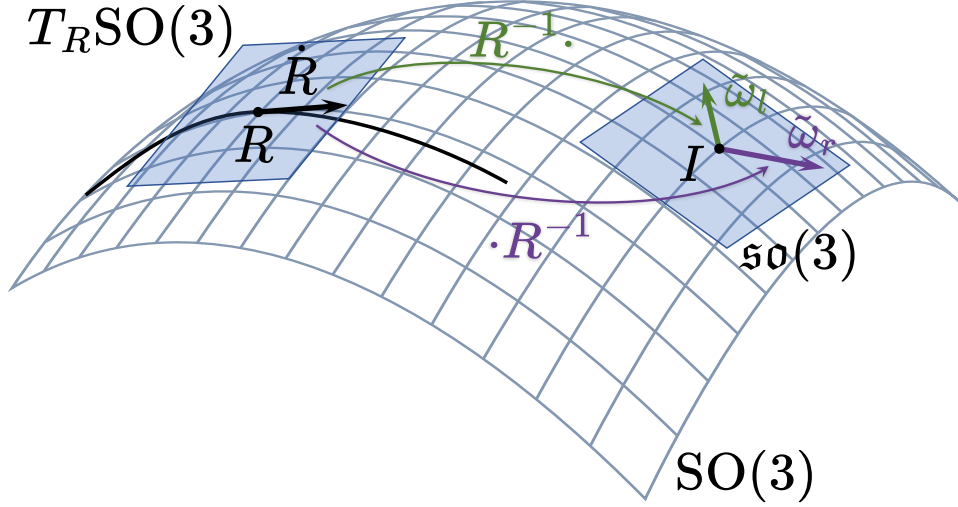


Figure 6.1: Left- and right translation of a tangent vector $\dot{R}$ at R to the Lie algebra $\mathfrak{so}(3)$.

What form does the matrix $\tilde{\omega}$ have? Let $R(t)$ be an arbitrary smooth curve on $\text{SO}(3)$. Then the following holds due to orthogonality of rotation matrices:

$$\begin{aligned} 0 = \dot{I} &= (\dot{R}R^\top) = \dot{R}R^\top + R\dot{R}^\top = \dot{R}R^{-1} + (\dot{R}R^{-1})^\top, \\ 0 = \dot{I} &= (R^\top \dot{R}) = \dot{R}^\top R + R^\top \dot{R} = (R^{-1}\dot{R})^\top + R^{-1}\dot{R}, \end{aligned}$$

which implies that $\dot{R}R^{-1} = -(\dot{R}R^{-1})^\top$ and $R^{-1}\dot{R} = -(R^{-1}\dot{R})^\top$. Therefore, the angular velocity matrix $\tilde{\omega} \in \mathfrak{so}(3)$ is skew-symmetric:

$$\tilde{\omega} = -\tilde{\omega}^\top,$$

which means that it has the form:

$$\tilde{\omega} = \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix} \quad \text{for some } \omega_x, \omega_y, \omega_z \in \mathbb{R}.$$

In fact, the Lie algebra $\mathfrak{so}(3)$ is precisely the space of all 3×3 skew-symmetric matrices. To see this, note that the left translation $(l_{R^{-1}})_{*,R}: T_R \text{SO}(3) \rightarrow \mathfrak{so}(3)$ is a linear isomorphism between the tangent vector space $T_R \text{SO}(3)$ and the Lie algebra $\mathfrak{so}(3)$. Consequently, for any element of $\mathfrak{so}(3)$ there exists a tangent vector $\dot{R} \in T_R \text{SO}(3)$ such that $R^{-1}\dot{R}$ equals to this Lie algebra element. Therefore, each element of $\mathfrak{so}(3)$ is a skew-symmetric matrix. We thus established that $\mathfrak{so}(3)$ is a subspace of the space of 3×3 skew-symmetric matrices, but is any 3×3 skew-symmetric matrix an element of $\mathfrak{so}(3)$? The Lie algebra $\mathfrak{so}(3)$, viewed as the tangent space to $\text{SO}(3)$, is a vector space of dimension three. This dimension matches the

space of all 3×3 skew-symmetric matrices. Therefore, the Lie algebra $\mathfrak{so}(3)$ is exactly the space of all 3×3 skew-symmetric matrices:

$$\mathfrak{so}(3) = \{\tilde{\omega} \in \mathbb{R}^{3 \times 3} \mid \tilde{\omega} = -\tilde{\omega}^\top\}.$$

As a vector space, $\mathfrak{so}(3)$ can be canonically identified with $\mathbb{R}^3$ by the *tilde operation*:

$$\begin{aligned} \tilde{\cdot}: \mathbb{R}^3 &\rightarrow \mathfrak{so}(3), \\ x = \begin{pmatrix} x^1 \\ x^2 \\ x^3 \end{pmatrix} &\mapsto \tilde{x} := \begin{pmatrix} 0 & -x^3 & x^2 \\ x^3 & 0 & -x^1 \\ -x^2 & x^1 & 0 \end{pmatrix}. \end{aligned} \quad (6.3)$$

One can verify that the tilde operation is a linear isomorphism between $\mathbb{R}^3$ and $\mathfrak{so}(3)$. This, for example, implies that the addition is preserved: $\tilde{\omega}_1 + \tilde{\omega}_2 = \widetilde{\omega_1 + \omega_2}$.

Therefore, the angular velocity of a rotating rigid body has two equivalent representations: as a column of three real numbers ω and as a 3×3 skew-symmetric matrix $\tilde{\omega}$:

$$\omega = \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix} \quad \text{and} \quad \tilde{\omega} = \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix}.$$

The tilde notation is used to distinguish between the two representations. Due to the canonical identification, we will sometimes write $\omega \in \mathfrak{so}(3)$ emphasising that the angular velocity ω is an element of the Lie algebra $\mathfrak{so}(3)$.

Remark 6.2. For two-dimensional rotations, the Lie algebra $\mathfrak{so}(2)$ of the Lie group $\text{SO}(2)$ is a one-dimensional vector space. In a matrix form, the Lie algebra $\mathfrak{so}(2)$ consists of all 2×2 skew-symmetric matrices of the form

$$\tilde{\omega} = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix} \in \mathfrak{so}(2).$$

An important thing to remember is that we are working in coordinates. The Lie group $\text{SO}(3)$ consists of rotation matrices, which are coordinate representations of abstract rotations. Therefore, the angular velocity $\tilde{\omega}$ we have defined is merely a coordinate representation of the abstract angular velocity, this is why it can be represented with three real numbers.

6.1.2 Geometric interpretation of the angular velocity

Definition 6.1 introduces the angular velocity ‘expressed in the body-fixed frame’ Ψ_b and ‘expressed in the reference frame’ Ψ_r . Yet, what is the coordinate-independent angular velocity that is being expressed in coordinates, and what does it mean for the angular velocity to be expressed in a frame in $\mathcal{E}$?

Theorem 6.3 (Instantaneous Euler’s rotation theorem). *Consider a rigid body in a 3-dimensional Euclidean space $\mathcal{E}$ rotating relative to a reference frame Ψ_r about*

a point $o \in \mathcal{E}$ fixed in Ψ_r ($\dot{o}^r = 0$). At any instant of time, there exists a line in $\mathcal{E}$ passing through o which is an instantaneous axis of the relative rotation, that is, a line of points of the body whose velocities relative to Ψ_r are all zero. In case the body is not at rest relative to Ψ_r , such a line is unique.

As the Euler's rotation theorem (Theorem 5.31) states, any finite rotation about a point in a 3-dimensional Euclidean space has a unique axis of rotation. This theorem is an instantaneous version of this fact.

Proof. Consider a rigid body rotating about a point $o \in \mathcal{E}$ and an arbitrary point $p(t) \in \mathcal{E}$ fixed to the body. Coordinates of the point $p(t)$ in the reference frame Ψ_r are given by $p^r(t) \in \mathbb{R}^3$ and in the body-fixed frame Ψ_b by constant $p^b(t) = p^b \in \mathbb{R}^3$. The change of coordinates is given by the rotation matrix $R_b^r(t) \in \text{SO}(3)$:

$$p^r(t) = R_b^r(t) p^b.$$

Taking the time derivative of this equation, we obtain that the velocity of the point $p(t)$ in the reference frame Ψ_r is given by the vector product of the angular velocity of the body with the coordinates of p in Ψ_r :

$$\dot{p}^r = \dot{R}_b^r p^b + R_b^r \overset{0}{\dot{p}^b} = \dot{R}_b^r p^b = \dot{R}_b^r R_r^b p^r = \tilde{\omega}_b^{r,r} p^r = \omega_b^{r,r} \times p^r. \quad (6.4)$$

The vector product used here is defined on $\mathbb{R}^3$ with the standard vector space structure, orientation and inner product.

Claim: the angular velocity $\omega_b^{r,r}$ specifies the instantaneous axis of the relative rotation. Since the equation holds for any body-fixed point p , we deduce that the angular velocity vector $\omega_b^{r,r} \in \mathbb{R}^3$ is perpendicular to all velocity vectors $\dot{p}^r \in \mathbb{R}^3$. Moreover, for any body-fixed point p lying on the line given by $\omega_b^{r,r}$ (i.e. $p^r = a\omega_b^{r,r}$ for some $a \in \mathbb{R}$), the velocity $\dot{p}^r$ is zero. From the definition of the vector product, aligning the thumb of the right hand with the direction of $\omega_b^{r,r}$, the fingers will curl in the direction of the velocity vectors $\dot{p}^r$, i.e. in the direction of the rotation of the body. $\square$

The norm $\|\omega_b^{r,r}\|_{\mathbb{R}^3}$ of the angular velocity with respect to the standard inner product on $\mathbb{R}^3$ indicates the *angular speed* of the relative rotation.

This construction works only in a 3-dimensional Euclidean space $\mathcal{E}(3)$. Consider, for example, rotations in a 2-dimensional Euclidean space $\mathcal{E}(2)$. The space of angular velocities $\mathfrak{so}(2)$ is one-dimensional, and it does not make any sense to talk about it as a vector in $\mathbb{R}^2$ or a line in $\mathcal{E}(2)$. Furthermore, a vector product cannot be defined in a 2-dimensional space.

6.1.3 Change of coordinates for angular velocity

The space of all oriented lines in $\mathcal{E}$ passing through the centre of rotation $o \in \mathcal{E}(3)$ with a magnitude $\mu \in \mathbb{R}_{>0}$ (together with a zero element $\{0\}$) forms the *coordinate-free space of angular velocities*. A choice of a coordinate frame Ψ_c centred at the

point o allows obtaining a coordinate representation $\tilde{\omega}_b^{c,r} \in \mathfrak{so}(3)$ (or $\omega_b^{c,r} \in \mathbb{R}^3$) – the angular velocity of Ψ_b relative to Ψ_r expressed in Ψ_c .

Explicitly, one goes to coordinates and back as follows:

- Given a coordinate-free angular velocity as an oriented line in $\mathcal{E}$ passing through o with a magnitude $\mu \in \mathbb{R}_{>0}$ and a coordinate frame Ψ_c centred at o , find a point $p \in \mathcal{E}$ lying on the line at the distance μ from o in the specified direction. The angular velocity $\omega^c \in \mathbb{R}^3$ expressed in Ψ_c is defined as the coordinates p^c of p in the frame Ψ_c or

$$\tilde{\omega}^c := \widetilde{\psi_c(p)} = \tilde{p}^c \in \mathfrak{so}(3).$$

Given the zero element $\{0\}$, the angular velocity is zero $\tilde{\omega}^c = 0 \in \mathfrak{so}(3)$.

- Given a coordinate frame Ψ_c centred at o and non-zero components $\omega^c = (\omega_x^c, \omega_y^c, \omega_z^c) \in \mathbb{R}^3$ (or equivalently $\tilde{\omega}^c \in \mathfrak{so}(3)$), construct a point $p = \psi_c^{-1} \omega^c \in \mathcal{E}$ such that $p^c = \omega^c$. Then the vector $w = p - o \in \vec{\mathcal{E}}$ determines an oriented line in $\mathcal{E}$ passing through o with the magnitude given by $\mu = \|w\|_{\mathcal{E}} = \|\omega^c\|_{\mathbb{R}^3}$. The zero components $\tilde{\omega}^c = 0 \in \mathfrak{so}(3)$ correspond to the zero element $\{0\}$.

Consider now another coordinate frame Ψ_d centred at o . The configuration of Ψ_c relative to Ψ_d is given by the rotation matrix $R_c^d \in \text{SO}(3)$ that represents the motion from Ψ_d to Ψ_c . Then the coordinates of any point $p \in \mathcal{E}(3)$ transform as follows:

$$p^d = R_c^d p^c$$

as discussed in Remark 5.37.

Consequently, the components of the angular velocity transform as

$$\omega^d = R_c^d \omega^c \tag{6.5}$$

under a change of frame from Ψ_c to Ψ_d . Note that $R_c^d \in \text{SO}(3)$ is an orthogonal matrix, hence the norm of the angular velocity is indeed the same in all frames.

Proposition 6.4. *In the matrix form, the change of coordinates for the angular velocity from frame Ψ_c to frame Ψ_d is given by:*

$$\tilde{\omega}^d = R_c^d \tilde{\omega}^c R_d^c. \tag{6.6}$$

Proof. To show that the equality

$$\tilde{\omega}^d = (\widetilde{R_c^d \omega^c}) \stackrel{?}{=} R_c^d \tilde{\omega}^c R_d^c$$

holds, we prove that for all $R \in \text{SO}(3)$ and $v \in \mathbb{R}^3$:

$$(\widetilde{Rv}) = R\tilde{v}R^{-1}. \tag{6.7}$$

Note that $R \in \text{SO}(3)$ is a positive orthogonal transformation of $\mathbb{R}^3$ with the standard inner product and orientation, hence it preserves norms, angles and orientation. Therefore, R preserves the vector product on $\mathbb{R}^3$:

$$Rv \times Rw = R(v \times w) \quad \forall v, w \in \mathbb{R}^3.$$

Now, for any $v, w \in \mathbb{R}^3$:

$$R\tilde{v}R^{-1}w = R(v \times (R^{-1}w)) = (Rv) \times w = (\widetilde{Rv})w,$$

and therefore, $(\widetilde{Rv}) = R\tilde{v}R^{-1}$, which completes the proof. $\square$

6.2 Twist – Rigid body velocity

In this section, we lift the assumption of the previous section on a fixed point of rotation, consider general rigid body motion and reproduce the derivation of the angular velocity in the general setting.

6.2.1 Definition

The special Euclidean group $\text{SE}(3)$ is the relative configuration space of a rigid body moving in $\mathcal{E}(3)$. An evolution of the body configuration in time is described by a smooth curve:

$$\begin{aligned} H_b^r &: \mathbb{R} \rightarrow \text{SE}(3) \\ t &\mapsto H_b^r(t). \end{aligned}$$

The velocity of the curve at each point $H_b^r(t)$ is a tangent vector to the Lie group given by the time derivative of the homogeneous matrix:

$$\dot{H}_b^r(t) = \frac{d}{dt}H_b^r(t) \in T_{H_b^r(t)}\text{SE}(3).$$

This tangent vector does represent the generalised velocity of the body, however, as in the case of angular velocity, one cannot add or compare velocities of the body at different configurations since the tangent space $T_{H_b^r}\text{SE}(3)$ at each different configuration H_b^r is a different vector space. We, therefore, translate these tangent vectors to a common tangent space $T_I\text{SE}(3)$ at the identity element – the Lie algebra $\mathfrak{se}(3)$ of $\text{SE}(3)$.

The push-forward of the left- and right translation at H_b^r maps vectors from the tangent space $T_{H_b^r}\text{SE}(3)$ to the Lie algebra $T_I\text{SE}(3) = \mathfrak{se}(3)$:

$$\begin{aligned} (l_{(H_b^r)^{-1}})_*, H_b^r &: T_{H_b^r}\text{SE}(3) \rightarrow \mathfrak{se}(3), & (r_{(H_b^r)^{-1}})_*, H_b^r &: T_{H_b^r}\text{SE}(3) \rightarrow \mathfrak{se}(3), \\ \dot{H}_b^r &\mapsto (H_b^r)^{-1}\dot{H}_b^r, & \dot{H}_b^r &\mapsto \dot{H}_b^r(H_b^r)^{-1}. \end{aligned}$$

Definition 6.5. The *generalised velocity* or *twist* of a rigid body with a body-fixed frame Ψ_b relative to a reference frame Ψ_r , expressed in the body-fixed frame Ψ_b is the Lie algebra element

$$\tilde{T}_b^{b,r} := (H_b^r)^{-1}\dot{H}_b^r \in \mathfrak{se}(3). \quad (6.8)$$

The *generalised velocity* or *twist* of a rigid body with a body-fixed frame Ψ_b relative to a reference frame Ψ_r , expressed in the reference frame Ψ_r is the Lie algebra element

$$\tilde{T}_b^{r,r} := \dot{H}_b^r(H_b^r)^{-1} \in \mathfrak{se}(3). \quad (6.9)$$

Recall that the inverse $(H_b^r)^{-1}$ of a homogeneous matrix H_b^r is the homogeneous matrix H_r^b , i.e. the configuration of the reference frame Ψ_r relative to the body-fixed frame Ψ_b . The twist is thus

$$\tilde{T}_b^{b,r} = H_r^b \dot{H}_b^r \quad \text{and} \quad \tilde{T}_b^{r,r} = \dot{H}_b^r H_r^b.$$

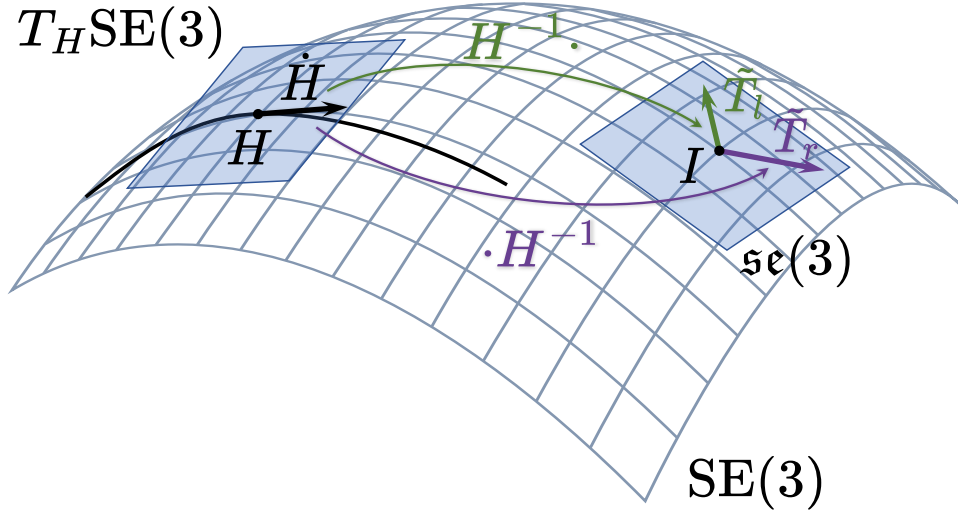


Figure 6.2: Left- and right translation of a tangent vector $\dot{H}$ at H to the Lie algebra $\mathfrak{se}(3)$.

The index notation we use for twists reads as follows:

$$\tilde{T}_a^{c,b} - \text{the twist of } \Psi_a \text{ relative to } \Psi_b \text{ expressed in } \Psi_c.$$

What form does a twist $\tilde{T} \in \mathfrak{se}(3)$ have? Let

$$H(t) = \begin{pmatrix} R(t) & s(t) \\ \mathbf{0} & 1 \end{pmatrix}$$

be an arbitrary relative configuration evolution (a smooth curve) on $\text{SE}(3)$. Then using the inversion formula for homogeneous matrices (5.8), the following holds:

$$\tilde{T}_l = H^{-1} \dot{H} = \begin{pmatrix} R^{-1} & -R^{-1}s \\ \mathbf{0} & 1 \end{pmatrix} \begin{pmatrix} \dot{R} & \dot{s} \\ \mathbf{0} & 0 \end{pmatrix} = \begin{pmatrix} R^{-1}\dot{R} & R^{-1}\dot{s} \\ \mathbf{0} & 0 \end{pmatrix} = \begin{pmatrix} \tilde{\omega} & v \\ \mathbf{0} & 0 \end{pmatrix},$$

where $\tilde{\omega} \in \mathfrak{so}(3)$ and $v \in \mathbb{R}^3$.

Since the left (and equivalently right-) translation $(l_{H^{-1}})_{*,H}: T_H\text{SE}(3) \rightarrow \mathfrak{se}(3)$ is a linear isomorphism between the tangent vector space $T_H\text{SE}(3)$ and the Lie algebra

$\mathfrak{se}(3)$, we can conclude that each element of $\mathfrak{se}(3)$ takes the form of a twist matrix. Therefore, the Lie algebra $\mathfrak{se}(3)$ is the space of all matrices of the form

$$\mathfrak{se}(3) = \left\{ \begin{pmatrix} \tilde{\omega} & v \\ \mathbf{0} & 0 \end{pmatrix} \in \mathbb{R}^{4 \times 4} \mid \tilde{\omega} \in \mathfrak{so}(3), v \in \mathbb{R}^3 \right\}.$$

As a vector space, $\mathfrak{se}(3)$ can be canonically identified with $\mathbb{R}^6$ by the *tilde operation for twists*:

$$\begin{aligned} \tilde{\cdot}: \mathbb{R}^3 \times \mathbb{R}^3 = \mathbb{R}^6 &\rightarrow \mathfrak{se}(3), \\ T = (\omega, v) = \begin{pmatrix} \omega \\ v \end{pmatrix} &\mapsto \tilde{T} := \begin{pmatrix} \tilde{\omega} & v \\ \mathbf{0} & 0 \end{pmatrix}. \end{aligned} \quad (6.10)$$

One can verify that the tilde operation is a linear isomorphism between $\mathbb{R}^6$ and $\mathfrak{se}(3)$. This, for example, implies that the addition is preserved: $\tilde{T}_1 + \tilde{T}_2 = \widetilde{T_1 + T_2}$.

Therefore, the twist of a rigid body has two equivalent representations: as a column of three six numbers T and as a 4×4 matrix $\tilde{T}$:

$$T = \begin{pmatrix} \omega \\ v \end{pmatrix} \quad \text{and} \quad \tilde{T} = \begin{pmatrix} \tilde{\omega} & v \\ \mathbf{0} & 0 \end{pmatrix}.$$

The tilde notation is used to distinguish between the two representations. Due to the canonical identification, we will sometimes write expressions like

$$T = \begin{pmatrix} \omega \\ v \end{pmatrix} \in \mathfrak{se}(3) \quad \text{or} \quad T = (\omega, v) \in \mathfrak{se}(3),$$

emphasising that the twist T is an element of the Lie algebra $\mathfrak{se}(3)$.

As in the case of angular velocities, we are working in coordinates. The Lie group $\text{SE}(3)$ consists of homogeneous matrices, which are coordinate representations of abstract rigid motions. Therefore, the twist $\tilde{T}$ we have defined is merely a coordinate representation of the abstract twist, this is why it can be represented with six real numbers.

6.2.2 Geometric interpretation of the twist

As in the case of the angular velocity, Definition 6.5 introduces the twist ‘expressed in the body-fixed frame’ Ψ_b and ‘expressed in the reference frame’ Ψ_r . Yet, what does it mean for the twist to be expressed in an orthonormal frame in $\mathcal{E}$? One possible approach to answer this question is to recall that the homogeneous matrix H_b^r is the coordinate representation of the abstract rigid motion h_b^r in Ψ_r or Ψ_b , however this approach would require some more advanced differential geometric arguments. Instead, we will provide a geometric coordinate-free representation of the twist as the instantaneous screw axis with a magnitude and a pitch and will define its coordinate representation in an arbitrary frame.

Theorem 6.6 (Instantaneous Mozzi–Chasles theorem). *Consider a rigid body moving relative to a reference frame Ψ_r in a 3-dimensional Euclidean space. At any*

instant of time, there exists a line in $\mathcal{E}$, which is the instantaneous screw axis of the relative motion, i.e. the instantaneous relative velocities of all body-fixed points are a result of a rotation about- and translation along this line. Furthermore, if the body is not at rest, the direction of such a line is unique, and if the instantaneous motion is not a pure translation, the line is unique.

This theorem is an instantaneous version of the Mozzi–Chasles theorem (Theorem 5.32).

Proof. Consider a rigid body with an arbitrary body-fixed point $p(t) \in \mathcal{E}$. Coordinates of the point $p(t)$ in the reference frame Ψ_r are given by $p^r(t) \in \mathbb{R}^3$ and in the body-fixed frame Ψ_b by constant $p^b(t) = p^b \in \mathbb{R}^3$. The change of coordinates is given by the homogeneous matrix $H_b^r(t) \in \text{SE}(3)$:

$$P^r(t) = H_b^r(t) P^b,$$

where $P^r(t) = \begin{pmatrix} p^r(t) \\ 1 \end{pmatrix}$ and $P^b = \begin{pmatrix} p^b \\ 1 \end{pmatrix}$ are the augmented coordinates. Taking the time derivative of this equation, we obtain that the velocity of the point $p(t)$ in the reference frame Ψ_r is given by:

$$\begin{aligned} \dot{P}^r &= (\dot{H}_b^r P^b) \\ &= \dot{H}_b^r P^b + H_b^r \overset{0}{\dot{P}^b} \quad (\text{chain rule}) \\ &= \dot{H}_b^r P^b \quad (p \text{ is body-fixed}) \\ &= \dot{H}_b^r H_r^b P^r \quad (\text{change of coordinates for } P^b) \\ &= \tilde{T}_b^{r,r} P^r \quad (\text{definition of the twist}). \end{aligned} \tag{6.11}$$

Therefore,

$$\begin{pmatrix} \dot{p}^r \\ 0 \end{pmatrix} = \dot{P}^r = \tilde{T}_b^{r,r} P^r = \begin{pmatrix} \tilde{\omega} & v \\ \mathbf{0} & 0 \end{pmatrix} \begin{pmatrix} p^r \\ 1 \end{pmatrix} = \begin{pmatrix} \tilde{\omega} p^r + v \\ 0 \end{pmatrix} = \begin{pmatrix} \omega \times p^r + v \\ 0 \end{pmatrix}, \tag{6.12}$$

and

$$\dot{p}^r = \omega \times p^r + v. \tag{6.13}$$

In case both ω and v are zero, the linear velocity of all body-fixed points is zero:

$$\dot{p}^r = 0,$$

which corresponds to a body at rest.

In case $\omega = 0$, but $v \neq 0$, we obtain that all body-fixed points move with the same linear velocity:

$$\dot{p}^r = v,$$

which corresponds to a pure translation in the direction uniquely determined by v .

Consider the case of $\omega \neq 0$. Claim: ω specifies the direction of the screw axis. To show this, we look for points p , whose velocity $\dot{p}^r$ is parallel to ω . This is equivalent to requiring that the vector product of ω with $\dot{p}^r = 0$ vanishes:

$$\dot{p}^r \parallel \omega \quad \Leftrightarrow \quad \omega \times \dot{p}^r = 0$$

We have that

$$\begin{aligned} \omega \times \dot{p}^r &= \omega \times (\omega \times p^r) + \omega \times v && \text{(plugging in (6.13))} \\ &= (\omega \cdot p^r) \omega - \|\omega\|^2 p^r + \omega \times v && \text{(triple product identity (2.7))} \\ &= -\|\omega\|^2 \left(p^r - \frac{(\omega \cdot p^r) \omega}{\|\omega\|^2} \right) + \omega \times v && (\omega \neq 0) \\ &= -\|\omega\|^2 p_{\perp\omega}^r + \omega \times v && \text{(orthogonal projection on } \omega), \end{aligned}$$

where $\omega \cdot p^r$ is the standard inner product of $\mathbb{R}^3$. So,

$$\dot{p}^r \parallel \omega \quad \Leftrightarrow \quad p_{\perp\omega}^r = \frac{\omega \times v}{\|\omega\|^2}.$$

Any vector p^r can be uniquely decomposed into the sum the component $p_{\perp\omega}^r$ orthogonal to ω and the component $p_{\parallel\omega}^r$ parallel to ω :

$$p^r = p_{\perp\omega}^r + p_{\parallel\omega}^r = \frac{\omega \times v}{\|\omega\|^2} + p_{\parallel\omega}^r.$$

Since the orthogonal component is independent of the point p , we conclude that the set of all points p whose velocity $\dot{p}^r$ is parallel to ω is the line in $\mathcal{E}$ parallel to ω and passing through the point $q = \psi_r^{-1}\left(\frac{\omega \times v}{\|\omega\|^2}\right)$ with coordinates $q^r = \frac{\omega \times v}{\|\omega\|^2}$.

Uniqueness of the screw axis now follows from (6.13), which states that the velocity of any body-fixed point is a result of a rotation about- and translation along the screw axis given by ω . $\square$

We infer from the proof that the ω component of the twist specifies the direction of the instantaneous screw axis and that its norm $\|\omega\|$ is the angular speed of the rotational component of the motion.

Physical meaning of v in the twist

Consider a body-fixed frame Ψ_b moving relative to a reference frame Ψ_r with the configuration evolution

$$H_b^r(t) = \begin{pmatrix} R_b^r(t) & o_b^r(t) \\ \mathbf{0} & 1 \end{pmatrix} \in \text{SE}(3),$$

where the form of the homogeneous matrix is as induced in (5.10).

Then (using (5.8)), the left-translated twist is given by

$$\begin{aligned}
\tilde{T}_b^{b,r} &= (H_b^r)^{-1} \dot{H}_b^r \\
&= \begin{pmatrix} (R_b^r)^{-1} & -(R_b^r)^{-1} o_b^r \\ \mathbf{0} & 1 \end{pmatrix} \begin{pmatrix} \dot{R}_b^r & \dot{o}_b^r \\ \mathbf{0} & 0 \end{pmatrix} \\
&= \begin{pmatrix} R_r^b \dot{R}_b^r & R_r^b \dot{o}_b^r \\ \mathbf{0} & 0 \end{pmatrix} \\
&= \begin{pmatrix} \tilde{\omega}_b^{b,r} & v_b^{b,r} \\ \mathbf{0} & 0 \end{pmatrix},
\end{aligned}$$

where we define

$$v_b^{b,r} := R_r^b \dot{o}_b^r \in \mathbb{R}^3, \quad (6.14)$$

and $\tilde{\omega}_b^{b,r} \in \mathfrak{so}(3)$ is the angular velocity in the body-fixed frame. Recall (from Section 5.3.1) that o_b^r is the coordinates of the origin of the body-fixed frame Ψ_b in the reference frame Ψ_r , and hence $\dot{o}_b^r$ is the components of the linear velocity of the origin of Ψ_b relative to Ψ_r expressed in Ψ_r . Since R_r^b changes the coordinates from Ψ_r to Ψ_b , we conclude that $v_b^{b,r} = R_r^b \dot{o}_b^r$ is the *linear velocity of the origin of Ψ_b relative to Ψ_r expressed in Ψ_b* .

Similarly, for the right-translated, we obtain

$$\begin{aligned}
\tilde{T}_b^{r,r} &= \dot{H}_b^r (H_b^r)^{-1} \\
&= \begin{pmatrix} \dot{R}_b^r & \dot{o}_b^r \\ \mathbf{0} & 0 \end{pmatrix} \begin{pmatrix} (R_b^r)^{-1} & -(R_b^r)^{-1} o_b^r \\ \mathbf{0} & 1 \end{pmatrix} \\
&= \begin{pmatrix} \dot{R}_b^r R_r^b & \dot{o}_b^r - \dot{R}_b^r R_r^b o_b^r \\ \mathbf{0} & 0 \end{pmatrix} \\
&= \begin{pmatrix} \tilde{\omega}_b^{r,r} & v_b^{r,r} \\ \mathbf{0} & 0 \end{pmatrix},
\end{aligned}$$

where we define

$$v_b^{r,r} := \dot{o}_b^r - \tilde{\omega}_b^{r,r} o_b^r \in \mathbb{R}^3, \quad (6.15)$$

and $\tilde{\omega}_b^{r,r} \in \mathfrak{so}(3)$ is the angular velocity in the reference frame. Note that $v_b^{r,r}$ is not the linear velocity of the origin of Ψ_b relative to Ψ_r expressed in Ψ_r , that quantity is given by $\dot{o}_b^r$. Rewriting (6.15) as

$$v_b^{r,r} = \dot{o}_b^r + \omega_b^{r,r} \times (-o_b^r),$$

and virtually extending the moving body, we infer that $v_b^{r,r}$ is the *instantaneous linear velocity of the body-fixed point currently at the origin of Ψ_r relative to Ψ_r expressed in Ψ_r* .

Corollary 6.7. *Any (coordinate) twist $T = (\omega, v) \in \mathfrak{se}(3)$ with $\omega \neq 0$ can be uniquely decomposed into the sum of a rotational component about ω and a translational component along ω :*

$$\begin{pmatrix} \omega \\ v \end{pmatrix} = \underbrace{\begin{pmatrix} \omega \\ s \times \omega \end{pmatrix}}_{\text{rotation}} + \underbrace{\begin{pmatrix} 0 \\ \lambda \omega \end{pmatrix}}_{\text{translation}}, \quad s \in \mathbb{R}^3, \lambda \in \mathbb{R}. \quad (6.16)$$

Proof. The vector v can be uniquely decomposed into the sum of a component parallel to ω and a component orthogonal to ω :

$$v = v_{\parallel\omega} + v_{\perp\omega} = \frac{v \cdot \omega}{\|\omega\|^2} \omega + \left(v - \frac{v \cdot \omega}{\|\omega\|^2} \omega \right) = \frac{\omega \cdot v}{\|\omega\|^2} \omega + \frac{\omega \times v}{\|\omega\|^2} \times \omega,$$

where the last equality follows from the triple product identity (2.7). Thus, by defining

$$\lambda = \frac{\omega \cdot v}{\|\omega\|^2} \in \mathbb{R} \quad \text{and} \quad s = \frac{\omega \times v}{\|\omega\|^2} \in \mathbb{R}^3, \quad (6.17)$$

we obtain the desired decomposition. $\square$

Note that the choice of the shift vector $s \in \mathbb{R}^3$ is not unique, however, the resulting vector product $s \times \omega$ is unique. Geometrically, the shift vector $s = \frac{\omega \times v}{\|\omega\|^2}$ connects the origin of the coordinate frame with the point on the screw axis closest to the origin. However, the choice of $s \in \mathbb{R}^3$ connecting the origin to any other point on the screw axis yields the same result of the vector product $s \times \omega$, known as the *moment of an oriented line with a magnitude at a point*.

6.2.3 Change of coordinates for twists

The *coordinate-free space of twists* in a 3-dimensional Euclidean space $\mathcal{E}$ is the union of the following three sets:

- If the body is at rest (instantaneously): $\{0\}$;
- otherwise, if the instantaneous motion is a pure translation: the set of all directions in $\mathcal{E}$ with a magnitude $\nu \in \mathbb{R}_{>0}$, or equivalently, $\vec{\mathcal{E}}$;
- otherwise, the set of all oriented lines in $\mathcal{E}$ with a magnitude $\mu \in \mathbb{R}_{>0}$ and a pitch $\lambda \in \mathbb{R}$.

A choice of a coordinate frame Ψ_c in $\mathcal{E}$ allows obtaining a coordinate representation $T_b^{c,r} \in \mathfrak{se}(3)$ – the twist of Ψ_b relative to Ψ_r expressed in Ψ_c .

Explicitly, given a coordinate-free twist of a body Ψ_b relative to Ψ_r and a coordinate frame Ψ_c , one obtains its coordinate representation $T_b^{c,r} \in \mathfrak{se}(3)$ as follows:

- If $\{0\}$, then $T_b^{c,r} = 0$, so $\omega_b^{c,r} = 0$ and $v_b^{c,r} = 0$;
- If the twist corresponds to a pure translation and is given by a direction $\hat{n} \in \vec{\mathcal{E}}$ with $\|\hat{n}\| = 1$ and a magnitude $\nu \in \mathbb{R}_{>0}$, then

$$\omega_b^{c,r} = 0 \quad \text{and} \quad v_b^{c,r} = \psi_c(\nu \cdot \hat{n})$$

- Otherwise, given an oriented line (directed by $\hat{n} \in \vec{\mathcal{E}}$) with a magnitude $\mu \in \mathbb{R}_{>0}$ and a pitch $\lambda \in \mathbb{R}$, set the angular velocity as

$$\omega_b^{c,r} := \psi_c(\mu \cdot \hat{n}).$$

Then choose any point $q \in \mathcal{E}$ on the line and set

$$v_b^{c,r} := q^c \times \omega_b^{c,r} + \lambda \omega_b^{c,r}.$$

Note that the last expression is independent of the choice of the point q on the line since $q^c \times \omega_b^{c,r}$ are the components of the *moment of an oriented line with a magnitude* at the origin of the coordinate frame Ψ_c .

Conversely, given a coordinate frame Ψ_c and a coordinate representation $T_b^{c,r} = (\omega_b^{c,r}, v_b^{c,r}) \in \mathfrak{se}(3)$ of a twist, one can recover the coordinate-free twist as follows:

- If $T_b^{c,r} = 0$: $\{0\}$;
- Otherwise, if $\omega_b^{c,r} = 0$, the direction is given by the vector $n = \psi_c^{-1}(v_b^{c,r})$ with magnitude $\mu = \|v_b^{c,r}\|$;
- Otherwise, ($\omega_b^{c,r} \neq 0$) the oriented line is directed by the vector $n = \psi_c^{-1}(\omega_b^{c,r})$, has magnitude $\mu = \|\omega_b^{c,r}\|$ and passes through the point

$$q = \psi_c^{-1} \left(\frac{\omega_b^{c,r} \times v_b^{c,r}}{\|\omega_b^{c,r}\|^2} \right).$$

The pitch is given by

$$\lambda = \frac{\omega_b^{c,r} \cdot v_b^{c,r}}{\|\omega_b^{c,r}\|^2}.$$

Expressing a coordinate-free twist in two coordinate frames Ψ_c and Ψ_d leads (with the details skipped) to the following coordinate transformation rule.

Proposition 6.8. *Let Ψ_c and Ψ_d be two coordinate frames in $\mathcal{E}$. Then the coordinate expressions of the twist of a body Ψ_b relative to Ψ_r in the frames Ψ_c and Ψ_d are related by*

$$\tilde{T}_b^{d,r} = H_c^d \tilde{T}_b^{c,r} H_d^c, \quad (6.18)$$

where $H_c^d \in \text{SE}(3)$ is the homogeneous matrix that represents the change of coordinates from Ψ_c to Ψ_d .

The expression for the velocity of the points of a rigid body (6.12) can now be generalised to an arbitrary coordinate frame Ψ_c :

$$\dot{P}^r = \tilde{T}_b^{r,r} P^r = H_c^r \tilde{T}_b^{c,r} H_r^c P^r \quad \implies \quad \dot{P}^c = \tilde{T}_b^{c,r} P^c. \quad (6.19)$$

The expression (6.18) is for twists in the matrix form. We now derive the change of coordinates for twists in the column form. Dropping the indices b and r for clarity,

we have

$$\begin{aligned}
\begin{pmatrix} \tilde{\omega}^d & v^d \\ \mathbf{0} & 0 \end{pmatrix} &= \tilde{T}^d \\
&= H_c^d \tilde{T}^c H_d^c \\
&= \begin{pmatrix} R_c^d & o_c^d \\ \mathbf{0} & 1 \end{pmatrix} \cdot \begin{pmatrix} \tilde{\omega}^c & v^c \\ \mathbf{0} & 0 \end{pmatrix} \cdot \begin{pmatrix} R_d^c & -R_d^c o_c^d \\ \mathbf{0} & 1 \end{pmatrix} \\
&= \begin{pmatrix} R_c^d \tilde{\omega}^c & R_c^d v^c \\ \mathbf{0} & 0 \end{pmatrix} \cdot \begin{pmatrix} R_d^c & -R_d^c o_c^d \\ \mathbf{0} & 1 \end{pmatrix} \\
&= \begin{pmatrix} R_c^d \tilde{\omega}^c R_d^c & -R_c^d \tilde{\omega}^c R_d^c o_c^d + R_c^d v^c \\ \mathbf{0} & 0 \end{pmatrix} \\
&= \begin{pmatrix} \widetilde{R_c^d \omega^c} & -(R_c^d \omega^c) \times o_c^d + R_c^d v^c \\ \mathbf{0} & 0 \end{pmatrix} \\
&= \begin{pmatrix} \widetilde{R_c^d \omega^c} & \tilde{o}_c^d R_c^d \omega^c + R_c^d v^c \\ \mathbf{0} & 0 \end{pmatrix},
\end{aligned}$$

where $\tilde{o}_c^d$ is the skew-symmetric matrix formed from the vector $o_c^d \in \mathbb{R}^3$ to encode the vector product.

Therefore,

$$\begin{pmatrix} \omega^d \\ v^d \end{pmatrix} = \begin{pmatrix} R_c^d \omega^c \\ \tilde{o}_c^d R_c^d \omega^c + R_c^d v^c \end{pmatrix} = \underbrace{\begin{pmatrix} R_c^d & 0 \\ \tilde{o}_c^d R_c^d & R_c^d \end{pmatrix}}_{=: \text{Ad}_{H_c^d}} \cdot \begin{pmatrix} \omega^c \\ v^c \end{pmatrix},$$

where we have defined the *adjoint matrix* of H_c^d as

$$\text{Ad}_{H_c^d} := \begin{pmatrix} R_c^d & 0 \\ \tilde{o}_c^d R_c^d & R_c^d \end{pmatrix} \in \mathbb{R}^{6 \times 6}. \quad (6.20)$$

As a result, the change of coordinates for twists in the column form is given by

$$T^d = \text{Ad}_{H_c^d} T^c. \quad (6.21)$$

As a change of coordinates, the adjoint matrix $\text{Ad}_{H_c^d}$ is always invertible, with the inverse given by

$$\text{Ad}_{H_c^d}^{-1} = \text{Ad}_{H_d^c}, \quad (6.22)$$

thus it is a linear isomorphism of $\mathbb{R}^6$.

6.2.4 Composition of twists

Proposition 6.9. *Consider two rigid bodies in $\mathcal{E}$ with body-fixed frames Ψ_1 and Ψ_2 , respectively, and a reference frame Ψ_0 , and let Ψ_c be an arbitrary coordinate frame in $\mathcal{E}$. Then the twist of the body Ψ_2 relative to Ψ_0 is given by the sum of the twist of Ψ_1 relative to Ψ_0 and the twist of Ψ_2 relative to Ψ_1 , i.e.*

$$T_2^{c,0} = T_1^{c,0} + T_2^{c,1}. \quad (6.23)$$

Proof.

$$\begin{aligned}
\tilde{T}_2^{c,0} &= H_0^c \tilde{T}_2^{0,0} H_c^0 \\
&= H_0^c \dot{H}_2^0 H_0^2 H_c^0 \\
&= H_0^c (\dot{H}_1^0 H_2^1) H_0^2 H_c^0 \\
&= H_0^c (\dot{H}_1^0 H_2^1 + H_1^0 \dot{H}_2^1) H_0^2 H_c^0 \\
&= H_0^c \dot{H}_1^0 H_2^1 H_0^2 H_c^0 + H_0^c H_1^0 \dot{H}_2^1 H_0^2 H_c^0 \\
&= H_0^c (\dot{H}_1^0 H_0^1) H_c^0 + H_1^c (\dot{H}_2^1 H_1^2) H_c^1 \\
&= H_0^c \tilde{T}_1^{0,0} H_c^0 + H_1^c \tilde{T}_2^{1,1} H_c^1 \\
&= \tilde{T}_1^{c,0} + \tilde{T}_2^{c,1}.
\end{aligned}$$

To conclude the proof, we recall that the tilde operation is a linear isomorphism between $\mathbb{R}^6$ and $\mathfrak{se}(3)$, that is, the matrix and column forms of the twist are equivalent. $\square$

By induction from Proposition 6.9, one can obtain the chain rule for twists.

Corollary 6.10. *Given a sequence of n rigid bodies with body-fixed frames Ψ_i , $i = 1, \dots, n$, a reference frame Ψ_0 , and a coordinate frame Ψ_c , the twist of the body Ψ_n relative to Ψ_0 is given by the sum:*

$$T_n^{c,0} = T_1^{c,0} + T_2^{c,1} + \dots + T_n^{c,(n-1)} = \sum_{i=1}^n T_i^{c,(i-1)}. \quad (6.24)$$

Corollary 6.11. *The twist of a body is independent of the choice of a body-fixed frame, i.e. given a rigid body with body-fixed frames Ψ_b and $\Psi_{b'}$, a reference frame Ψ_r , and a coordinate frame Ψ_c :*

$$T_b^{c,r} = T_{b'}^{c,r}. \quad (6.25)$$

Proof. Since $H_{b'}^b$ is constant, we have

$$\tilde{T}_{b'}^{c,r} = \tilde{T}_{b'}^{c,b} + \tilde{T}_b^{c,r} = H_b^c (\overset{0}{\cancel{\dot{H}_{b'}^b}} H_b^{b'}) H_c^b + \tilde{T}_b^{c,r} = \tilde{T}_b^{c,r}.$$

$\square$

Similarly, the twist of a body is independent of the choice of a reference frame as long as the reference frames are at rest relative to each other.

Corollary 6.12. *The twist of a frame Ψ_b relative to a frame Ψ_r is the negative of the twist of the frame Ψ_r relative to the frame Ψ_b expressed in any coordinate frame Ψ_c :*

$$T_b^{c,r} = -T_r^{c,b}.$$

Proof.

$$0 = T_b^{c,b} = T_b^{c,r} + T_r^{c,b}$$

□

Remark 6.13. We have first defined the twist expressed in the body-fixed $T_b^{b,r}$ and in the reference frame $T_b^{r,r}$ as the left-, and respectively right, translation of the velocity vector $\dot{H}_b^r$ to the Lie algebra $\mathfrak{se}(3)$. We then defined a coordinate-free twist as a screw axis in the 3-dimensional Euclidean space $\mathcal{E}$ to define the twist expressed in an arbitrary coordinate frame Ψ_c as a numerical expression of the screw axis in that frame, and then the coordinate transformation for twists followed from this definition. The chain rule for twists (Proposition 6.9) was then derived as a consequence of the transformation rule.

One could instead define the twist in an arbitrary coordinate frame Ψ_c in terms of the left- and right-translated twists by postulating the chain rule:

$$T_b^{c,r} := T_c^{c,r} + T_b^{c,c},$$

that is, by defining the twist $T_b^{c,r}$ of a body Ψ_b relative to a reference frame Ψ_r expressed in a coordinate frame Ψ_c as the sum of the twist $T_c^{c,r}$ of the coordinate frame Ψ_c relative to the reference frame Ψ_r and the twist $T_b^{c,c}$ of the body Ψ_b relative to the coordinate frame Ψ_c both expressed in Ψ_c . The coordinate transformation for twists would then follow from this definition without the need to introduce the coordinate-free twist and screw theory considerations which are applicable to the 3-dimensional case only.

Chapter 7

Kinematics of Serial Chain Robots

7.1 Exponential map

Consider a body-fixed frame Ψ_b moving relative to a reference frame Ψ_r . From the definition of the twist, we obtain

$$\tilde{T}_b^{r,r} = \dot{H}_b^r H_r^b \quad \Rightarrow \quad \dot{H}_b^r = \tilde{T}_b^{r,r} H_b^r,$$

which is a first-order linear ordinary differential equation (ODE) for the body configuration $H_b^r(t)$. If the twist $\tilde{T}_b^{r,r}$ is constant in time, then the solution to this ODE is given by

$$H_b^r(t) = e^{\tilde{T}_b^{r,r} t} H_b^r(0), \quad (7.1)$$

where $H_b^r(0)$ is the initial configuration of the body at time $t = 0$ and $e^{\tilde{T}_b^{r,r} t}$ is the matrix exponential of the twist $\tilde{T}_b^{r,r} t \in \mathfrak{se}(3)$ (see Chapter 4).

Note that the matrix exponential of a twist $e^{\tilde{T}_b^{r,r} t}$ must be valued in the Lie group $\text{SE}(3)$, so it can be seen as a map

$$\begin{aligned} e: \mathfrak{se}(3) &\rightarrow \text{SE}(3) \\ \tilde{T} &\mapsto e^{\tilde{T}}. \end{aligned} \quad (7.2)$$

Intuitively, the exponential $e^{\tilde{T}}$ of a twist $\tilde{T}$ returns the configuration of the body after it has moved with the twist $\tilde{T}$ for a unit time starting from the identity configuration.

Remark 7.1. The matrix exponential used here is a particular case of the exponential map of a Lie group. Let G be a Lie group with the Lie algebra $\mathfrak{g} = T_I G$. The exponential is a map

$$\exp: \mathfrak{g} \rightarrow G$$

defined such that the tangent vector to the curve

$$\begin{aligned} \gamma: \mathbb{R} &\rightarrow G \\ t &\mapsto \gamma(t) := \exp(tX) \end{aligned}$$

at any point $\gamma(t)$ along the curve is the left translation of $X \in \mathfrak{g}$:

$$\dot{\gamma}(t) = (l_{\gamma(t)})_{*,I} X \in T_{\gamma(t)} G.$$

Similarly, given a rigid body Ψ_b rotating about a fixed point with a constant angular velocity $\omega_b^{r,r}$ relative to Ψ_r , the solution to the ODE

$$\tilde{\omega}_b^{r,r} = \dot{R}_b^r R_r^b \quad \Rightarrow \quad \dot{R}_b^r = \tilde{\omega}_b^{r,r} R_b^r$$

is given by

$$R_b^r(t) = e^{\tilde{\omega}_b^{r,r} t} R_b^r(0), \quad (7.3)$$

where $R_b^r(0)$ is the initial orientation of the body at time $t = 0$ and $e^{\tilde{\omega}_b^{r,r} t}$ is the matrix exponential:

$$\begin{aligned} e: \mathfrak{so}(3) &\rightarrow \mathrm{SO}(3) \\ \tilde{\omega} &\mapsto e^{\tilde{\omega}}. \end{aligned}$$

Exercise 7.2. Prove that the matrix exponentials $e: \mathfrak{so}(3) \rightarrow \mathrm{SO}(3)$ and $e: \mathfrak{se}(3) \rightarrow \mathrm{SE}(3)$ are surjective, but not injective. Physically, surjectivity means that starting from the identity configuration, any other configuration of a rigid body can be reached by a continuous rotation with a constant angular velocity (continuous motion with a constant twist). Non-injectivity means that there exist configurations that can be reached with two different constant angular velocities (twists) under continuous motion from the identity configuration. Hint: use Theorems 5.31 and 5.32.

The exponential of a general square matrix $A \in \mathbb{R}^{n \times n}$ is defined as the power series

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}.$$

However, the matrix exponential of an angular velocity $\tilde{\omega} \in \mathfrak{so}(3)$ and a twist $\tilde{T} \in \mathfrak{se}(3)$ can be computed more efficiently using the Rodrigues' formula.

Proposition 7.3 (Rodrigues¹ formula). *The matrix exponential of a 3×3 skew-symmetric matrix $\tilde{\omega} = \theta \hat{\omega} \in \mathfrak{so}(3)$ with $\theta \in \mathbb{R}$ and $\|\hat{\omega}\|_{\mathbb{R}^3} = 1$ is given by*

$$e^{\tilde{\omega}} = e^{\theta \hat{\omega}} = I + \sin \theta \hat{\omega} + (1 - \cos \theta) \hat{\omega}^2, \quad (7.4)$$

Proof. The matrix exponential is given by a power series. We can, however, express

¹Olinde Rodrigues (1795–1851), French banker and mathematician.

all the powers of $\tilde{\omega}$ in terms of its first and second powers.

$$\begin{aligned}
\tilde{\omega}^3 &= \tilde{\omega} \cdot \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix} \\
&= \tilde{\omega} \cdot \begin{pmatrix} -\omega_z^2 - \omega_y^2 & \omega_x \omega_y & \omega_x \omega_z \\ \omega_x \omega_y & -\omega_z^2 - \omega_x^2 & \omega_y \omega_z \\ \omega_x \omega_z & \omega_y \omega_z & -\omega_y^2 - \omega_x^2 \end{pmatrix} \\
&= \tilde{\omega} \cdot (\omega \cdot \omega^\top - \|\omega\|^2 I) \\
&= (\omega \times \omega) \cdot \omega^\top - \|\omega\|^2 \tilde{\omega} \\
&= -\|\omega\|_{\mathbb{R}^3}^2 \tilde{\omega} = -\theta^2 \tilde{\omega}
\end{aligned}$$

This implies that all odd and even powers of $\tilde{\omega}$ are of the form

$$\tilde{\omega}^{2k-1} = (-1)^k \theta^{2k-1} \tilde{\omega} \quad \text{and} \quad \tilde{\omega}^{2k} = (-1)^{k-1} \theta^{2k} \tilde{\omega}^2 \quad \forall k \in \mathbb{N}.$$

Thus, the matrix exponential of $\tilde{\omega}$ is given by

$$\begin{aligned}
e^{\tilde{\omega}} &= I + \tilde{\omega} + \frac{\tilde{\omega}^2}{2!} + \frac{\tilde{\omega}^3}{3!} + \frac{\tilde{\omega}^4}{4!} + \frac{\tilde{\omega}^5}{5!} + \dots \\
&= I + \theta \tilde{\omega} + \frac{\theta^2 \tilde{\omega}^2}{2!} - \frac{\theta^3 \tilde{\omega}}{3!} - \frac{\theta^4 \tilde{\omega}^2}{4!} + \frac{\theta^5 \tilde{\omega}}{5!} + \dots \\
&= I + \tilde{\omega} \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \right) + \tilde{\omega}^2 \left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \dots \right) \\
&= I + \tilde{\omega} \sin \theta + \tilde{\omega}^2 (1 - \cos \theta).
\end{aligned}$$

□

Exercise 7.4. Consider a rigid body rotating with a constant angular velocity $\omega = (0, 0, 1) \in \mathfrak{so}(3)$ about the z -axis. Starting from the identity configuration $I \in \text{SO}(3)$ at time $t = 0$, compute the configuration of the body at time t using (7.3) and the Rodrigues' formula. Do you recognise the resulting matrix?

Exercise 7.5. Formulate and prove the two-dimensional version of Proposition 7.3 for an angular velocity $\tilde{\omega} = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix} \in \mathfrak{so}(2)$.

Proposition 7.6. The matrix exponential of a twist $\tilde{T} = (\tilde{\omega}, v) \in \mathfrak{se}(3)$ is given by:

- if $\omega = 0$,

$$e \begin{pmatrix} \mathbf{0} & v \\ \mathbf{0} & 0 \end{pmatrix} = \begin{pmatrix} I & v \\ \mathbf{0} & 1 \end{pmatrix}, \quad (7.5)$$

- otherwise ($\omega \neq 0$),

$$e \begin{pmatrix} \tilde{\omega} & v \\ \mathbf{0} & 0 \end{pmatrix} = \begin{pmatrix} e^{\tilde{\omega}} & \frac{1}{\|\omega\|_{\mathbb{R}^3}^2} ((I - e^{\tilde{\omega}}) \tilde{\omega} v + (\omega^\top v) \omega) \\ \mathbf{0} & 1 \end{pmatrix}, \quad (7.6)$$

where $e^{\tilde{\omega}}$ can be computed using the Rodrigues' formula.

Proof. In case $\omega = 0$, we note that

$$\begin{pmatrix} \mathbf{0} & v \\ \mathbf{0} & 0 \end{pmatrix}^2 = \begin{pmatrix} \mathbf{0} & v \\ \mathbf{0} & 0 \end{pmatrix} \cdot \begin{pmatrix} \mathbf{0} & v \\ \mathbf{0} & 0 \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & 0 \end{pmatrix},$$

and consequently, all the higher powers will be zero as well. Thus, plugging in the twist matrix into the definition of the matrix exponential yields

$$e^{\begin{pmatrix} \mathbf{0} & v \\ \mathbf{0} & 0 \end{pmatrix}} = I + \begin{pmatrix} \mathbf{0} & v \\ \mathbf{0} & 0 \end{pmatrix} + \mathbf{0} + \mathbf{0} + \cdots = \begin{pmatrix} I & v \\ \mathbf{0} & 1 \end{pmatrix}.$$

In case $\omega \neq 0$, the idea of the proof is to transform the twist $\tilde{T}$ into another coordinate frame placed at the screw axis of the twist. Specifically, if the given twist is expressed in the frame Ψ_a and is decomposed using (6.16) as $T^a \equiv T = (\omega, v) = (\omega, s \times \omega + \lambda \omega)$, we consider the frame Ψ_b shifted from Ψ_a by the translation s to be placed at the screw axis of the twist. In this frame, the twist is given by

$$T^b = \begin{pmatrix} \omega \\ \mathbf{0} \times \omega \end{pmatrix} + \begin{pmatrix} \mathbf{0} \\ \lambda \omega \end{pmatrix} = \begin{pmatrix} \omega \\ \lambda \omega \end{pmatrix},$$

using the decomposition (6.16). Then,

$$e^{\tilde{T}^a} = e^{H_b^a \tilde{T}^b H_a^b} = H_b^a e^{\tilde{T}^b} H_a^b,$$

where $H_b^a = \begin{pmatrix} I & s \\ \mathbf{0} & 1 \end{pmatrix}$ is the homogeneous relating the frames Ψ_a and Ψ_b . The matrix exponential of $\tilde{T}^b$ can now be computed as

$$\begin{aligned} e^{\tilde{T}^b} &= e^{\begin{pmatrix} \tilde{\omega} & \lambda \omega \\ \mathbf{0} & 0 \end{pmatrix}} \\ &= I + \begin{pmatrix} \tilde{\omega} & \lambda \omega \\ \mathbf{0} & 0 \end{pmatrix} + \frac{1}{2!} \begin{pmatrix} \tilde{\omega} & \lambda \omega \\ \mathbf{0} & 0 \end{pmatrix}^2 + \frac{1}{3!} \begin{pmatrix} \tilde{\omega} & \lambda \omega \\ \mathbf{0} & 0 \end{pmatrix}^3 + \cdots \\ &= I + \begin{pmatrix} \tilde{\omega} & \lambda \omega \\ \mathbf{0} & 0 \end{pmatrix} + \frac{1}{2!} \begin{pmatrix} \tilde{\omega}^2 & \mathbf{0} \\ \mathbf{0} & 0 \end{pmatrix} + \frac{1}{3!} \begin{pmatrix} \tilde{\omega}^3 & \mathbf{0} \\ \mathbf{0} & 0 \end{pmatrix} + \cdots \\ &= \begin{pmatrix} e^{\tilde{\omega}} & \lambda \omega \\ \mathbf{0} & 1 \end{pmatrix}. \end{aligned}$$

As a result, the matrix exponential of the twist $\tilde{T} \in \mathfrak{se}(3)$ is given by

$$e^{\begin{pmatrix} \tilde{\omega} & v \\ \mathbf{0} & 0 \end{pmatrix}} = \begin{pmatrix} I & s \\ \mathbf{0} & 1 \end{pmatrix} \begin{pmatrix} e^{\tilde{\omega}} & \lambda \omega \\ \mathbf{0} & 1 \end{pmatrix} \begin{pmatrix} I & -s \\ \mathbf{0} & 1 \end{pmatrix} = \begin{pmatrix} e^{\tilde{\omega}} & (I - e^{\tilde{\omega}})s + \lambda \omega \\ \mathbf{0} & 1 \end{pmatrix}.$$

Plugging in the expressions

$$\lambda = \frac{\omega^\top v}{\|\omega\|^2} \quad \text{and} \quad s = \frac{\tilde{\omega} v}{\|\omega\|^2}$$

for the pitch and shift from (6.17), we prove the proposition. $\square$

Exercise 7.7. Formulate and prove the two-dimensional version of Proposition 7.6 for a twist $\tilde{T} = (\tilde{\omega}, v) \in \mathfrak{se}(2)$.

7.2 Kinematic pairs

Definition 7.8. A *one-degree-of-freedom (1-DoF) kinematic pair* is a pair of rigid bodies connected by a joint that constrains the relative motion of the bodies to a single degree of freedom.

Remark 7.9. This constrains the relative configuration space of the bodies to be a 1-dimensional submanifold of $\text{SE}(3)$, which is topologically isomorphic either to the real line $\mathbb{R}$ or to the circle S^1 .

Definition 7.10. A *simple kinematic pair* or a *simple joint* is a 1-DoF kinematic pair that constrains the relative motion of the bodies to a single type of motion: translation along a fixed direction, or a screw motion about a fixed axis with constant pitch λ (zero pitch for a pure rotation), where ‘fixed’ means fixed relative to both bodies.

Definition 7.11. A *unit (coordinate-free) translational twist* is a direction in the Euclidean space $\mathcal{E}$ with a unit magnitude $\nu = 1$.

A *unit (coordinate-free) screw twist* is an oriented line in the Euclidean space $\mathcal{E}$ with a unit magnitude $\mu = 1$ and a pitch $\lambda \in \mathbb{R}$.

Consequently, a *coordinate representation of a unit translational twist* is an element of the Lie algebra $\hat{T}_{\text{tr}} \in \mathfrak{se}(3)$ with

$$\hat{T}_{\text{tr}} = \begin{pmatrix} \mathbf{0} \\ \hat{v} \end{pmatrix}, \quad \text{such that } \|\hat{v}\|_{\mathbb{R}^3} = 1,$$

and a *coordinate representation of a unit screw twist* is an element of the Lie algebra $\hat{T}_{\text{scr}} \in \mathfrak{se}(3)$ with

$$\hat{T}_{\text{scr}} = \begin{pmatrix} \hat{\omega} \\ s \times \hat{\omega} \end{pmatrix} + \begin{pmatrix} \mathbf{0} \\ \lambda \hat{\omega} \end{pmatrix}, \quad \text{such that } \|\hat{\omega}\|_{\mathbb{R}^3} = 1.$$

A *unit rotational twist* is a unit screw twist with a zero pitch.

$$\hat{T}_{\text{rot}} = \begin{pmatrix} \hat{\omega} \\ s \times \hat{\omega} \end{pmatrix}, \quad \text{such that } \|\hat{\omega}\|_{\mathbb{R}^3} = 1.$$

Since the screw axis / direction of the translation in a simple joint is fixed relative to both bodies, the relative twist of the bodies expressed in a frame attached to either of the bodies during continuous motion can change only by a scale factor depending on the speed of the motion.

Proposition 7.12. Let Ψ_a and Ψ_b be frames fixed to the two bodies connected by a simple joint. Then for any possible relative motion of the bodies there exist unique (up to a sign) constant unit twists $\hat{T}_a^{b,b}$ and $\hat{T}_a^{a,b}$ expressed in the body-fixed frames Ψ_b and Ψ_a , respectively, such that

$$T_a^{b,b}(t) = \alpha(t) \hat{T}_a^{b,b} \quad \text{and} \quad T_a^{a,b}(t) = \beta(t) \hat{T}_a^{a,b},$$

where $\alpha(t) \in \mathbb{R}$ and $\beta(t) \in \mathbb{R}$ are scale factors depending on the speed of the motion.

This unique (up to a sign) unit twist $\hat{T}_a^{\bullet,b}$ is called the *unit twist of the joint*. It would remain a unit twist independently of the choice of the coordinate frame, however, it is constant only if the coordinate frame is fixed to one of the bodies.

Consider the configuration space of the joint given by a 1-dimensional manifold $\mathcal{Q}$. It can be (locally) parametrised by a single coordinate $q \in \mathbb{R}$, so that $H_a^b(q)$ is a curve in $\text{SE}(3)$ that describes the relative configuration of the joint as a function of the coordinate q , and $\frac{d}{dq}H_a^b(q) \in T_{H_a^b(q)}\text{SE}(3)$ is the tangent vector to the curve at the point $H_a^b(q)$. The coordinate q can always be rescaled such that

$$\left(\frac{d}{dq}H_a^b(q)\right)H_b^a(q) = \tilde{T}_a^{b,b} \in \mathfrak{se}(3)$$

is the unit twist of the joint. Then the solution to the ODE for the relative configuration of the joint is given by

$$H_a^b(q) = e^{\tilde{T}_a^{b,b} q} H_a^b(0), \quad (7.7)$$

where $H_a^b(0) = H_a^b(q=0)$ is the reference configuration of the joint at $q=0$.

7.3 Direct kinematics

Definition 7.13. A *serial chain mechanism* is a sequence of rigid bodies, called *links*, connected by simple joints in a chain. The final link is called the *end-effector* of the mechanism.

Definition 7.14. The *direct* or *forward kinematics* of a serial chain mechanism is the problem of finding the configuration of the end-effector of the mechanism given the joint configurations.

The configuration space $\mathcal{Q}$ of a serial chain mechanism with n joints is the Cartesian product of the configuration spaces Q_i of its joints:

$$\mathcal{Q} = Q_1 \times Q_2 \times \cdots \times Q_n.$$

Therefore, solving the direct kinematics problem is equivalent to finding the map

$$\begin{aligned} L: \mathcal{Q} &\rightarrow \text{SE}(3) \\ q &\mapsto H_n^0(q), \end{aligned}$$

where H_n^0 is the configuration of the end-effector relative to the base frame Ψ_0 .

Since simple joints have one degree of freedom, the dimension of the configuration space of a serial chain mechanism with n simple joints is n : $\dim(\mathcal{Q}) = n$. Hereafter, we neglect the fact that generally the configuration space $\mathcal{Q}$ is covered by multiple local coordinate charts and work with a single coordinate chart implicitly identifying points $q \in \mathcal{Q}$ with their coordinates $q = (q^1, \dots, q^n) \in \mathbb{R}^n$.

Theorem 7.15 (Brockett²'s Product of Exponentials Formula). *Consider a serial chain mechanism with n simple joints. Let Ψ_i be a frame fixed to the i -th link for $i \in \{0, \dots, n\}$, and let $q = (q^1, \dots, q^n) \in \mathbb{R}^n$ denote local coordinates on the joint configuration space $\mathcal{Q}$. The direct kinematics of the mechanism is locally given by*

$$H_n^0(q) = e^{\tilde{T}_1^{0,0}(0)q^1} \cdot e^{\tilde{T}_2^{0,1}(0)q^2} \cdot \dots \cdot e^{\tilde{T}_n^{0,(n-1)}(0)q^n} \cdot H_n^0(0) = \prod_{i=1}^n e^{\tilde{T}_i^{0,(i-1)}(0)q^i} \cdot H_n^0(0), \quad (7.8)$$

where $\tilde{T}_i^{0,(i-1)}(0)$ is the unit twist of the i -th joint at the reference configuration $q = 0$ expressed in the frame Ψ_0 .

Proof. From the joint relation (7.7), the relative configuration of the i -th and $(i-1)$ -th links is given by

$$H_i^{i-1}(q) = H_i^{i-1}(q^i) = e^{\tilde{T}_i^{(i-1),(i-1)}q^i} H_i^{i-1}(0),$$

where $\tilde{T}_i^{(i-1),(i-1)}$ is the constant unit twist of the i -th joint expressed in the frame Ψ_{i-1} . By the chain rule for homogeneous matrices (5.12), the relative configuration of the end-effector is given by

$$\begin{aligned} H_n^0(q) &= H_1^0(q) \cdot H_2^1(q) \cdot \dots \cdot H_n^{(n-1)}(q) \\ &= e^{\tilde{T}_1^{0,0}q^1} H_1^0(0) \cdot e^{\tilde{T}_2^{1,1}q^2} H_2^1(0) \cdot \dots \cdot e^{\tilde{T}_n^{(n-1),(n-1)}q^n} H_n^{(n-1)}(0) \\ &= e^{\tilde{T}_1^{0,0}q^1} \cdot H_1^0(0) e^{\tilde{T}_2^{1,1}q^2} H_2^1(0) \cdot H_2^0(0) \dots e^{\tilde{T}_n^{(n-1),(n-1)}q^n} H_n^{(n-1)}(0) \cdot H_n^0(0) \\ &= e^{\tilde{T}_1^{0,0}q^1} \cdot e^{H_1^0(0)\tilde{T}_2^{1,1}H_1^0(0)^{-1}q^2} \cdot \dots \cdot e^{H_{(n-1)}^0(0)\tilde{T}_n^{(n-1),(n-1)}H_{(n-1)}^0(0)^{-1}q^n} \cdot H_n^0(0) \\ &= e^{\tilde{T}_1^{0,0}(0)q^1} \cdot e^{\tilde{T}_2^{0,1}(0)q^2} \cdot \dots \cdot e^{\tilde{T}_n^{0,(n-1)}(0)q^n} \cdot H_n^0(0), \end{aligned}$$

where we used the coordinate transformation rule (6.18) for the twists:

$$H_{i-1}^0(0) \cdot \tilde{T}_i^{(i-1),(i-1)} \cdot H_{i-1}^{i-1}(0) = \tilde{T}_i^{0,(i-1)}(0).$$

□

7.4 Differential kinematics – geometric Jacobian

Definition 7.16. The *direct differential kinematics* of a serial chain mechanism is the problem of finding the (generalised) velocity of the end-effector of the mechanism given the joint velocities.

The direct differential kinematics problem is equivalent to finding the map

$$\begin{aligned} J(q): T_q\mathcal{Q} &\rightarrow \mathfrak{se}(3) \\ \dot{q} &\mapsto T_n^{0,0}(q) \end{aligned} \quad (7.9)$$

for each $q \in \mathcal{Q}$.

²Roger W. Brockett (1938–2023), American control theorist.

Proposition 7.17 (Differential joint relation). *Consider a simple joint with frames Ψ_a and Ψ_b attached to the two bodies connected by the joint. Let $\hat{T}_a^{b,b}$ be the unit twist of the joint and $q \in \mathbb{R}$ be the local coordinate on the joint configuration space $\mathcal{Q}$. Then for each $(q, \dot{q})$, the differential joint relation is given by*

$$T_a^{b,b} = \hat{T}_a^{b,b} \dot{q}.$$

Proof. Taking the time derivative of the joint relation (7.7) gives

$$\begin{aligned} \dot{H}_a^b(q) &= \frac{d}{dt} e^{\tilde{T}_a^{b,b} q} H_a^b(0) \\ &= \tilde{T}_a^{b,b} \dot{q} e^{\tilde{T}_a^{b,b} q} H_a^b(0) \\ &= \tilde{T}_a^{b,b} \dot{q} H_a^b(q). \end{aligned}$$

Therefore,

$$\tilde{T}_a^{b,b} = \dot{H}_a^b(q) H_b^a(q) = \tilde{T}_a^{b,b} \dot{q}.$$

□

Theorem 7.18 (Differential Kinematics). *Consider a serial chain mechanism with n simple joints. Let Ψ_i be a frame fixed to the i -th link for $i \in \{0, \dots, n\}$, and let $q = (q^1, \dots, q^n) \in \mathbb{R}^n$ denote local coordinates on the joint configuration space $\mathcal{Q}$. The differential kinematics of the mechanism is given by*

$$T_n^{0,0} = J(q) \dot{q}, \quad (7.10)$$

where $J(q) \in \mathbb{R}^{6 \times n}$ is the geometric Jacobian³ of the mechanism defined for each $q \in \mathcal{Q}$ as

$$J(q) := \begin{pmatrix} \left. \hat{T}_1^{0,0}(q) \right| & \left. \hat{T}_2^{0,1}(q) \right| & \cdots & \left. \hat{T}_n^{0,(n-1)}(q) \right| \\ \left| \right. & \left| \right. & & \left| \right. \end{pmatrix}, \quad (7.11)$$

where $\hat{T}_i^{0,(i-1)}(q)$ is the unit twist of the i -th joint in the current configuration q expressed in the frame Ψ_0 .

Remark 7.19. Note that we implicitly identify vectors in the tangent space $\dot{q} \in T_q \mathcal{Q}$ with their components $\dot{q} = (\dot{q}^1, \dots, \dot{q}^n) \in \mathbb{R}^n$ in the coordinate chart.

Proof. From the chain rule for twists Corollary 6.10, the twist of the end-effector relative to the base frame Ψ_0 is given by the sum of the twists of the joints:

$$\begin{aligned} T_n^{0,0} &= T_1^{0,0} + T_2^{0,1} + \dots + T_n^{0,(n-1)} \\ &= \text{Ad}_{H_0^0(q)} T_1^{0,0} + \text{Ad}_{H_1^0(q)} T_2^{1,1} + \dots + \text{Ad}_{H_{n-1}^0(q)} T_n^{(n-1),(n-1)} \\ &= \text{Ad}_{H_0^0(q)} \hat{T}_1^{0,0} \dot{q}_1 + \text{Ad}_{H_1^0(q)} \hat{T}_2^{1,1} \dot{q}_2 + \dots + \text{Ad}_{H_{n-1}^0(q)} \hat{T}_n^{(n-1),(n-1)} \dot{q}_n \\ &= J(q) \cdot \dot{q}. \end{aligned}$$

□

³Carl Gustav Jacob Jacobi (1804–1851), German mathematician.

Part III

Dynamics

Chapter 8

Rigid Body Dynamics

8.1 Dynamics of a point mass

Consider a point particle of mass $m \in \mathbb{R}_{>0}$ in a Euclidean space $(\mathcal{E}, g)$ moving with velocity $v \in T_q\mathcal{E}$ at a point $q \in \mathcal{E}$.

Definition 8.1. The *kinetic energy* of the particle is defined as

$$K := \frac{1}{2}m \|v\|^2 = \frac{1}{2}m g(v, v) \in \mathbb{R}_{\geq 0}.$$

Definition 8.2. We define the *mass tensor* of the particle as the $\binom{0}{2}$ -tensor

$$M := m g \in \text{Sym}_2^0 T_q\mathcal{E},$$

so that the kinetic energy can be expressed as

$$K = \frac{1}{2}M(v, v).$$

In particular, in any g -orthonormal basis of $T_q\mathcal{E}$, the components of the mass tensor are given by the diagonal matrix

$$\mathbf{M} = m I.$$

Definition 8.3. The *momentum* of the particle is defined as the covector

$$p := M(v, \cdot) \equiv M^\flat(v) \in T_q^*\mathcal{E}.$$

The kinetic energy can now be expressed as the dual pairing

$$K = \frac{1}{2}\langle p \mid v \rangle.$$

Definition 8.4. A *force* acting on the particle is an element of the cotangent space (a covector) $f \in T_q^*\mathcal{E}$. The total force acting on the particle is defined as the sum of all forces acting on it:

$$f_{\text{total}} := \sum_i f^i \in T_q^*\mathcal{E}.$$

Postulate 8.5 (Newton's¹ second law). *The time derivative of the momentum of a point particle is equal to the total force acting on it:*

$$\dot{p} = f_{\text{total}}.$$

The dual pairing $\langle f | v \rangle$ between a force acting on the particle and its velocity is called the *power* or *instantaneous work* produced by the force on the particle.

Upon choosing an orthonormal coordinate frame Ψ_c in $\mathcal{E}$, we represent the components of the velocity vector v (in the corresponding basis of $T_q\mathcal{E}$) by a column $v^c \in \mathbb{R}^3$ and the components of the covectors p and f (in the corresponding dual basis of $T_q^*\mathcal{E}$) by rows $p^c, f^c \in \mathbb{R}^3$, so that

$$\langle p | v \rangle = \left(\begin{array}{ccc} - & p^c & - \end{array} \right) \cdot \left(\begin{array}{c} | \\ v^c \\ | \end{array} \right) \quad \text{and} \quad \langle f | v \rangle = \left(\begin{array}{ccc} - & f^c & - \end{array} \right) \cdot \left(\begin{array}{c} | \\ v^c \\ | \end{array} \right).$$

8.2 Inertia tensor and momentum of a rigid body

Definition 8.6. A *rigid body* in a three-dimensional Euclidean space $\mathcal{E}$ at each instant of time is defined as the set of points $V \subset \mathcal{E}$ it occupies and a mass density function $\rho: V \rightarrow \mathbb{R}_{>0}$ over that set.

Definition 8.7. The *kinetic energy* of a rigid body is defined as the integral

$$K := \frac{1}{2} \int_V \|\dot{q}\|^2 \rho(q) \, dV,$$

where $\dot{q} \in T_q\mathcal{E}$ is the velocity of the point $q \in V$ of the body and dV is an infinitesimal volume element.

We have learned that the velocities of all the points of a rigid body are determined by a single quantity – the twist of the body. In the following, we shall express the kinetic energy of a rigid body in terms of its twist.

Let Ψ_0 be a fixed inertial orthonormal frame in $\mathcal{E}$, Ψ_b be a body-fixed frame, and Ψ_c be an arbitrary coordinate frame. Then the components $\dot{q}^c$ of the velocity of a point q of the body expressed in Ψ_c are given by

$$\dot{q}^c = \tilde{\omega}_b^{c,0} q^c + v_b^{c,0} = \left(-\tilde{q}^c \quad I \right) T_b^{c,0}.$$

Using the fact that the frame Ψ_c is orthonormal, we express

$$\|\dot{q}\|^2 = (\dot{q}^c)^\top \dot{q}^c = (T_b^{c,0})^\top \begin{pmatrix} \tilde{q}^c \\ I \end{pmatrix} \left(-\tilde{q}^c \quad I \right) T_b^{c,0} = (T_b^{c,0})^\top \begin{pmatrix} -\tilde{q}^c \tilde{q}^c & \tilde{q}^c \\ -\tilde{q}^c & I \end{pmatrix} T_b^{c,0}.$$

Therefore, the kinetic energy of the rigid body can be expressed as

$$K = \frac{1}{2} (T_b^{c,0})^\top \left(\int_V \begin{pmatrix} -\tilde{q}^c \tilde{q}^c & \tilde{q}^c \\ -\tilde{q}^c & I \end{pmatrix} \rho(q) \, dV \right) T_b^{c,0} = \frac{1}{2} (T_b^{c,0})^\top \mathcal{I} T_b^{c,0},$$

¹Isaac Newton (1643–1727), English mathematician and physicist.

where we defined (the components of) the *inertia tensor* of the rigid body expressed in the frame Ψ_c as

$$\mathcal{I}^c := \int_V \begin{pmatrix} -\tilde{q}^c \tilde{q}^c & \tilde{q}^c \\ -\tilde{q}^c & I \end{pmatrix} \rho(q) dV \in \mathbb{R}^{6 \times 6}.$$

It may be regarded as a symmetric $\binom{0}{2}$ -tensor on the Lie algebra $\mathfrak{se}(3)$: $\mathcal{I}^c \in \text{Sym}_2^0 \mathfrak{se}(3)$, so that the kinetic energy is $K = \frac{1}{2} \mathcal{I}^c(T_b^{c,0}, T_b^{c,0})$.

To obtain the coordinate transformation rule for the inertia tensor, we express the twist in another orthonormal frame Ψ_d ,

$$K = \frac{1}{2} (T_b^{d,0})^\top \text{Ad}_{H_d^c}^\top \mathcal{I}^c \text{Ad}_{H_d^c} T_b^{d,0} = \frac{1}{2} (T_b^{d,0})^\top \mathcal{I}^d T_b^{d,0},$$

and hence

$$\mathcal{I}^d = \text{Ad}_{H_d^c}^\top \mathcal{I}^c \text{Ad}_{H_d^c}.$$

Note that in any body-fixed frame Ψ_b , the coordinates q^b of the points of the body are constant in time, and hence the inertia tensor $\mathcal{I}^b$ in that frame is constant in time as well.

Proposition 8.8. *There always exists a body-fixed orthonormal frame Ψ_b in which the inertia tensor matrix is diagonal of the form*

$$\mathcal{I}^b = \begin{pmatrix} j_{xx} & 0 & 0 & 0 & 0 & 0 \\ 0 & j_{yy} & 0 & 0 & 0 & 0 \\ 0 & 0 & j_{zz} & 0 & 0 & 0 \\ 0 & 0 & 0 & m & 0 & 0 \\ 0 & 0 & 0 & 0 & m & 0 \\ 0 & 0 & 0 & 0 & 0 & m \end{pmatrix},$$

where m is the total mass of the body. Furthermore, in such a frame, the origin of the frame Ψ_b corresponds to the centre of mass of the body.

The frame Ψ_b is called a *principal inertia frame* of the rigid body, its basis vectors are called the *principal axes of inertia*, and the diagonal elements j_{xx}, j_{yy}, j_{zz} are called the *principal moments of inertia* of the body. The proposition implies that any rigid body with any mass distribution inertially behaves like a uniform ellipsoid with semi-axes aligned with the principal axes of inertia.

Definition 8.9. The (*generalised*) *momentum* of a rigid body expressed in a coordinate frame Ψ_c is an element of the dual space to the Lie algebra $\mathfrak{se}(3)$ defined as

$$P^c := (\mathcal{I}^c T_b^{c,0})^\top \in \mathfrak{se}(3)^* \cong \mathbb{R}^6,$$

where we implicitly identify $\mathfrak{se}(3)^* \cong \mathbb{R}^6$ and represent elements of $\mathfrak{se}(3)^*$ by rows of six numbers.

Remark 8.10. There is no matrix representation of $\mathfrak{se}(3)^*$ which would be consistent with the matrix representation of $\mathfrak{se}(3)$ and the Lie algebra operations.

The kinetic energy of the rigid body can now be expressed as the dual pairing

$$K = \frac{1}{2} \langle P^c \mid T_b^{c,0} \rangle = \frac{1}{2} \left(\begin{array}{cc} - & P^c & - \end{array} \right) \cdot \left(\begin{array}{c} | \\ T_b^{c,0} \\ | \end{array} \right).$$

Given another coordinate frame Ψ_d , the momentum expressed in that frame is given by

$$P^d = (\mathcal{I}^d T_b^{d,0})^\top = (\text{Ad}_{H_d^c}^\top \mathcal{I}^c T_b^{c,0})^\top = P^c \text{Ad}_{H_d^c},$$

which written in column form reads

$$(P^d)^\top = \text{Ad}_{H_d^c}^\top (P^c)^\top.$$

Note that the same adjoint matrix $\text{Ad}_{H_d^c}$ transforms twists from Ψ_c to Ψ_d and momenta in the opposite direction – from Ψ_d to Ψ_c – corresponding to the basis transformation of vectors and covectors, respectively (see Section 2.2.1).

8.3 Wrenches – generalised forces

Recall that to a twist of a rigid body $T_b^{c,0} \in \mathfrak{se}(3)$ corresponds the field of velocities of all the points of the body (the Lie algebra action of $\mathfrak{se}(3)$ on $\mathcal{E}$). Fixing an orthonormal coordinate frame Ψ_c in $\mathcal{E}$, the velocity of a point $q \in \mathcal{E}$ of the body expressed in that frame is obtained by applying the linear map

$$\begin{aligned} a_q^c: \mathfrak{se}(3) &\rightarrow \mathbb{R}^3 (\simeq_{\Psi_c} T_q \mathcal{E}) \\ T_b^{c,0} &\mapsto \dot{q}^c = \tilde{\omega}_b^{c,0} q^c + v_b^{c,0}. \end{aligned} \tag{8.1}$$

The dual map (see Section 2.2.2)

$$\begin{aligned} (a_q^c)^*: (T_q^* \mathcal{E} \simeq_{\Psi_c}) \mathbb{R}^3 &\rightarrow \mathfrak{se}(3)^* \\ f^c &\mapsto (a_q^c)^*(f^c) =: W^c \end{aligned} \tag{8.2}$$

maps (the components of) a force $f \in T_q^* \mathcal{E}$ applied at the point q to the dual space $\mathfrak{se}(3)^*$ to the Lie algebra $\mathfrak{se}(3)$.

In components, the map (8.1) can be expressed as

$$\dot{q}^c \mapsto \begin{pmatrix} -\tilde{q}^c & I \end{pmatrix} T_b^{c,0},$$

and therefore the dual map (8.2) is represented by

$$f^c \mapsto f^c \cdot \begin{pmatrix} -\tilde{q}^c & I \end{pmatrix} = \begin{pmatrix} -f^c \tilde{q}^c & f^c \end{pmatrix} = \left((q^c \times (f^c)^\top)^\top \quad f^c \right),$$

where we implicitly identify $\mathfrak{se}(3)^* \cong \mathbb{R}^6$ and represent elements of $\mathfrak{se}(3)^*$ by rows of six numbers.

Definition 8.11. The *generalised force* or *wrench* induced by a force $f \in T_q^* \mathcal{E}$ acting on a point q of a rigid body expressed in a coordinate frame Ψ_c is defined as the element $W^c \in \mathfrak{se}(3)^*$ of the dual space to the Lie algebra $\mathfrak{se}(3)$ given by

$$W^c := (-f^c \tilde{q}^c \quad f^c) = (\tau^c \quad f^c) \in \mathfrak{se}(3)^*.$$

The element $\tau^c := -f^c \tilde{q}^c = (q^c \times (f^c)^\top)^\top \in (\mathbb{R}^3)^* \cong \mathfrak{so}(3)^*$ is called the *moment* or *torque* of the force f^c about the origin of the frame Ψ_c .

By the defining property of the dual map, the power produced by the force can be expressed as

$$\langle f^c \mid \dot{q}^c \rangle = \langle W^c \mid T_b^{c,0} \rangle = \tau^c \omega_b^{c,0} + f^c v_b^{c,0} \in \mathbb{R},$$

representing the sum of angular and linear power components. Note that the pairing $\langle W^c \mid T_b^{c,0} \rangle$ produces power if and only if the wrench W^c and the twist $T_b^{c,0}$ are expressed in the same coordinate frame.

Let $W^d \in \mathfrak{se}(3)^*$ be the wrench produced by the force f acting on the point q expressed in another coordinate frame Ψ_d . Then

$$\langle W^d \mid T_b^{d,0} \rangle = \langle f^d \mid \dot{q}^d \rangle = \langle f^c \mid \dot{q}^c \rangle = \langle W^c \mid T_b^{c,0} \rangle = W^c \text{Ad}_{H_d^c} T_b^{d,0},$$

and hence

$$W^d = W^c \text{Ad}_{H_d^c},$$

which in column form reads

$$(W^d)^\top = \text{Ad}_{H_d^c}^\top (W^c)^\top.$$

Note that the same adjoint matrix $\text{Ad}_{H_d^c}$ transforms twists in one direction and wrenches in the opposite direction in accordance with the basis transformation of vectors and covectors, respectively (see Section 2.2.1).

Definition 8.12. The *total generalised force* or *total wrench* acting on a rigid body expressed in a coordinate frame Ψ_c is defined as the sum of the wrenches induced by all the forces acting on the body:

$$W_{\text{total}}^c := \sum_i (W^c)^i \in \mathfrak{se}(3)^*.$$

Definition 8.13. A system of forces acting on a rigid body is called a *load*. Two loads are called *equivalent* if they induce the same total wrench.

Remark 8.14. Using the metric g of the Euclidean space, one can identify the covector space of forces $T_q^* \mathcal{E}$ with $T_q \mathcal{E} \cong \vec{\mathcal{E}}$ and hence picture a force f^c as a vector in $\vec{\mathcal{E}}$ with components in Ψ_c given by $(f^c)^\top \in \mathbb{R}^3$. Using such a geometric representation of a force, we formulate the following proposition.

Proposition 8.15. *A force translated along its line of action produces an equivalent load.*

Proof. Let $f \in T_q^* \mathcal{E}$ be a force acting on a rigid body at a point $q \in \mathcal{E}$. Then the wrench induced by the force translated along its line of action takes the form

$$\begin{pmatrix} -f^c(\widetilde{q^c + \alpha f^c}) & f^c \end{pmatrix} = \begin{pmatrix} -f^c \widetilde{q^c} & f^c \end{pmatrix} = W^c$$

for some $\alpha \in \mathbb{R}$. □

Proposition 8.16. *Let $f_p \in T_p^* \mathcal{E}$ and $f_q \in T_q^* \mathcal{E}$ be two forces acting on a rigid body at points $p, q \in \mathcal{E}$ such that in some coordinate frame Ψ_c they satisfy $f_p^c = -f_q^c$, i.e. they act in opposite directions with the same magnitude. Then the total wrench induced by these two forces expressed in any coordinate frame Ψ_c is of the form*

$$W_{\text{total}}^c = \begin{pmatrix} \tau^c & 0 \end{pmatrix} = \begin{pmatrix} -f_p^c(\widetilde{p^c - q^c}) & 0 \end{pmatrix},$$

Such a pair of forces inducing a pure moment is called a *couple*.

Remark 8.17. In the three-dimensional Euclidean space, the dimension of the space of moments $\mathfrak{so}(3)^*$ happens to coincide with the dimension of the space itself. One can therefore picture also a moment τ^c as a vector in $\vec{\mathcal{E}}$ with components in Ψ_c given by $(\tau^c)^\top \in \mathbb{R}^3$. Under this identification, we formulate the following theorem.

Theorem 8.18 (Poinsot's² theorem). *Any load acting on a rigid body in the three-dimensional Euclidean space is equivalent to a single force and a couple in the direction of that force.*

Proof. Let $W^c = \begin{pmatrix} \tau^c & f^c \end{pmatrix} \in \mathfrak{se}(3)^*$ be the total wrench produced by the load. If $f^c = 0$, then the load is already a pure moment. Otherwise, the wrench can be decomposed as

$$W^c = \underbrace{\begin{pmatrix} -f^c \tilde{r} & f^c \end{pmatrix}}_{\text{force}} + \underbrace{\begin{pmatrix} \lambda f^c & 0 \end{pmatrix}}_{\text{pure moment}},$$

where

$$r = \frac{(f^c)^\top \times (\tau^c)^\top}{\|f^c\|^2} \in \mathbb{R}^3 \quad \text{and} \quad \lambda = \frac{f^c (\tau^c)^\top}{\|f^c\|^2} \in \mathbb{R}.$$

□

8.4 Equations of motion of a rigid body

Theorem 8.19 (Newton's second law for a rigid body). *The time derivative of the momentum of a rigid body expressed in an inertial frame Ψ_0 is equal to the total wrench acting on it expressed in the same frame:*

$$\dot{P}^0 = W_{\text{total}}^0. \tag{8.3}$$

In the following we formulate the equation of motion of a rigid body in an arbitrary coordinate frame Ψ_c . To do so, we first prove the following technical lemma.

²Louis Poinsot (1777–1859), French mathematician and physicist.

Lemma 8.20 (Derivative of Ad_H). *Let $H_i^j(t) \in \text{SE}(3)$ be a time-dependent homogeneous matrix. Then*

$$\frac{d}{dt} \text{Ad}_{H_i^j} = \text{ad}_{T_i^{j,j}} \text{Ad}_{H_i^j}, \quad (8.4)$$

where

$$\text{ad}_T := \begin{pmatrix} \tilde{\omega} & 0 \\ \tilde{v} & \tilde{\omega} \end{pmatrix} \in \mathbb{R}^{6 \times 6} \quad \text{for } T = (\omega, v) \in \mathfrak{se}(3)$$

is the adjoint representation of the Lie algebra $\mathfrak{se}(3)$ on itself.

Proof. Using the definition of the adjoint representation of the Lie group $\text{SE}(3)$, (6.20), we express

$$\begin{aligned} \frac{d}{dt} \text{Ad}_{H_i^j} &= \begin{pmatrix} \dot{R}_i^j & 0 \\ \dot{\tilde{o}}_i^j R_i^j + \tilde{o}_i^j \dot{R}_i^j & \dot{R}_i^j \end{pmatrix} \\ &= \begin{pmatrix} \tilde{\omega}_i^{j,j} R_i^j & 0 \\ (\tilde{o}_i^j + \tilde{o}_i^j \tilde{\omega}_i^{j,j}) R_i^j & \tilde{\omega}_i^{j,j} R_i^j \end{pmatrix} \\ &= \begin{pmatrix} \tilde{\omega}_i^{j,j} R_i^j & 0 \\ \tilde{v}_i^{j,j} R_i^j + \tilde{\omega}_i^{j,j} \tilde{o}_i^j R_i^j & \tilde{\omega}_i^{j,j} R_i^j \end{pmatrix} \\ &= \begin{pmatrix} \tilde{\omega}_i^{j,j} & 0 \\ \tilde{v}_i^{j,j} & \tilde{\omega}_i^{j,j} \end{pmatrix} \begin{pmatrix} R_i^j & 0 \\ \tilde{o}_i^j R_i^j & R_i^j \end{pmatrix} \\ &= \text{ad}_{T_i^{j,j}} \text{Ad}_{H_i^j}, \end{aligned}$$

where we used the definition (6.14), $v_i^{j,j} = \dot{o}_i^j - \tilde{\omega}_i^{j,j} o_i^j$, and the Jacobi identity for the vector product in the form $(\tilde{\omega} o) = \tilde{\omega} \tilde{o} - \tilde{o} \tilde{\omega}$. $\square$

Theorem 8.21 (Lie–Poisson³ equation for a rigid body). *In an arbitrary coordinate frame Ψ_c , the dynamics of a rigid body is governed by the equation*

$$\dot{P}^c - P^c \text{ad}_{T_c^{c,0}} = W_{\text{total}}^c,$$

or in column form,

$$(\dot{P}^c)^\top - \text{ad}_{T_c^{c,0}}^\top (P^c)^\top = (W_{\text{total}}^c)^\top,$$

where

$$\text{ad}_{T_c^{c,0}} = \begin{pmatrix} \tilde{\omega}_c^{c,0} & 0 \\ \tilde{v}_c^{c,0} & \tilde{\omega}_c^{c,0} \end{pmatrix} \in \mathbb{R}^{6 \times 6}.$$

Proof. Applying the coordinate transformation to momentum and wrench in (8.3) and using (8.4), we get

$$\begin{aligned} W_{\text{total}}^c \text{Ad}_{H_0^c} &= \frac{d}{dt} (P^c \text{Ad}_{H_0^c}) \\ &= \dot{P}^c \text{Ad}_{H_0^c} + P^c \dot{\text{Ad}}_{H_0^c} \\ &= \dot{P}^c \text{Ad}_{H_0^c} + P^c \text{ad}_{T_0^{c,c}} \text{Ad}_{H_0^c}. \end{aligned}$$

³Siméon Denis Poisson (1781–1840), French mathematician and physicist.

Multiplying both sides by $\text{Ad}_{H_c^0}$ from the right and using the fact that $T_0^{c,c} = -T_c^{c,0}$, we obtain

$$\dot{P}^c - P^c \text{ad}_{T_c^{c,0}} = W_{\text{total}}^c.$$

□

Chapter 9

Dynamics of Serial Chain Robots

In this chapter, we continue studying serial-chain mechanisms introduced in Chapter 7, now focusing on their dynamics. We consider a serial-chain robot with n simple joints and the configuration space $\mathcal{Q}$, so that $\dim \mathcal{Q} = n$.

9.1 Dual of the Jacobian map

Recall the differential kinematics map (7.9)

$$\begin{aligned} J(q) : T_q \mathcal{Q} &\rightarrow \mathfrak{se}(3) \\ \dot{q} &\mapsto J(q) \dot{q} = T_n^{0,0} \end{aligned}$$

that linearly maps joint velocities of a serial-chain robot $\dot{q}$ to the corresponding twist of the end-effector $T_n^{0,0}$ expressed in the base frame, where $J(q)$ is the geometric Jacobian of the robot at the configuration q . Its dual linear map

$$\begin{aligned} (J(q))^* : \mathfrak{se}(3)^* &\rightarrow T_q^* \mathcal{Q} \\ W^0 &\mapsto (J(q))^*(W^0) = W^0 J(q) =: \tau \end{aligned} \tag{9.1}$$

maps a wrench W^0 applied to the end-effector expressed in the base frame to the *generalised joint forces* τ induced by this wrench. If a wrench W^0 is applied to the end-effector at the configuration $q \in \mathcal{Q}$, it induces the generalised forces at the joints of the robot given by

$$\tau = W^0 J(q) \in T_q^* \mathcal{Q}.$$

Here, we implicitly identify the cotangent space $T_q^* \mathcal{Q}$ with $\mathbb{R}^n$ and represent the generalised forces as a row of n components. In column notation, the map reads

$$\tau^\top = J^\top(q) (W^0)^\top.$$

By the property of the dual map, for any $W^0 \in \mathfrak{se}(3)^*$ and any $\dot{q} \in T_q \mathcal{Q}$, we have

$$\langle W^0 \mid T_n^{0,0} \rangle = \langle \tau \mid \dot{q} \rangle,$$

which constitutes the two equivalent representations of the power: in the end-effector (task) space and in the joint space.

It is important to note that, given an arbitrary wrench $W^0 \in \mathfrak{se}(3)^*$, the map (9.1) generally does not produce the generalised forces $\tau = W^0 J(q)$ applying which at the joints, for instance using the robot actuators, would make the end-effector exert the wrench W^0 to the environment. Instead, applying $\tau = W^0 J(q)$ at the joints, is dynamically equivalent to applying the wrench W^0 to the end-effector link. One may use this property to design control schemes for robot manipulators. In such a scheme, one comes up with a wrench W^0 that would act on the end-effector as a (six-dimensional) virtual spring pulling the end-effector towards a desired configuration. Then, the application of the corresponding generalised forces $\tau = W^0 J(q)$ at the joints emulates the effect of the virtual spring pulling the end-effector towards the desired configuration. Remarkably, this approach does not require computing the inverse kinematics of the robot or the inverse of the Jacobian and works even in the presence of singularities, redundancies or for underactuated robots.

9.2 Equations of motion of a serial-chain robot

The dynamics of a mechanical system with configuration space $\mathcal{Q}$ is governed by the *Euler–Lagrange*¹ equations:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial q^i} = (\tau_{\text{total}})_i \quad \text{for } i \in \{1, \dots, n\},$$

where $L(q): T_q \mathcal{Q} \rightarrow \mathbb{R}$ is the *Lagrangian* of the system defined as the difference between its kinetic and potential energies

$$L(q, \dot{q}) = K(q, \dot{q}) - V(q),$$

and $\tau_{\text{total}} = (\tau_1, \dots, \tau_n) \in T_q^* \mathcal{Q}$ are the total generalised forces acting on the system. For a serial-chain robot, τ_{total} may include the generalised forces produced by the actuators at the joints, forces induced by interactions with the environment, friction, and all other external forces acting on the robot.

Proposition 9.1. *The total kinetic energy of a serial-chain robot is given by*

$$K(q, \dot{q}) = \frac{1}{2} \dot{q}^\top M(q) \dot{q},$$

where the mass matrix $M(q)$ is defined as the sum of the contributions of individual links

$$M(q) := \sum_{i=1}^n M_{(i)}(q), \quad \text{with } M_{(i)}(q) := J_{(i)}^\top(q) \mathcal{I}_{(i)}^0(q) J_{(i)}(q) \in \mathbb{R}^{n \times n},$$

where $\mathcal{I}_{(i)}^0(q) \in \mathbb{R}^{6 \times 6}$ is the inertia tensor of the i -th link expressed in the inertial frame Ψ_0 , and $J_{(i)}(q) \in \mathbb{R}^{6 \times n}$ is the concatenated geometric Jacobian of the i -th link

¹Joseph-Louis Lagrange (born Giuseppe Luigi Lagrangia, 1736–1813), Italian-French mathematician and physicist.

defined as

$$J_{(i)}(q) := \begin{pmatrix} \begin{array}{c} | \\ \hat{T}_1^{0,0}(q) \\ | \end{array} & \cdots & \begin{array}{c} | \\ \hat{T}_i^{0,(i-1)}(q) \\ | \end{array} & \begin{array}{c} | \\ 0 \\ | \end{array} & \cdots & \begin{array}{c} | \\ 0 \\ | \end{array} \end{pmatrix} \in \mathbb{R}^{6 \times n}.$$

The mass matrix $M(q)$ is symmetric and positive definite for all $q \in \mathcal{Q}$.

Proof. The twist of the i -th link can indeed be represented using the concatenated geometric Jacobian as

$$T_i^{0,0} = J_{(i)}(q) \dot{q}.$$

Then the kinetic energy of the i -th link is given by

$$K_{(i)} = \frac{1}{2} (T_i^{0,0})^\top \mathcal{I}_{(i)}^0 T_i^{0,0} = \frac{1}{2} \dot{q}^\top J_{(i)}^\top \mathcal{I}_{(i)}^0 J_{(i)} \dot{q} = \frac{1}{2} \dot{q}^\top M_{(i)} \dot{q},$$

from which the total kinetic energy follows by summation over all links. The symmetry of the mass matrix follows from the symmetry and positive definiteness of the inertia tensors $\mathcal{I}_{(i)}^0$ and the fact that the geometric Jacobian is composed of *unit* twists. $\square$

Remark 9.2. At each point of the configuration space $q \in \mathcal{Q}$, the mass matrix $M(q)$ comprises the components of a symmetric, non-degenerate $\binom{0}{2}$ -tensor on the tangent space $T_q \mathcal{Q}$, thus defining a Riemannian metric on the configuration space $\mathcal{Q}$.

Theorem 9.3. Consider a serial-chain robot with configuration space $\mathcal{Q}$, mass matrix $M(q)$, generalised joint forces τ , and potential energy $V: \mathcal{Q} \rightarrow \mathbb{R}$. Its motion is governed by the following equation:

$$M(q) \ddot{q} + C(q, \dot{q}) \dot{q} + G(q) = \tau^\top, \quad (9.2)$$

where $G(q) := \frac{\partial V}{\partial q}$, and

$$C_{ij}(q, \dot{q}) := \sum_{k=1}^{\dim \mathcal{Q}} \Gamma_{ijk}(q) \dot{q}^k \quad \text{with} \quad \Gamma_{ijk}(q) := \frac{1}{2} \left(\frac{\partial M_{ij}}{\partial q^k} + \frac{\partial M_{ik}}{\partial q^j} - \frac{\partial M_{kj}}{\partial q^i} \right).$$

Proof. The two terms of the Euler–Lagrange equations

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial q^i} = \tau_i$$

with the Lagrangian

$$L(q, \dot{q}) = K(q, \dot{q}) - V(q) = \frac{1}{2} \sum_{i,j=1}^n M_{ij}(q) \dot{q}^i \dot{q}^j - V(q)$$

are computed as

$$\begin{aligned}
\frac{d}{dt} \frac{\partial L}{\partial \dot{q}^i} &= \frac{d}{dt} \left(\sum_{j=1}^n M_{ij}(q) \dot{q}^j \right) \\
&= \sum_{j=1}^n (M_{ij}(q) \ddot{q}^j + \dot{M}_{ij}(q) \dot{q}^j) \\
&= \sum_{j=1}^n \left(M_{ij}(q) \ddot{q}^j + \sum_{k=1}^n \frac{\partial M_{ij}}{\partial q^k} \dot{q}^k \dot{q}^j \right) \\
&= \sum_{j=1}^n M_{ij}(q) \ddot{q}^j + \frac{1}{2} \sum_{j,k=1}^n \left(\frac{\partial M_{ij}}{\partial q^k} \dot{q}^k \dot{q}^j + \frac{\partial M_{ik}}{\partial q^j} \dot{q}^j \dot{q}^k \right)
\end{aligned}$$

and

$$\frac{\partial L}{\partial q^i} = \frac{1}{2} \sum_{j,k=1}^n \frac{\partial M_{jk}}{\partial q^i} \dot{q}^j \dot{q}^k - \frac{\partial V}{\partial q^i}.$$

□

Remark 9.4. Setting the generalised forces τ and the potential energy V to zero, the equation (9.2) describes a geodesic curve on the configuration space $\mathcal{Q}$ with the Riemannian metric $M(q)$.

Bibliography

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